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Physics 7B

Worksheets

T-1. Ideal Gases: The Ideal Gas Law and Internal Energy

Part 1: The Ideal Gas Law and the p-V Diagram

Questions for discussion (Part 1)

1. An ideal gas confined to a box exerts pressure on the walls of the box. Where does this pressure come from? (In other words, what is going on microscopically?)

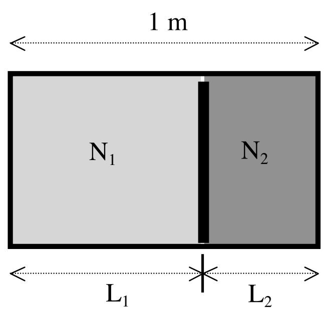
2. How could you measure the pressure of a gas? ("Use a pressure gauge" is not an answer, unless you also explain how a pressure gauge works. The same goes for barometers, etc.)

3. If you cause an ideal gas to contract, does the temperature go up or down? Explain.

Problems (Part 1)

You should complete your work for the "Problems" on *separate* sheets of paper. Do not work in the margins below; you'd like to be able to make sense of your work later when you review for the exams!

1. A box of length 1 meter and cross-sectional area A has a moveable partition inside it. There is some gas on either side of the partition.



The number of particles on the left is $N_1 = 3{\times}10^{23}$. The number of particles on the right is $N_2 = 2{\times}10^{23}$. The gas on both sides is in thermal equilibrium at the same temperature T. When the partition settles down to its final position, find the lengths L_1 and L_2 of the left and right sides of the box. ❖

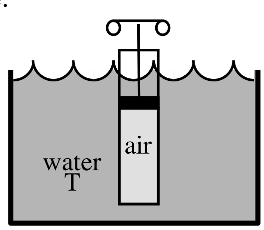
2. Your baby brother is toddling around the house, playing with a toy balloon. You decide to teach him something about physics, so you take the balloon away from him and hide it in the freezer. (His cries are enough to convince you that he is excited about this learning experience.)

When you put the balloon in the refrigerator, it will shrink. (Try it and see!)

a) If the temperature inside your house is 25° Celsius, while the temperature inside the freezer is -10° Celsius, then by what percentage will the volume of the balloon change? We will take the pressure inside the balloon as constant.

c) Sketch a p-V diagram for the gas inside the balloon as it cools inside the regrigerator. Label the axes as completely as possible according to the given information. ❖

3. A canister with thin metal walls is immersed in water with temperature T. (See figure.) Initially the canister holds air at atmospheric pressure.



Then, using the piston that forms the lid of the canister, you compress the air until it occupies half of its original volume.

a) Assuming that the water maintains the air at temperature T throughout the process, what will be the final air pressure in the canister?

b) Sketch this process on a p-V diagram. Label the axes as completely as possible according to the given information. ❖

4. Now would be a good time to look back over your answers to the Discussion Questions above.

a) Do any of the diagrams on this worksheet suggest answers to Discussion Question 1?

b) Does your answer to Discussion Question 3 still make sense, in light of your answers to Problems 3 and 4? ❖❖

Part 2: Energy Contained in an Ideal Gas

Summary

The *total energy of a system* of particles is called the "internal energy" of the system, $E_{internal}$.

<u>Equipartition theorem</u>: If a system is in thermal equilibrium at temperature T, then each independent quadratic term (or *degree of freedom*) in its energy has average value equal to (1/2)kT.

For a *single* particle <u>of any type</u>

$$\langle KE_{\text{translational}}\rangle = \langle \frac{1}{2} m v_x^2\rangle + \langle \frac{1}{2} m v_y^2\rangle + \langle \frac{1}{2} m v_z^2\rangle$$

$$\langle KE_{\text{translational}}\rangle = \frac{3}{2} kT$$

(*d* degrees of freedom)

$$E_{internal} = \frac{d}{2} NkT$$

For a *system* of <u>monatomic</u> particles

$$E_{internal} = N\langle KE_{translational}\rangle$$

$$E_{internal} = \frac{3}{2} NkT$$

For a *system* of <u>diatomic</u> particles

$$E_{internal} = N\langle KE_{translational}\rangle + N\langle \frac{1}{2} I_x \omega_x^2\rangle + N\langle \frac{1}{2} I_y \omega_y^2\rangle$$

(at medium temperature)

$$E_{internal} = \frac{5}{2} NkT$$

For a *system* of particles with *d* degrees of freedom

$$E_{internal} = \frac{d}{2} NkT$$

Questions for discussion (Part 2)

1. A box with total volume V_o is divided in half by a partition. On the left-hand side of the partition, there is a sample of monatomic ideal gas with initial pressure p_o and initial temperature T_o. On the right-hand side of the partition, the box is empty.

The partition is then suddenly removed, and the gas expands freely to fill the entire box. Soon the gas is in thermal equilibrium again.[1]

a) What is the final temperature of the gas? Explain.

b) What is the final pressure of the gas?

c) Can you explain why the pressure has changed? (Note: "pV = NkT" is not an explanation!) Hint: Think about where the pressure comes from: When the box suddenly doubles in size, what can you say about the particles' collisions with the walls?

2. I have two samples of ideal gas, identical except that sample A is at temperature $T_A = 100$ K, while sample B is at temperature $T_B = 400$ K. (Both samples are in thermal equilibrium at their respective temperatures.) If you could somehow become microscopically tiny, and could see the gas particles close up, then what difference would you observe between the particles of A and the particles of B? (Try to answer both qualitatively *and* quantitatively.)

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[1] Note: For further discussion of this situation, see "Free Expansion of an Ideal Gas," in the Supplementary Material at the end of the workbook.

3. Again we have two samples of ideal gas A and B, each in thermal equilibrium. But this time the two gases are at the same temperature. And this time, the gases are of different kinds, with the particles of A being twice as massive as the particles of B.

a) Compared with the particles in sample A, are the particles in sample B moving faster, slower, or at the same speed, on average?

b) If faster or slower, then by what factor?

4. Consider once again two samples of ideal gas A and B, each in thermal equilibrium. The samples are both at room temperature. Furthermore, a given particle of A has the same mass as a given particle of B. The difference this time is in the *structure* of the particles: the particles of A are *monatomic*, whereas the particles of B are *diatomic*.

a) Compared with the particles in sample A, are the particles in sample B moving faster, slower, or at the same speed, on average? If faster or slower, then by what factor?

b) Compared with the particles in sample A, does a typical particle in sample B have more kinetic energy, less kinetic energy, or the same kinetic energy, on average? If more or less, then by what factor?

5. If an ideal gas expands or contracts isothermally, how does the total internal energy E_{int} change? Explain.

6. Consider once again two samples of ideal gas A and B, each in thermal equilibrium. Both samples are at the same temperature T. Both samples A and B are made up of diatomic molecules of the same mass. However, in addition to the translational kinetic degrees of freedom and the rotational degrees of freedom, the molecules in sample B can *vibrate* like a spring. Compared with sample A, do the particles in sample B have a larger, smaller, or the same averate total energy, on average? If different, then by what factor?

Problems (Part 2)

1. A gas of $N = 7\times10^{25}$ diatomic particles initially has a pressure $p = 1.65$ atm and a volume $V_1 = 3.7\ m^3$. The gas contracts at constant pressure until it has a volume $V_2 = 2.9\ m^3$.

a) Sketch this process on a p-V diagram.

b) Determine the initial and final temperatures of the gas.

c) Determine the change in the internal energy of the gas ΔE_{int} for this process.

d) How could an engineer make a gas contract at constant pressure? ❖

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2. A diatomic ideal gas has initial pressure p_1 and initial volume V_1. The gas then undergoes a series of three transformations:

- First, a bunsen burner causes the gas to expand, at constant pressure, to volume $7V_1$.
- Next, the volume is held constant while an ice bath lowers the pressure to $p_1/4$.
- Finally, a water bath allows the gas to be compressed along a straight line in the p-V plane, until the pressure and the volume return to their initial values.

a) Sketch this cycle of transformations on a p-V diagram.

b) Find the temperature at all three "corners" of the cycle. Express all three temperatures in terms of p_1, V_1, and N.

c) Find ΔE_i, the change in the internal energy of the gas during transformation (i). Likewise, find ΔE_{ii} and ΔE_{iii}. (Express all three answers in terms of p_1 and V_1.)

d) Add up the three changes in internal energy: $\Delta E_i + \Delta E_{ii} + \Delta E_{iii}$. Why do you get zero for the total change in internal energy over the cycle? ❖❖

T-2. Thermal Expansion, Kinetic Theory, and Calorimetry

Part 1: Thermal Expansion

Questions for discussion (Part 1)

1. Imagine a metal disk with a hole cut out of it (an annulus). If you increased the temperature, does the hole get bigger or smaller? Explain.

2. Metal has a larger coefficient of linear expansion than glass. With that in mind, you wish to remove a metal lid from a glass jar. Do you run it under cold water or hot water? Explain?

Problems (Part 1)

1. Suppose you have a ring of a metal with $a_1 = 2 \times 10^{-6}$ $(C^0)^{-1}$. You wish to put it around a pipe made of metal with $a_2 = 3 \times 10^{-6}$ $(C^0)^{-1}$. If at 25 C^0 the inner radius of the ring is 10.0 cm and the outer radius of the pipe is 10.001 cm, what is the temperature that will allow you to slip the ring around the pipe?

2. You have a metal sphere of $a_1 = 2 \times 10^{-6}$ $(C^0)^{-1}$. How many degrees do you need to increase the temperature to increase the volume of the sphere by 1%?

Part 2: Kinetic Theory

The Maxwell Distribution describes the *distribution of speeds* of individual particles at a given temperature T. The function is:

$$f(v) = 4\pi N\left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{\frac{-mv^2}{2kT}}$$

where v is the speed, N is the total number of molecules, T is the temperature, m is the mass of each particle, and k is Boltzmann's constant.

Questions for discussion (Part 2)

1. Why do puddles evaporate, even if the temperature is much colder than the boiling point of water? *(Hint: to become a vapor, a water molecule needs to be moving fast enough to escape the surface tension of the water.)* Why do sealed jars never evaporate?

2. Plot a typical Maxwell Distribution for some value of N and T. What would it look like if you increased the temperature, keeping N constant? What would it look like if you increased the number of molecules, but kept T constant?

Problems (Part 2)

1. An ideal gas consists of N particles in thermal equilibrium at temperature T. We wish to show that the equipartition theorem is consistent with the Maxwell distribution of speeds.

a) On average, how many particles will have a speed between v and v+dv?

b) Write an expression for the average of a function of v, g(v), for the Maxwell Distribution (Hint: Think of how you would compute the average if we had a finite distribution such as N_1 particles of speed v_1, N_2 particles of speed v_2, etc.)

c) Verify that your formula is correct by finding that the average of the function g(v) = 1 is, indeed, 1.

d) Find the average value of the kinetic energy of the particles, $g(v) = \frac{1}{2}mv^2$.

e) Does your answer to part (d) agree with what you would have expected from the equipartition theorem?

The following integrals will be useful:

$$\int_0^\infty u^2 e^{-u^2}\,du = \frac{\sqrt{\pi}}{4}$$

$$\int_0^\infty u^4 e^{-u^2}\,du = \frac{3\sqrt{\pi}}{8}$$ ❖

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2. You have 10 diatomic gas molecules in a box. At one moment, two have a speed of 10 m/s, four have a speed of 12 m/s, two have a speed of 14 m/s, one has a speed of 15 m/s, and one has a speed of 17 m/s. The gas molecules have a total mass m.

a) Calculate the average speed and the rms speed.

b) Using its strict definition, what would the "temperature" be for this theoretical distribution? Leave your answer in terms of m and k_B.

c) What would the total internal energy be, if the kinetic theory, ideal gas law, and equipartition theory hold?

d) Why would they probably not hold in this problem? ❖

Part 3: Calorimetry

Questions for discussion (Part 3)

1. Is it possible for H_2O to be in liquid form at 0^oC?

2. Suppose you have a glass of water at 0^oC. Is it possible to draw heat out of the water without lowering its temperature? If heat energy *can* leave the system without a corresponding decrease in temperature, then where is that heat energy coming from?

3. Why can you get a more severe burn from steam at 100 ^oC than from water at 100 ^oC?

4. Why do coastal regions tend to have a more moderate climate than inland regions?

Problems (Part 3)

1. A copper sample (c_{Cu} = 387 J/kg·K) of mass m_c = 75 g and temperature T_c = 312^0C is dropped into a glass beaker that contains a mass of water m_w = 220 g (c_w = 4190 J/kg·K).

The *heat capacity* (the specific heat times the mass) of the beaker is C_b' = 190 J/K.

The initial temperature of the water and the beaker is $T_{w,b}$ = 12.0^0C.

What is the final temperature of the copper, beaker, and water? ❖

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2. What mass of steam at 100^0C must be mixed with 150 g of ice at -10^0C, in a thermally insulated container, to produce liquid water at 50^0C?

Data: L_v = 2256 kJ/kg
L_f = 333 kJ/kg
c_w = 4190 J/kg·K
c_{ice} = 2220 J/kg·K ❖❖

T-3. Heat Transfer: Conduction and Radiation

Part 1: Heat Flow by Conduction

<u>Summary</u>
The rate of heat flow by conduction, H, through an object of cross-sectional area A, length ¬, and thermal conductivity k is given by

$$\frac{dQ}{dt} = \text{H} = \frac{kA}{\ell}(\text{T}_1 - \text{T}_2),$$

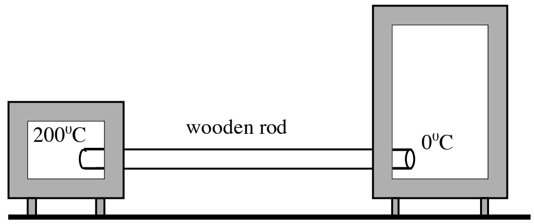
where T_1 and T_2 are the temperatures of the two ends of the object. In MKS, H has units of J/s. [Giancoli writes this rate H as $\Delta Q/\Delta t$.] In Physics 7B, we will only consider situations where the system has reached *steady state*: the rate of heat flow through the object is constant in time, so that the temperature at each point in the object is constant in time.

Questions for discussion (Part 1)

1. You are standing in your bathroom with bare feet, one foot on the tile floor, and the other on a rug. You notice that the tile feels colder than the rug. Are they not at the same temperature? Explain.

2. You are able to reach into a hot oven without getting burned, but you will be burned if your hand brushes the metal rack or a baking dish inside. Explain.

3. A wooden rod has length L and cross-sectional area A. One end of the rod is maintained at 200^{0}C by an oven. The other end of the rod is maintained at 0^{0}C by a refrigerator.

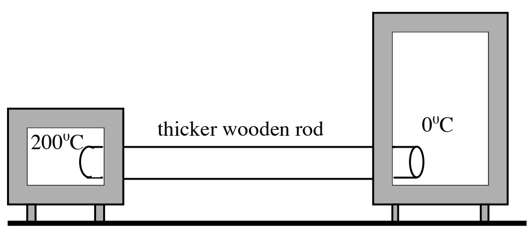


Because of the temperature difference across the rod, heat flows through the rod. As usual, we assume that everything has settled into a steady state, so that the temperature distribution in the rod is not changing with time.

Numerically, we suppose that the rate of heat flow is 50 Joules per second.

a) Will the temperature of the rod at its midpoint be greater than 100^{0}C, less than 100^{0}C, or equal to 100^{0}C? Explain.

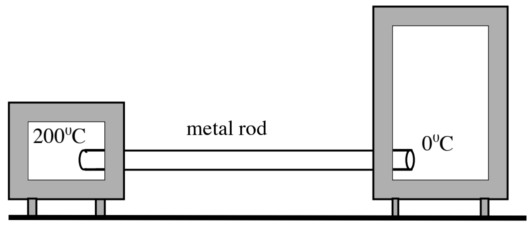
4. Now suppose that the original wooden rod is replaced by another wooden rod, with the same length L, but with cross-sectional area 2A.



a) At what rate will heat flow through this new rod? (Give a numerical answer.)

b) Will the temperature of this rod at its midpoint be greater than 100^{0}C, less than 100^{0}C, or equal to 100^{0}C?

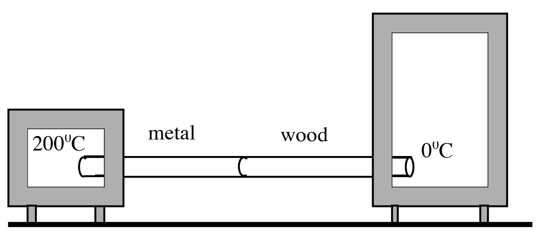
5. Next, suppose that the second wooden rod is replaced by a *metal* rod, with cross-sectional area A and length L.



a) Will the temperature of this rod at its midpoint be greater than 100°C, less than 100°C, or equal to 100°C?

b) Will the rate of heat flow through the metal rod be greater than 50 J/sec, less than 50 J/sec, or equal to 50 J/sec? Explain.

6. Finally, suppose that the metal rod is replaced by a composite rod, with cross-sectional area A and length L. One half of the rod is wood, and the other half is metal.



a) How does the rate of heat flowing *into* the junction of the composite rod compare to the rate of heat flowing *out of* the junction?

b) Will the rate of heat flow across the composite rod be greater than 50 J/sec, less than 50 J/sec, or equal to 50 J/sec? Explain.

c) How will the rate of heat flow through the *wood* portion of the rod compare to the rate of heat flow through the *metal* portion of the rod? Explain.

d) Will the temperature of this rod at its midpoint be greater than 100^0C, less than 100^0C, or equal to 100^0C? Explain.

Problems (Part 1)

1. For a composite rod like shown above in Discussion Question 6, each section has length 0.75 m and a cross-sectional area 4 cm^2. Suppose that the thermal conductivities of wood and metal are k_w = 1 and k_m = 14, in MKS units.

a) Find the temperature at the midpoint of the composite rod.

b) Find the rate of heat flow across the composite rod. ❖

2. A container of water has been outdoors in cold weather until a 5.0-cm thick slab of ice has formed on its surface. The air above the ice is at -10^0C.

Calculate the rate of formation of ice (in cm/hr) on the bottom surface of the slab.

Data:
k_{ice} = 1.7 W/m·K
ρ_{ice} = 0.92 g/cm^3
$L_{f,ice}$ = 333 kJ/kg

Assume that the walls of the container are thermally insulating. ❖

Part 2: Heat Transfer by Radiation

<u>Summary</u>

The rate of heat lost to radiation, P_{out}, through an object of *surface* area A and *emissivity* ε is

$$\frac{dQ}{dt} = P_{out} = \varepsilon\sigma A T^4$$

where T is the temperature of the object. In MKS, P has units of J/s. The emissivity is a number between 0 and 1 which tells how good of an emitter of radiation the object is. An object with an emissivity of 1 is a perfect black body. The symbol $\sigma = 5.67\times10^{8}\ \mathrm{W/m^2 \cdot K^4}$ is the Stefan-Boltzmann constant.

If radiation is hitting an object, then the rate of heat gained by absorption, P_{in}, is

$$\frac{dQ}{dt} = P_{in} = \varepsilon A S$$

where ε is again the emissivity, A is the *cross-sectional* area of the object as seen by the incoming radiation, and S is the *power flux*, or the amount of heat due to radiation perpendicularly incident on a unit area of the object per unit time. When the incoming radiation is from the sun, S is known as the *solar constant*. The solar constant for the earth is approximately $S \approx 1350\ \mathrm{W/m^2}$.

Questions for discussion (Part 2)

1. Given a spherical blackbody, what can you say about the distribution of emitted radiation?

2. Consider a sheet of metal. How does the power emitted on one side compare to the power emitted on the other? What about it you paint one side black and the other side white?

3. Why is it better to wear a white shirt rather than a black shirt on a hot day out in the sun?

4. Given a system in a steady state situation, how does the power absorbed compare to the power emitted?

5. In calculating the power absorbed by the earth from the sun, what area should we use? Why?

6. Given two bodies giving off the same total power, how do their temperatures compare if one body has four times the emissivity of the other? How about if one body has twice the radius of the other?

7. A body at temperature T_1 is immersed in a heat bath at temperature T_2. What is the net rate of heat loss due to radiation in this case?

Problems (Part 2)

1. Consider a simplified version of the Earth-Sun system in which both bodies are perfect blackbodies at uniform temperatures and in a steady-state situation.

a) Given the radius of Earth's orbit, r_0, the radius of the sun, r_S, and the temperature of the sun, *S*, find the solar constant.

b) Using T_S=5770K, r_E=149.6×10^6 km, and r_0=6.96×10^5 km, find a numerical value for *S*.

c) Why is your answer for *S* different from the one quoted at the beginning of this section?

d) With your expression for *S* as a function of T_S, r_S, and r_0, and the radius of the Earth (r_E = 6.38×10^3 km, calculate the average temperature of the Earth (consider only radiative effects). ❖

2. An object with surface area A is placed in an oven which is maintaining temperature T_{oven}. The object has a specific heat c, a mass m, and an initial temperature T_0. In this problem, ignore any heat transfer by conduction or convection, and assume the oven doesn't lose any heat to the outside environment. Also assume that the time for the objects to reach a steady state is much faster than any other time scales in this problem (so we will always assume our system is in a 'steady state') At time t, the object has temperature T(t).

a) At time t, what is the net rate of heat gain by the object?

b) Find the temperature of the object at a time t (You need only write out a differential equation. Only actually solve this if you are overly ambitious or have access to a table of integrals). ❖❖

T-4. The First Law of Thermodynamics

The First Law of Thermodynamics

$\Delta E_{int} = Q_{into\ gas} - W_{by\ gas}$

$$W_{1\rightarrow 2,\text{by gas}} = \int_{V_1}^{V_2} p\,dV$$

Questions for discussion

In the following, you may assume that all of the processes described are reversible.

1. When an ideal gas undergoes adiabatic expansion[1], the temperature

_____ goes up
_____ goes down
_____ stays the same
_____ may do any of these.

Explain your reasoning.

Would your answer differ if the gas underwent adiabatic *compression*?

2. When an ideal gas undergoes isothermal compression, the internal energy of the gas

_____ increases
_____ decreases
_____ stays the same
_____ may do any of these.

Explain your reasoning.

Would your answer differ if the gas underwent isothermal *expansion*?

[1] Note: For a derivation of the adiabatic expansion formula, see the challenge problem at the end of this worksheet.

3. When an ideal gas undergoes isothermal expansion,

_____ heat flows into the gas
_____ heat flows out of the gas
_____ there is no heat flow in or out
_____ any of these is possible.

Explain your reasoning.

Would your answer differ if the gas underwent isothermal *compression*?

4. If heat is added to a gas while the gas is held at constant volume, then the temperature of the gas

_____ must increase
_____ must decrease
_____ must stay the same
_____ may do any of these.

Explain your reasoning.

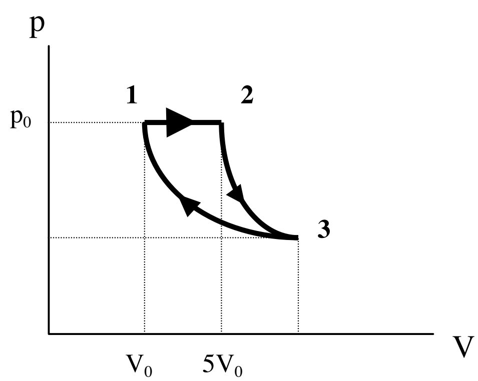
Suppose instead that the same amount of heat is added to the gas while the gas is held at constant *pressure* and the volume is allowed to vary. In this case, it turns out that the *qualitative* effect on the temperature is the same: namely, the temperature increases. But will it increase by the same amount? Explain.

5. When you let air out of a tire, the air feels cool. Explain. [Hint: air is a poor thermal conductor, so this process is approximately adiabatic.]

Problems

1. An ideal gas of N diatomic molecules ($\gamma = 1 + 2/5 = 7/5$) undergoes three consecutive transformations, as diagrammed below.

- The transformation $1 \rightarrow 2$ is *isobaric*.
- The transformation $2 \rightarrow 3$ is *adiabatic*.
- The transformation $3 \rightarrow 1$ is *isothermal*.



a) In terms of p_0, V_0, and N, find the temperatures T_1, T_2, T_3 at all three "corners" of the cycle.

b) In terms of V_0 and γ, find the volume V_3 at point 3 of the cycle.

c) In terms of p_0 and γ, find the pressure p_3 at point 3 of the cycle.

d) For each of the transformations $1 \rightarrow 2$, $2 \rightarrow 3$, $3 \rightarrow 1$, find the changes in internal energy $\Delta E_{1\rightarrow2}$, $\Delta E_{2\rightarrow3}$, $\Delta E_{3\rightarrow1}$.

- First hint: One of these is zero. Why?
- Second hint: What should the three changes $\Delta E_{1\rightarrow2}$, $\Delta E_{2\rightarrow3}$, $\Delta E_{3\rightarrow1}$ add up to?

e) For each transformation, find the work done by the gas on its environment. Express your answer in terms of p_0 and V_0.

- Hint: You should be able to write down $W_{2\rightarrow3}$ without doing any new calculations.

f) For each transformation, find the amount of heat flow into or out of the gas. ❖

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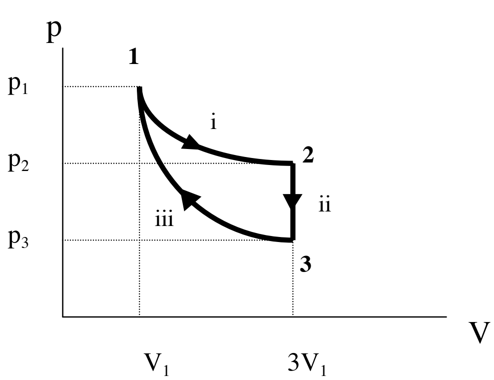
2. One mole of a monatomic gas ($\gamma = 5/3$) undergoes the following cycle:

(i) It is expanded isothermally from V_1 to $3V_1$.
(ii) Its pressure is decreased from p_2 to p_3 at constant volume $3V_1$.
(iii) It is compressed adiabatically back to its initial state.



a) Find p_2 and p_3 in terms of p_1, V_1, and γ.

b) Find T_1, T_2, and T_3 in terms of p_1, V_1, and γ.

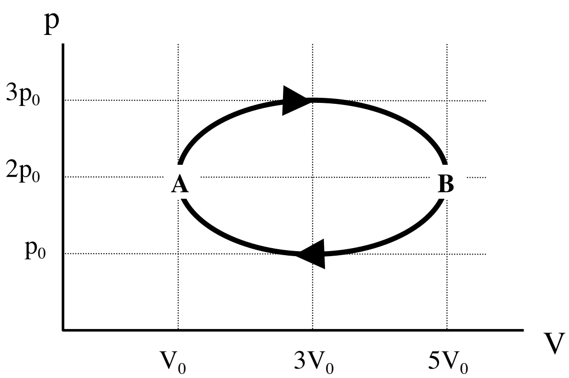
c) What is the direction of heat flow in step (i) (into or out of the gas) ? How about steps (ii) and (iii) ?

d) Find the amount of heat flow into or out of the gas during step (i), in terms of p_1, V_1, and γ. Then do likewise for step (ii).

e) Is the net work done by the gas during a complete cycle positive or negative? Explain. Is the net heat added to the gas during a complete cycle positive or negative? Explain.

f) Do you think this cycle represents a *heat engine* or a *refrigerator*? Why? ❖

3. A monatomic ideal gas undergoes a cyclic transformation as shown.



a) When the gas goes from A to B, what is the change in its internal energy $\Delta E_{A\to B}$?

b) When the gas goes from A to B, how much work $W_{A\to B}$ is done by it on its environment?

- Hint: The area of an ellipse with semimajor axis v and semiminor axis w is given by pvw.

c) How much heat $Q_{A\to B}$ flows into the gas during the transformation A→B?

d) Answer the same questions for the return transformation B→A.

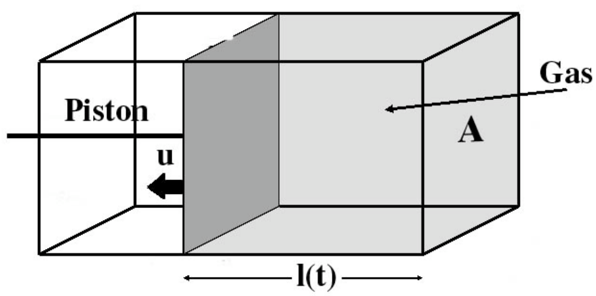
e) What is the *net* work done by the gas on its environment over the cycle? ❖❖

Now would be a good time to turn to the supplimentary worksheet "T-S1. Ideal Gas Transformations" and fill in what you can for future reference.

T-4 Challenge Problem

Adiabatic Expansion of an Ideal Gas - Derivation

In the Kinetic Theory and Ideal Gases worksheets, we discussed the origins of the Ideal Gas Law using a molecular picture of gas, with the pressure arising from collisions and the temperature arising from the random motions of the gas particles. We considered a gas is in a box with a partiton. Upon quickly removing the partition, we found that the temperature remained constant. That process was called free expansion and is our primary example of an irreversible process - that is, a process which cannot proceed the other way. In thermodynamics, and in particular heat engines, we will primarily be interested in reversible processes. One of the most important reversible process is the adiabatic expansion or compression of a gas. An adiabatic process is one in which our system does not exchange heat with the outside environment. For an ideal gas that is adiabatically changed from pressure P1 and volume V1 to a pressure P2 and volume V2, we have the relation $P_1V_1^\gamma = P_2V_2^\gamma$. In this challenge problem, we will derive this result, along with an expression for γ, using the same model that was used to derive the ideal gas law.



Consider a rectangular box with a movable piston, as shown in the figure above. The box has a cross-sectional area, A, and at a time, t, the piston is at a length $\ell(t)$ from the edge of the box. The box is filled with N particles of an ideal gas which has *d* degrees of freedom. First, some preliminaries.

a) If the piston is being pulled out with a constant speed, u, and at time t = 0 has a length ℓ_0, what is $\ell(t)$ and V(t)?

b) When the gas is at a temperature T, what is the total internal energy of the gas and what is the average value of v_x^2, the velocity of the gas particles in the x-direction squared? If the temperature is changed by an amount dT , what is the change in the energy, dE?

Now consider a single gas particle, which as a mass m and is initially moving towards the piston with velocity v_i (for the next few parts, we will just consider the one dimension that is shown in the diagram, so the velocity mentioned is really the x-component of velocity, and the energy will be the energy associated with motion in the x-direction rather than the total energy). The particle collides *elastically* with the piston and recoils with a velocity v_f. Since the piston is much more massive than the gas particles, in the piston's rest frame, an elasic collision means that the particle will recoil with the same velocity it was incident on the piston with.

c) In terms of v_i and the velocity of the piston, u, what is v_f?

d) What is the change in energy of the particle, $\Delta E = E_f - E_i$, in terms of v_i, m, and u?

The key feature of our setup that ensures we have an adiabatic expansion rather than a free expansion is that we are pulling the piston out slowly! We can see from our expression for ΔE what we mean by slowly: The speed of the piston should be much less than the velocity of the gas particles! Symbolically, $u \ll v_i$.

e) Use the fact that the piston is moving slowly to express the ΔE to lowest order in u.

Now, we've found the change in energy for a single particle. How about the whole gas, though? Consider an initial state at time t, and a final state an infinitessimally small time later, at t + dt.

f) What is the infinitessimal change in volume, dV?

g) Calling the average speed in the x-direction v_i, what volume must a particle be within to hit the wall sometime between times t and t + dt?

h) How many particles will hit the wall in this time interval? (There is a factor of 1/2 in this expression - why?)

i) What is the total change in energy of the gas in this time interval?

j) Use parts (a), (b), and (f) to write your answer from part (i) in terms of the variables *d*, V, dV, N, T, and dT .

k) Use separation of variables to solve the differential equation from part (j) using the inital points V_i and T_i and the final points V_f and T_f .

ℓ) Use the Ideal Gas Law to massage your answer to part (k) into the form of the equation we presented at the beginning. What is γ? ❖❖

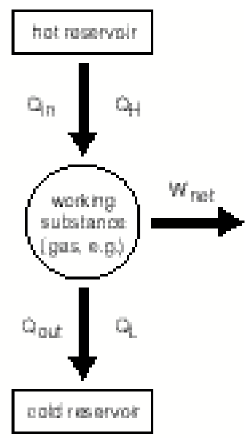
T-5. Engines and Efficiency

Questions for discussion

1. In common-sense language, efficiency is $\dfrac{\text{"what you want"}}{\text{"what you have to put in"}}$. Let's see how this common-sense notion of efficiency applies to heat engines and refrigerators.

a) A heat engine, such as a steam engine, takes advantage of the everyday fact that "heat wants to flow from hot to cold." The engine "siphons off" some of this flowing heat energy in the form of useful work. This is shown in the schematic diagram at right.

Keeping in mind the common-sense meaning of efficiency, how would you define the efficiency of a heat engine? Your definition of $e_{\text{heat engine}}$ should involve the quantities W_{net}, Q_H, and/or Q_L.

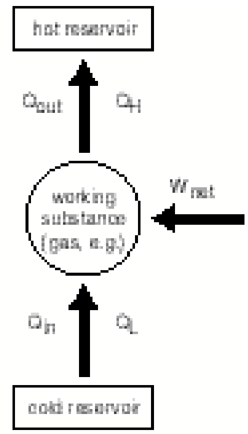


Schematic diagram of a heat engine, showing energy flow.

b) Use the First Law of Thermodynamics, together with the fact that the engine runs through a *cycle*, to show that your definition is equivalent to $e_{\text{heat engine}} = 1 - Q_L/Q_H$.

c) Your refrigerator forces heat energy to flow "against the grain," from the cold icebox to the warm kitchen. Naturally this requires an input of work, as shown in the schematic diagram (below).

Keeping in mind the common-sense definition of efficiency, how would you define the efficiency of a refrigerator? We actually call this the *coefficient of performance*, $K_{refrigerator}$. Your definition of $K_{refrigerator}$ should involve the quantities W_{net}, Q_H, and/or Q_L.



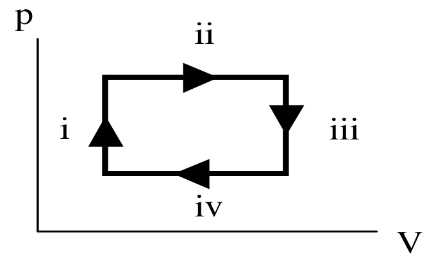
Schematic diagram of a refrigerator, showing energy flow.

d) In the winter, a heat pump uses energy to extract heat from the cold outdoors and pump it into your warm house. (So a heat pump is like a refrigerator for the outside air!) How would you define the efficiency, or coefficient of performance, for a heat pump? Your definition of K_{hp} should involve the quantities W_{net}, Q_H, and/or Q_L.

2. What does it mean for an engine to operate on the Carnot cycle?

3. Does the working substance of a Carnot engine have to be an ideal gas? If a Carnot engine uses a different substance, then can we still find the efficiency using the formula $e_C = 1 - T_L/T_H$?

4. A cyclic heat engine uses an ideal gas for its working substance. The engine operates on the following four-step process.



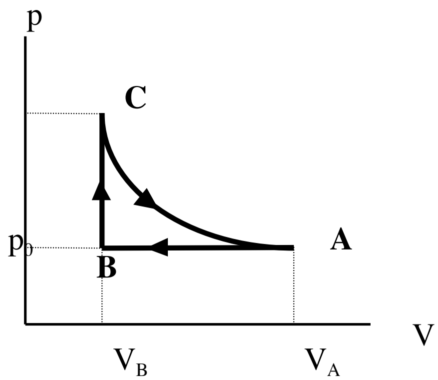
a) Think about what is going on during the first step of the cycle. Then decide: Is heat flowing into the gas or out of the gas during this step? (Think about the First Law. You should not need to calculate anything in order to decide.)

b) Is the efficiency of this engine given by $e = 1 - T_L/T_H$? If not, how could you calculate the efficiency?

Problems

1. A sample of monatomic ideal gas undergoes the cycle shown in the figure.

- A → B is *isobaric*
- B → C is *isochoric*
- C → A is *isothermal*.



a) Before doing any calculations, let's try to understand the energy flow in this cycle. For example, during the step A→B, does heat flow into the gas or out of the gas? (You should be able to answer without doing any calculations.) How about during steps B→C and C→A ?

b) Calculate the change in the gas's internal energy during each step of the cycle. (You may find it worthwhile to enter your results in a table like that shown below.) Answer in terms of p_0, V_A, and V_B.

Next, calculate the work done by the gas during each step of the cycle. What is the net work done on the gas during the cycle? Answer in terms of p_0, V_A, and V_B.

Finally, calculate the amount of heat flow into or out of the gas during each step. Again, answer in terms of p_0, V_A, and V_B.

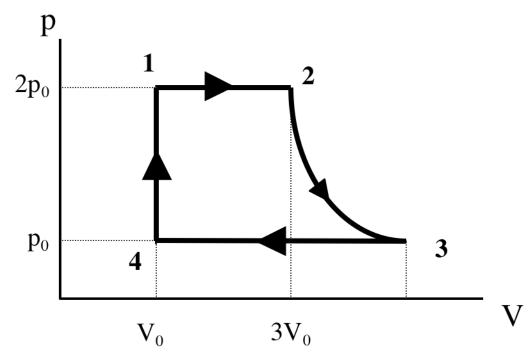
Step	ΔE_{int}	W	Q	
A → B				
B → C				
C → A				
		↑ W_{net}		↑ Q_{in}

c) Does this cycle represent a heat engine, or does it represent a refrigerator / heat pump? Explain.

d) Find the efficiency of this cycle. ❖

2. A heat engine uses an ideal gas of N monatomic particles as its working substance. The engine runs on the following four-step cycle.

- Transformation 1→2 is *isobaric*
- Transformation 2→3 is *adiabatic*
- Transformation 3→4 is *isobaric*
- Transformation 4→1 is *isochoric*



a) Find the efficiency of this engine. You may find it helpful to organize your work as follows:

Step	ΔE_{net}	W	Q	
1 → 2				
2 → 3				
3 → 4				
4 → 1				
		↑ W_{net}		↑ Q_{in}

b) Where in this cycle does the gas reach its highest and lowest temperatures? Find these extreme temperatures. (Your answers should be in terms of p_0, V_0, and N.)

c) What would be the efficiency of a *Carnot* engine operating between these two temperatures?

d) Are your answers consistent with the fact that the Carnot engine is the most efficient engine possible? ❖❖

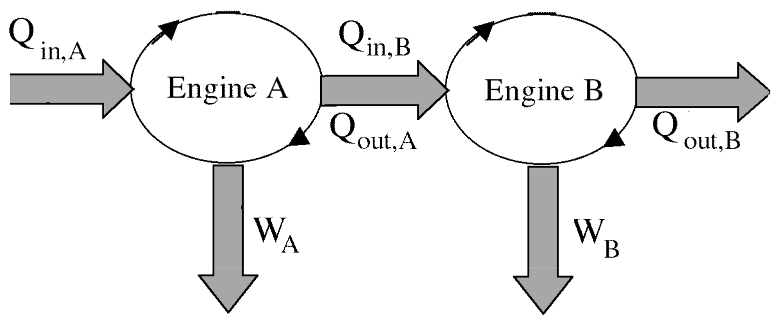
Now would be a good time to turn to the supplimentary worksheets "T-S1. Ideal Gas Transformations" and "T-S2. Efficiency of the Carnot Engine, The Long Way" and work through the problems.

T-5 Challenge Problem

Net Efficiency of Two Engines

In this challenge problem we will see how efficiencies add together. Since the total efficiency of an engine can never be greater than 1, then efficiencies obviously won't just be trivial to add together (otherwise, we could feed two efficiency 2/3 engines into eachother and get an efficiency of 4/3!

Consider two heat engines, Engine A and Engine B, with efficiencies e_A and e_B. We will create a composite engine, Engine C, by letting the heat output from Engine A be the heat input for Engine B, as shown schematically below.



a) If a heat $Q_{in,A}$ is fed into Engine A, what is the net work output and the total heat output from Engine A, W_A and $Q_{out,A}$ in terms of $Q_{in,A}$ and e_A?

b) If the heat input for Engine B is equal to the heat output of Engine A ($Q_{out,A} = Q_{in,B}$), what is the net work output and the total heat output from Engine B, W_B and $Q_{out,B}$ in terms of $Q_{in,A}$ and e_A?

c) What is the total work that is output from both engines as a result of feeding the engines the heat $Q_{in,A}$?

d) What is the net efficiency, e_c, of the combined engine system?

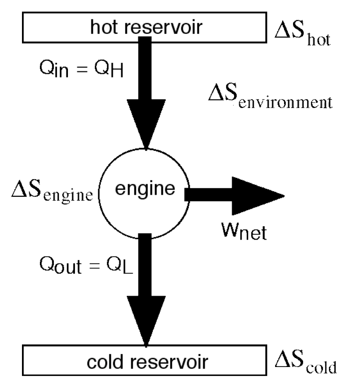
e) Show that if both $e_A<1$ and $e_B<1$, then $e_C<1$.

f) Suppose Engine A is a Carnot engine operating between temperatures T_H and T_M and Engine B is a Carnot engine operating between temperatures T_M and T_C ($T_H>T_M>T_C$). Show that the net efficiency, e_C, is just the efficiency of a Carnot engine operating between temperatures T_H and T_C. ❖❖

T-6. Entropy and the Second Law

Questions for discussion

1. Shown below are schematic figures for the energy and entropy of a heat engine.



ENERGY SCHEMATIC FIGURE

a) What is the relation between ΔS_{in} and ΔS_{out}? Why is it not possible to develop a cyclic engine that converts heat entirely into work? Why <u>must</u> some heat (Q_{out}) be ejected from the system?

b) In a Carnot engine, Q_{in} enters the system only along the isotherm T_H, and Q_{out} leaves the system only along the isotherm T_L. Use your result of (a) to find Q_{out}/Q_{in} in terms of T_H and T_L.

[*] **c)** An efficient engine converts as much heat as possible into work, ejecting as little as possible. Can you explain why a Carnot gives the greatest possible efficiency? [Hint: Q_{in} enters the system only at the highest temperature of the cycle. Q_{out} leaves the system only at the lowest temperature of the cycle.]

[*] More challenging questions or problems will be marked with a *. Your GSI will guide you as to whether you need to to complete them.

2. A cyclic heat engine uses an ideal gas as its working substance. Which of the following are true?

_____ For a complete cycle, the change in entropy of the gas is zero ($\Delta S_{gas} = 0$).

_____ For a complete cycle, the change in entropy of the gas is zero ($\Delta S_{gas} = 0$), but only if the engine operates reversibly. If the engine operates *irreversibly*, then $\Delta S_{gas} > 0$.

_____ For a complete cycle, the change in entropy of the universe is zero ($\Delta S_{universe} = \Delta S_{gas} + \Delta S_{environment} = 0$).

_____ For a complete cycle, the change in entropy of the universe is zero ($\Delta S_{universe} = \Delta S_{gas} + \Delta S_{environment} = 0$), but only if the engine operates reversibly. If the engine operates *irreversibly*, then $\Delta S_{universe} > 0$.

3. A box with total volume V_o is divided in half by a partition. On the left-hand side of the partition, there is a sample of ideal gas with initial pressure P_o and initial temperature T_o. On the right-hand side of the partition, the box is empty.

The partition is then suddenly removed, and the gas expands freely to fill the entire box. Soon the gas is in thermal equilibrium again.

a) Intuitively, what do you think happens to the entropy of the gas when it expands freely? Does the entropy increase, decrease, or stay the same? Justify your answer.

b) Suppose that two students, Carolina and Susan, are asked to find the change in the gas's entropy for this process.

- Carolina wants to find the change in entropy as follows:

$$\begin{aligned}\Delta S &= \int_{initial}^{final} \frac{dQ}{T} \\ &= \int_{initial}^{final} \frac{0}{T} \quad \text{(since no heat flows in or out of the gas during the free expansion)} \\ &= 0.\end{aligned}$$

- Susan, on the other hand, wants to find the change in entropy like so[†]:

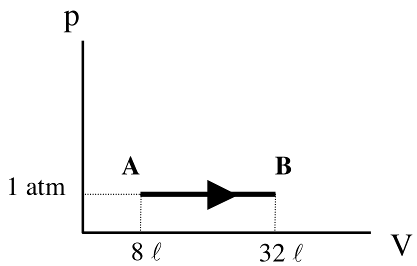
$$\begin{aligned}\Delta S_{\text{ideal gas}} &= \frac{d}{2}Nk\ln\frac{T_f}{T_i} + Nk\ln\frac{V_f}{V_i}\\ &= 0 + Nk\ln\frac{V_f}{V_i} \qquad (\text{since } T_f = T_i)\\ &= Nk\ln 2.\end{aligned}$$

Whose method is correct? Why?

[†] For a derivation of this result, see "Entropy of the Ideal Gas" in the Supplementary Material at the end of the workbook. It's a standard midterm-type problem.

Problems

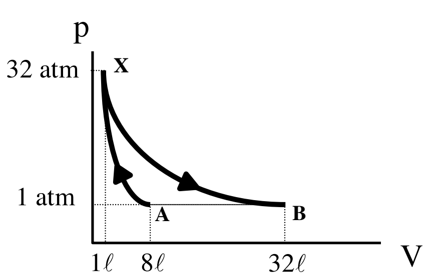
1. Two moles of monatomic ideal gas, under a constant pressure of 1 atmosphere, expand from an initial volume of 8 liters to a final volume of 32 liters. (This is a reversible transformation.)



a) Is heat flowing into the gas or out of the gas during this transformation? (You needn't calculate anything in detail; just decide whether the heat flow is in or out.)

b) What is the change in entropy of the gas during this transformation? Is the sign of your answer consistent with your answer from part (a) ?

Now consider a *different* path from A to B, this time via point X. The point X has been chosen so that the process A→X is *adiabatic*, and the process X→B is *isothermal*.



c) Find $\Delta S_{A\to X}$ using the basic rule for reversible processes,

$$\Delta S_{A\to X} = \int_A^X \frac{dQ}{T}\ .$$

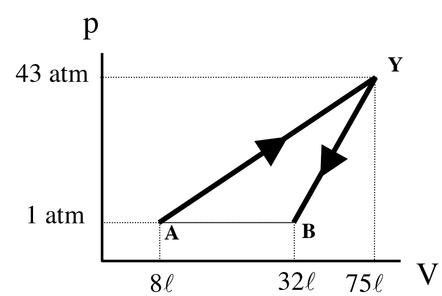
d) Next find $\Delta S_{X\to B}$, again using the basic rule for reversible processes:

$$\Delta S_{X\to B} = \int_X^B \frac{dQ}{T}\ .$$

e) Now add your answers for $\Delta S_{A\to X}$ and $\Delta S_{X\to B}$ to find the total change in entropy $\Delta S_{A\to X\to B}$.

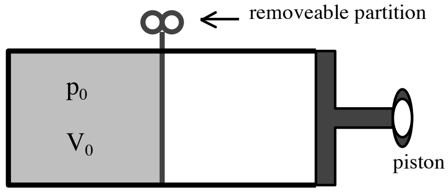
f) How do your answers for $\Delta S_{A\to B}$ and $\Delta S_{A\to X\to B}$ compare? Why is this?

Suppose we consider yet another path from A to B, this time via point Y.



g) What is the total change in entropy $\Delta S_{A\to Y\to B}$ for the total path A→Y→B? ❖

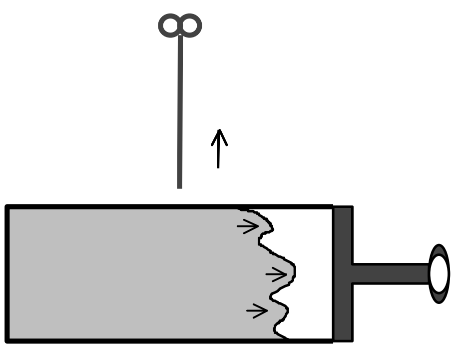
2. The device shown below consists of a chamber with volume $2V_0$. This chamber has a removeable partition in the middle. (Notice that the right-hand wall of the chamber is actually a piston.)



A

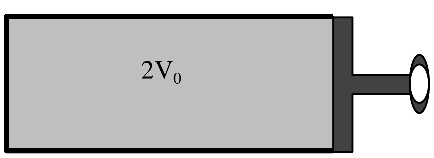
As shown in figure **A** above, the partition is initially in place, and an ideal diatomic gas is confined to the left-hand side, occupying volume V_0. The gas is under an initial pressure p_0. Meanwhile, the right-hand side of the chamber is vacuum.

In the first step of the process, the partition is suddenly removed. As a result, the gas expands freely to fill the chamber. This is shown below in figure **A** → **B**.



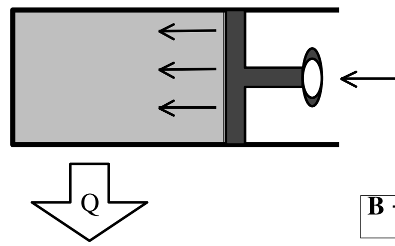
A → B

Soon the gas is once again in equilibrium, but now at volume $2V_0$. This is figure **B**.



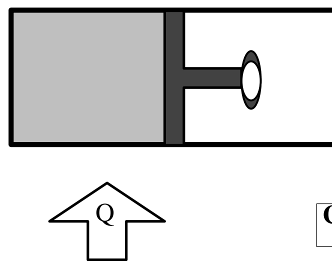
B

As the next step, the piston compresses the gas back down to the original volume V_0, but heat is drawn out of the gas also, so that the pressure remains constant during this process.



B → C

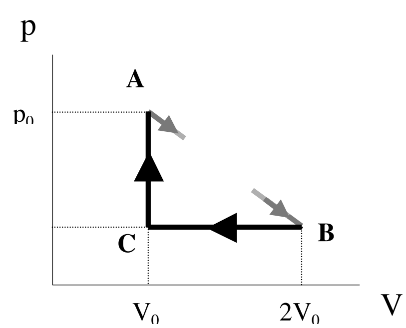
As the third and final step, heat is added to the gas at constant volume, until the pressure returns to the initial value p_0.



C → A

At this point the partition can be re-inserted, and the piston can be drawn back to its initial position. We are now ready to repeat the cycle.

Here is a p-V diagram for this cycle.



a) Explain why the temperature of the gas at B is the same as the temperature at A.

b) Using this fact, find the pressure at B.

c) Why is this device *not* an engine? (Hint: What can you say about W_{net}?)

d) How much entropy is added to the *gas* during each step? (i.e. find $\Delta S_{A\to B}$, $\Delta S_{B\to C}$, $\Delta S_{C\to A}$ for the gas)

e) What do these entropy changes add up to? Why?

f) How much entropy is added to the *environment* during each step? (i.e. find $\Delta S_{A\to B}$, $\Delta S_{B\to C}$, $\Delta S_{C\to A}$ for the environment)

g) What is $\Delta S_{universe} = \Delta S_{gas} + \Delta S_{environment}$ for the whole cycle?

h) Are these results consistent with your answers for Discussion Question 2 above? ❖

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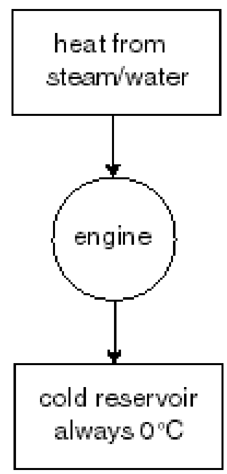
3. You have 50 kg of steam at 100° C, but no other heat source to maintain it in that condition. You also have a cold reservoir at 0° C that will stay 0° C at all times.

Suppose you operate a reversible heat engine with this system: the steam condenses and then cools until it reaches 0° C, and the heat released in this process is used to run the engine. (The steam/water itself remains in the upper container at all times.)

For water: L_v = 2256 kJ/kg, c_w = 4190 J/kg•C]



a) Calculate the total entropy change of the steam as it condenses to water and cools to 0° C.

b) Find the total amount of work that the engine can do. Explain your reasoning in a few sentences, in addition to carrying out any calculations. [Hint: How much heat must the engine expel to the low temperature reservoir?] ❖❖

Now would be a good time to turn to the supplimentary worksheet "T-S2. Efficiency of the Carnot Engine" and work through Part 2: Carnot Efficiency the Easy Way.

T-6 Challenge Problem

Equivalent Statements of the Second Law

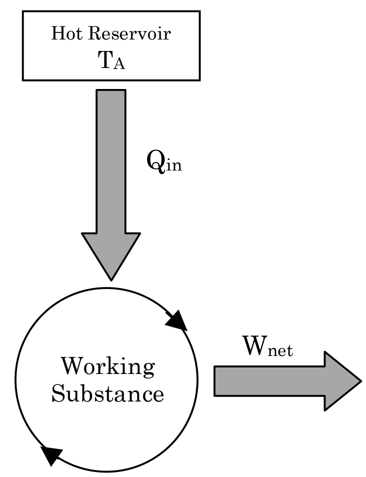
In this challenge problem we will prove that the following three different statements of the second law of thermodynamics are equivalent:

(i) $\Delta S_{universe} \geq 0$
(ii) An heat engine cannot convert heat directly into work (there is always some 'waste' heat).
(iii) The Carnot Engine is the most efficient engine that can operate between two temperatures.

We will first show that (ii) follows from (i) using *proof by contradiction*. Assume the converse of statement (ii). That is, assume we have an engine that does convert heat directly into work. Call this engine 'Engine A'. The heat intake comes from a heat reservoir at temperature T_A.

a) What are the elements of the 'universe' for Engine A? Compute the entropy change for each element as the engine goes through one complete cycle.

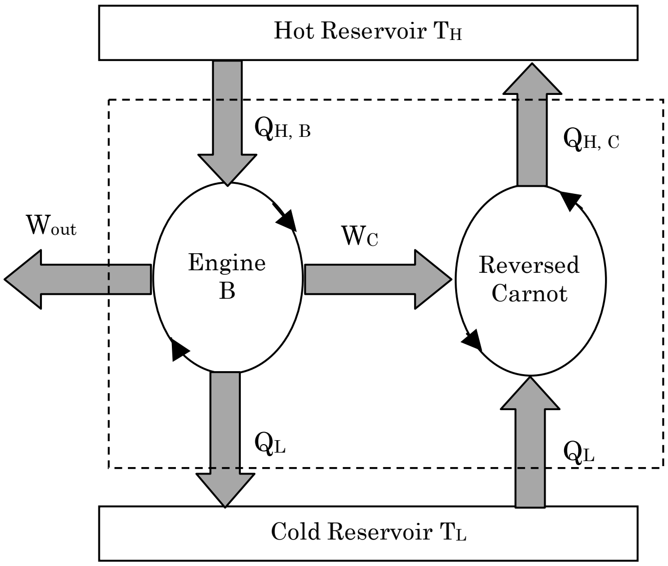
b) What is the change in entropy of the universe for one complete cycle of Engine A?



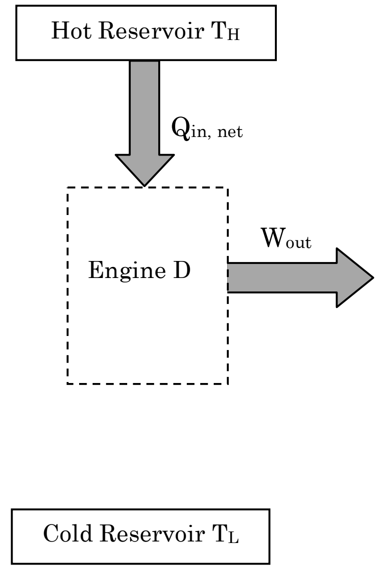
Schematic of Engine A

Your answer to part **(b)** should have been *negative*. That is, if we take statement (i) of the second law to be true, then our assumption that we can have an engine that can convert heat directly into work must have been incorrect. Therefore, statement (ii) follows from statement (i).

Now we will show that statement (iii) follows from statement (ii), again by contradiction. Assume we have an engine (not necessarily reversible) that operates between high and low temperature heat reservoirs, at temperatures T_H and T_L, with an efficiency *greater* than a Carnot engine operating between the same two reservoirs. Call this Engine B. Since we know Carnot engines are reversible, consider making a composite engine (Engine D) made out of one reversed Carnot engine and one engine of type B such that we use some of the work produced by Engine B to run the reversed Carnot and furthermore such that the heat expelled to the cold reservoir from Engine B is precisely the same as the heat intake from the cold reservoir for the reversed Carnot engine, as detailed below.



Full Schematic of Engine D



Net Energy Schematic of D

The schematic on the right shows the net effect of Engine D, where the dashed box indicates everything inside the dashed box in the full schematic on the left.

c) In terms of $Q_{H,\,B}$, e_C (the efficiency of a forward-running Carnot engine operating between the temperatures shown), and e_B (the efficiency of Engine B), determine $Q_{H,\,C}$ and W_{out}.

d) Show that, if $e_B > e_C$, as we supposed, then the arrows on the right-hand-side of the above figure are pointing in the correct direction (i.e. a net heat is *entering* the engine and a net work is *output* from the engine).

e) Finally, using your results from parts **(b)** and **(d)**, argue that statement (iii) of the second law must follow from statement (ii) and, therefore, from statement (i). ❖❖

T-7. Entropy: Other Topics

Part 1: Entropy with Calorimetry

Questions for discussion (Part 1)

1. The melting point of lead is 327.5 ^{o}C. To melt one kilogram of lead at this temperature, you must add about 25,000 J of heat. When you do this, does the entropy of the lead change? (See if you can answer based on the qualitative idea of "order vs. disorder.")

2. In Discussion Question 1, if you said that the entropy of the system changes, then by how much? (Give a numerical answer.)

Problems (Part 1)

1. A small lead block, of mass M and initial temperature 3T, is placed in contact with a large lead block, of mass 4M and initial temperature T. The system as a whole is isolated, so that no heat is lost to the surroundings.

a) Will the final temperature of the two-block system be less than 2T, greater than 2T, or equal to 2T?

b) Find the final temperature T_f in terms of T. Was your answer to part (a) correct?

c) Find the change in the entropy of the small block during this process. (Denote the specific heat of lead by c_L.) Is your answer consistent with the Second Law?

d) Next, find the change in entropy of the large block during this process. Is your answer consistent with the Second Law?

e) Finally, find the change in entropy of the two-block system. Is your answer consistent with the Second Law? ❖

---

2. A lead block of mass M is at initial temperature T_L. The block is placed within a thermally insulated canister of water. The water has mass m and is initially at temperature T_W.

If M = 12 kg, m = 75 g, and T_W = 30 ^oC, then what must T_L be (at least) if we want all the water to vaporize?

(Use c_w = 4190 J/kg$\cdot$^{0}C, c_L = 129 J/kg$\cdot$^{0}C, $L_V^{H_20}$ = 2260 kJ/kg) ❖

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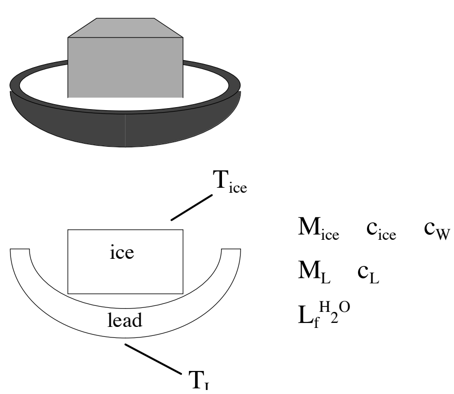
3. A cold block of ice is placed in a hot bowl made of lead.



After a while, we have bowl full of water, with both the bowl and the water at the same temperature T_f.

a) Find T_f in terms of the symbols shown in the diagram.

b) Find the change in entropy of the bowl for this process. (Leave your answer in terms of T_f.)

c) Next, find the change in entropy of the H_2O for this process. (Again, leave your answer in terms of T_f.)

d) For the combined bowl+H_2O system, will the total change in entropy ΔS_{sys} be positive, negative, or zero? Why?

e) Find ΔS_{sys} (in terms of T_f). ❖❖

Part 2: Microstates

Each possible way that a microscopic system can be configured is called a *microstate*. E.g. the microstates for a 2-coin system are "head-head", "head-tail", "tail-head", and "tail-tail". Given macroscopic quantities, such as Volume, Pressure, total Energy or Temperature, etc. only certain microstates are possible. The number of microstates available given macroscopic conditions is the quantity Ω. In thermodynamic equilibrium each of these microstates has an equal probability of 1/Ω of occurring.

Labeling microstates by i and the probability for being in a particular microstate as i_i, the entropy is defined as $S = -k_B \sum_i p_i \ln p_i$, where the sum is over *all* microstates. (Microstates that have zero probability of occurring don't affect the sum since 0 ln 0 = 0.)

For systems in equilibrium, the entropy is maximized, meaning that each microstate has an equal probability of occurring, giving $p_i = 1/\Omega$. When we plug this into the entropy formula, we get the microscopic definition of the entropy of equilibrium systems:

$$S = k_B \ln \Omega$$

Questions for discussion (Part 2)

1. Entropy is sometimes said to be a 'measure of disorder.' Why are systems that are 'disordered' said to be more entropic than systems that are 'ordered'? Hint: Consider your room. Each item in your room can be placed anywhere in the room. How does the number of ways for the room to be 'disordered' (messy) compare to the number of ways that your room to be 'ordered' (clean)?

2. Consider a system of N coins, each of which can land on heads (H) or tails (T) when flipped.

a) How many microstates are there in the flipped-coin system?

b) Suppose the coins are unweighted, so that the odds of a particular coin landing on heads are the same as the odds of that coin landing on tails. What is the probability for landing on a particular microstate? What is the entropy of this flipped-coin system?

c) Suppose all of the coins are double-head coins, so that each coin will invariably land on heads when flipped. What is the entropy of this flipped-coin system?

d) What is the entropy of the flipped-coin system if the first (N-1) are known to land on heads?

e) What is the entropy of the flipped-coin system if at least (N-1) coins are known to land on heads?

3. Entropy is sometimes said to be a measure of *ignorance* about a system. Why are systems that we know everything about less entropic than systems we know nothing about?

4. Prove that in the case where we have Ω microstates, and the probability for each microstate, p_i, is *equal*, then the formula $S = - k_B \sum p_i \ln(p_i)$ reduces to $S = k_B \ln\Omega$.

Problems (Part 2)

1. Consider a "gas" of 8 balls in a volume of 16 cubes arranged in a 2 by 2 by 4 grid, where a maximum of one ball is allowed in each cube.

a) What is the entropy of the system if all of the balls are known to be in the 8 leftmost cubes?

b) What is the entropy of the system if it is known that *n* of the balls are in the 8 leftmost cubes and (8-*n*) balls are in the 8 rightmost cubes?

c) Which configurations (*n* balls on the left, (8-*n*) on the right) have the least amount of entropy? Which has the greatest entropy?

d) Why does entropy increase in a free expansion? That is, why is a free expansion irreversible?

e) Suppose there is no constraint on the positions of the balls and the 'gas' is allowed to reach equilibrium. At any instant, what is the probability of finding 4 balls on the left and 4 on the right? What is the probability of finding all of the balls on the left? ❖

---

2. A system consists of 2 particles, particle a and particle b, each of which can have one of three possible magnetic moments: +M, 0, or -M. The total magnetic moment of the system is taken by adding the magnetic moments from each particle: $M_{tot} = M_a + M_b$.

a) List all nine of the possible microstates, *i*, of the two-particle system, and find the total magnetic moment for each state, M_i.

b) Compute the average value of the total magnetic moment in each of the cases listed to the right.

c) Compute the entropy of the system for each of the cases listed to the right. ❖❖

i) All microstates can occur and are equally likely.

ii) Microstates with $M_i \neq 0$ have zero probability of occurring. All other microstates are equally likely.

iii) Microstates where the magnetic moments of particles a and b are equal have zero probability of occurring. All other microstates are equally likely.

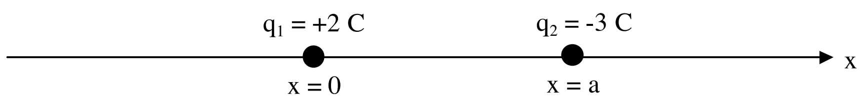
E-1. Coulomb's Law

Questions for discussion

1. Coulomb's Law for the electrostatic force between two point charges is $F = qQ/4\pi\varepsilon_0 r^2$. ($\varepsilon_0$ is constant, equal to 8.85×10^{-12} $C^2/N\cdot m^2$, which determines the relative strength of the electric force. Some texts use the constant $k_C = 1/4\pi\varepsilon_0$, called Coulomb's constant instead, which makes Coulomb's Law look even more like Newton's law of gravitation!) This looks a lot like Newton's law for the gravitatational force between two point masses: $F = G\,mM/r^2$. And indeed, because both forces are "$1/r^2$ forces," there are some mathematical similarities between them.

However, all similarities aside, can you think of any important *differences* between electrostatic forces and gravitational forces?

2. Two point charges q_1 = +2 C and q_2 = -3 C are fixed in place along the x-axis, as shown.



You have in your hand another point charge q, and you want to place it somewhere on the x-axis. But you want to place it at a point where it will *stay*. That is, you want to place it at a point where it will feel *no force* due to the fixed charges q_1 and q_2. (Hint: Think about what the force looks like very close to each charge and what it looks like very far away from both charges.)

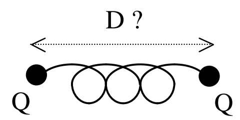
a) Is there any point on the x-axis *in between* the two fixed charges, where you could place your charge q and it would remain at rest? Explain. Does you answer depend on whether your charge q is positive or negative? Why or why not?

b) Is there any point on the x-axis to the *right* of the two fixed charges, where you could place your charge q and it would remain at rest? Explain. Does you answer depend on whether your charge q is positive or negative?

c) Is there any point on the x-axis to the *left* of the two fixed charges, where you could place your charge q and it would remain at rest? Explain. Does you answer depend on whether your charge q is positive or negative?

Problems

1. A spring with spring constant k_s and rest length L has positive charges Q attached to either end, as shown.



a) Find an equation that will determine the length D of the spring, once the charges have come to rest.

b) Repeat part (a), this time assuming that the charges on either end are both *negative*.

c) Repeat again, this time assuming that the charges on either end have *opposite* signs. ❖

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2. Returning to the situation described in Discussion Question 2 above, find the point(s) on the x-axis where your point charge q would remain at rest. ❖

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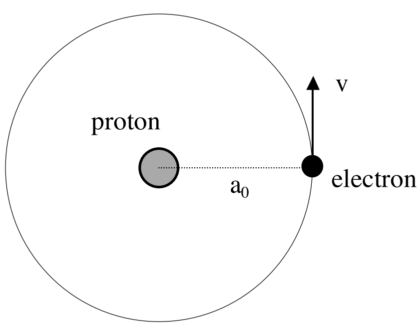
3. A hydrogen atom consists of a massive proton with a much lighter electron orbiting around it. In the "ground state" of the atom, the electron orbits the proton at a distance a_0 = 5.3×10^{-11} m.



$m_p = 1.7\times10^{-27}$ kg
$m_e = 9.1\times10^{-31}$ kg
$q_p = 1.6\times10^{-19}$ C [usually denoted *e*]
$q_e = -1.6\times10^{-19}$ C [usually denoted $-e$]
$G = 6.67\times10^{-11}$ Nm^2/kg^2

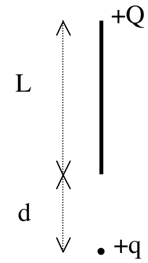
a) At this separation, what is the magnitude of the *gravitational* force between the proton and the electron?

b) At this separation, what is the magnitude of the *Coulomb* force between the proton and the electron?

c) Which of these two forces is negligible in comparison with the other?

d) In the ground state, how many times per second does the electron orbit the proton? ❖

4. A point charge +q is located a distance d from one end of a uniformly charged rod. The rod has total charge +Q and length L. (The rod and the point charge are each held fixed in place.)



a) What is the force on the point charge due to the rod?

b) What is the force on the rod due to the point charge? ❖

---

5. The setup is similar to that of Problem 4, except that the rod now has a <u>nonuniform</u> linear charge distribution $\lambda(x) = \lambda_o\, x/L$, where λ_o is a constant.

a) Calculate the total charge on the rod, in terms of λ_0 and L.

b) Set up an integral to find the force on the point charge due to the rod. (You do not need to evaluate the integral.) ❖

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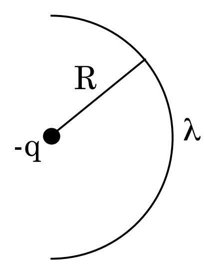
6. A rod with a uniform linear charge density λ is bent into a half-circle of radius R. A point charge -q is placed at the center of the circle. (The rod and the point charge are each held fixed in place.)



a) What is the net charge on the half-circle?

b) Set up an integral to find the force on the point charge due to the half-circle. (Remember that force is a *vector*.)

c) In which direction does the force point? How can you tell this without doing any calculation?

d) Evaluate the integral and find the force on the point charge. It might help to rewrite your vectors in terms of cartesion unit vectors $\hat{x}$ and $\hat{y}$ (Some books use $\hat{i}$ and $\hat{j}$ for the unit vectors). ❖❖

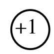
E-2. Electric Fields

<u>Summary</u>
An electric <u>force</u> requires two charges.
An electric <u>field</u> is produced by a single charge.

Questions for discussion

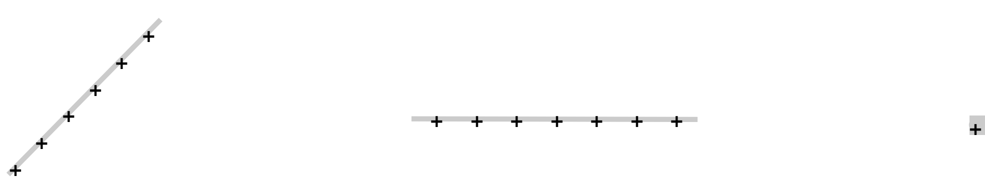
1. Sketch the electric field created by each of the following point charges.

a) q = +1 C



b) q = -2 C

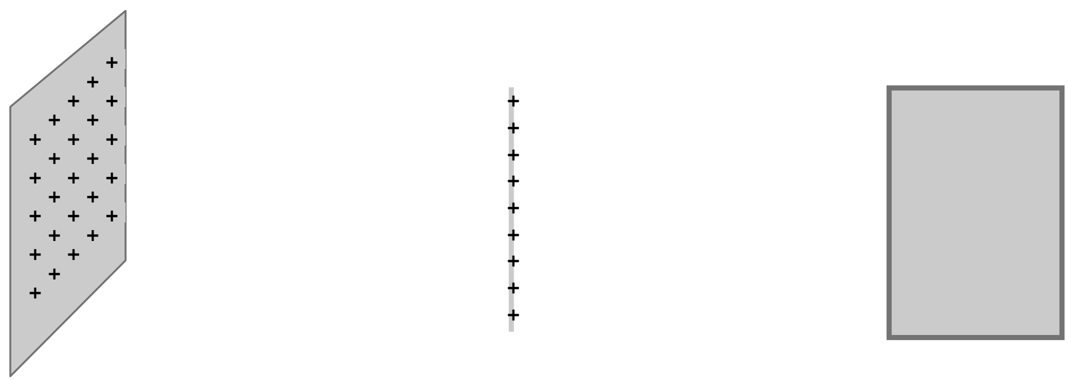
2. A long straight piece of fishing line has been sprayed evenly with positively charged paint. This creates a uniform *line charge distribution* (as opposed to a *point charge*). Sketch the electric field created by this line of charge.



perspective view side view end view

How would your picture look if the line were negatively charged?

3. Next, a large sheet of plastic has been sprayed evenly with positively charged paint. (See figure next page.) This creates a uniform *surface charge distribution*. Sketch the electric field created by this surface charge.



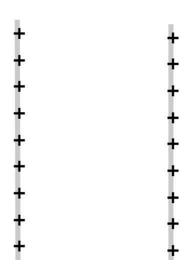
perspective view side view face view (surface charge suppressed)

How would your picture look if the sheet were negatively charged?

4. This time, two charged sheets are placed near each other. One has uniform positive surface charge, while the other has uniform negative surface charge. Sketch the (net) electric field created by the sheets.

perspective view side view

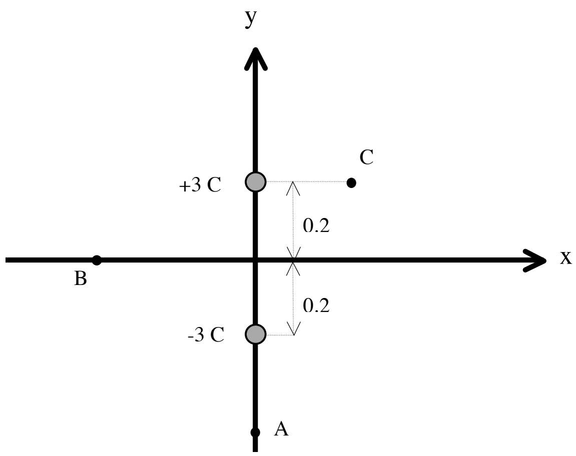
5. Repeat Discussion Question 4 if the sheets are *both* positively charged.



Side view

Problems

1. Two point charges q_1 = +3 C and q_2 = -3 C are fixed in place, as shown. They are separated by 0.4mm, and are centered at the origin.



The points A, B, and C in the figure are *not* point charges. They are merely *locations* in the x-y plane.

a) Is the electric field due to q_1 and q_2 at the origin zero? If so, explain why. If not, find the magnitude of the electric field at the origin.

b) What is the magnitude of the electric field at point A, at y = -0.5 mm?

c) What is the magnitude of the electric field at y = +0.5 mm?

d) What is the magnitude of the electric field at point B, at x = -0.5 mm?

e) Is there any location at which the electric field created by the point charges q_1 and q_2 is zero?

f) Sketch qualitatively the electric field lines due to q_1 and q_2. ❖

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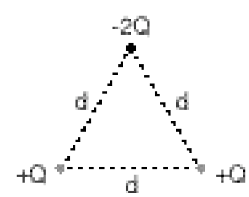
2. Three point charges are equidistant from one another and fixed in place.

a) Sketch the *net* electric field vector at the center of the triangle.

b) Calculate the electric field at this point. ❖



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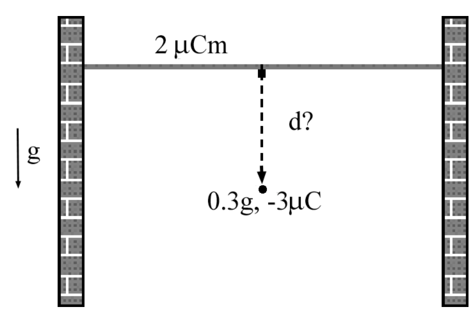
3. A long line of positive charge is strung horizontally from one wall to another, like a straight clothesline. The line of charge has uniform linear charge density 2μC/m. Beneath this line of charge, a point mass m = 0.3g with charge -3μC hangs motionless, its weight counteracted by its attraction to the line of charge.

a) In order for the point charge to be suspended in this way, how far below the line must it be?

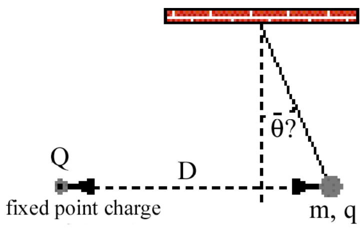
b) If you nudge the point charge downwards, what will happen to it? Describe its motion after you nudge it.

c) What if you had nudged the point charge upwards instead? ❖



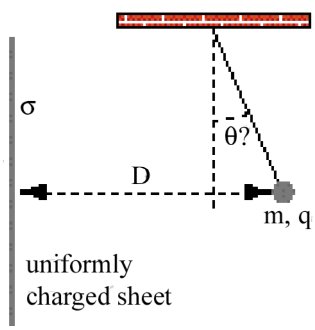
4. A point mass m with positive charge q is suspended from the ceiling by a thread. Nearby, a charged object causes the point charge q to deflect from the vertical. Our task will be to find the angle of deflection in various cases. (**Refer to Table at right**.)

a) First suppose that the nearby object is *another* positive point charge Q, located a horizontal distance D away.



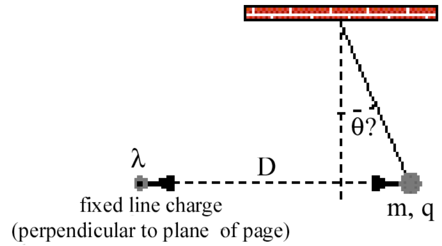
Find the angle of deflection θ.

b) Next suppose that the nearby object is a large positively charged sheet, with uniform surface charge density σ, located a perpendicular distance D away.



Find the angle of deflection θ.

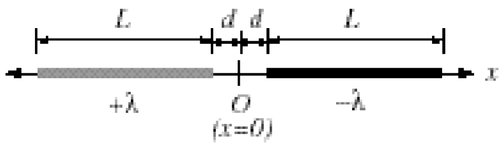
c) Finally, suppose that the nearby object is a long positively charged line, with uniform linear charge density λ, located a horizontal distance D away.



Find the angle of deflection θ. ❖

Electric fields created by various charged objects		
Shape of charged object	Strength of E-field greated by object	Note
Point charge Q	$E(r) = \dfrac{Q}{4\pi\varepsilon_0 r^2}\hat{r}$	"r" refers to distance from the point charge.
Infinite straight line of charge, *uniform* linear charge density λ	$E(r) = \dfrac{\lambda}{2\pi\varepsilon_0 r}\hat{r}$	"r" refers to perpendicular distance from the line charge.
Infinite flat sheet of charge, *uniform* surface charge density σ	$E(r) = \dfrac{\sigma}{2\varepsilon_0}\hat{r}$	E-field strength is independent of distance from the sheet and the direction points away from the sheet.

5. A thin rod of length L and uniform positive charge per unit length $-\lambda$ is positioned on the x–axis with one end at x = +d. A second rod, also of length L but with uniform charge per unit length $+\lambda$, is positioned with one end at x = –d.



Find the electric field at the origin *O* (x = 0). If the electric field is zero at this point, explain why. Otherwise find both the magnitude and direction of the field at the origin. ❖

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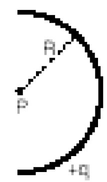
6. A semicircle of radius R has charge +q spread uniformly on it.

a) Sketch the electric field vector at point P. Explain your reasoning.

b) Find the strength of the electric field at point P. ❖



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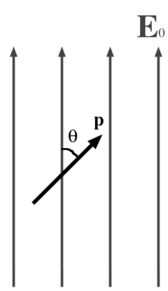
7. An *electric dipole* with dipole moment p is placed in an external, uniform electric field of strength E_0. The dipole is centered at the origin and is made by placing two charges of charge +q and -q a fixed distance d apart, with p=qd. The dipole moment is a vector, and the direction of **p** will be the unit vector pointing from the negative charge to the positive charge. Let θ be the angle between **p** and **E**.



dipole in an electric field

a) If **E** points in the x-direction, and the dipole is in the xy-plane, what are the coordinates for the positive charge and the negative charge, in terms of d and θ?

b) What force does each charge feel? What is the net force, $\mathbf{F}_{net}$, on the dipole?

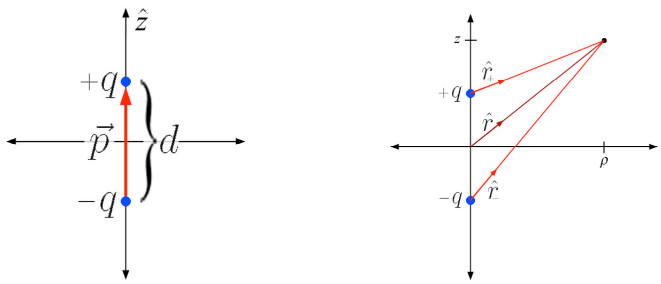
c) What is the net torque on the dipole, taking the torque about the center of the dipole? (Remember: $\tau_{net}=\Sigma\mathbf{r}\times\mathbf{F}$)

d) Show that your answer is consistent with the standard formula for the torque on a dipole, $\boldsymbol{\tau}=\mathbf{p}\times\mathbf{E}$. ❖❖

E-2 Challenge Problem

Electric Field of a Dipole

In this challenge problem we derive the form of the electric field very far from an electric dipole. Start with two charges, +q and -q, oriented along the z-direction in cylindrical coordinates and placed a distance d apart, centered on the origin. The dipole moment vector for this configuration of charges will be $\vec{\mathbf{p}} = qd\,\hat{z}$. We will find the electric field at a radial distance ρ from the axis and a distance z along the axis, in cylindrical coordinates. (ρ is used for the radial coordinate to distinguish it from the distance from the origin, r)



a) Call the vector pointing from the positive/negative charge to the point where we want to find the electric field $\mathbf{r}_\pm$ and the vector to the origin $\mathbf{r}$. Write down what $\hat{\mathbf{r}}_\pm$ (the unit vectors) and $r_\pm$ (the magnitudes) are in terms of the cylindrical coordinates and the unit vectors $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$.

b) In terms of d, q, z, and r and the unit vectors $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$, write down the exact electric field at the point we are considering. Simplify your expression by replacing the combination $\rho^2 + z^2$ with the distance of the point we would like to find the E-field from the origin of coordinates, r^2.

Now we will look at the *far field* of the dipole - the field at distances very large compared to the separation between the charges. In that case, the exact structure of the dipole becomes irrelevant and the electric field will approximately only depend on the dipole moment vector, **p**, and the position vector, **r** (just like for a single charge, or *monopole*, the field only depends on the charge and the position vector). The far field is found by taking the limit of the exact formula for the electric field as d goes to zero, while keeping p=qd constant. Since our formulas only have the first power of q in them, any time we have a d^2, we can throw it out since only one d will be able to combine with the q and the term will go to zero as the other d goes to zero. This is know as the *first order approximation* in d.

c) Expand your answer to part (b) to first order in d. Use the approximation formula $(1+\varepsilon)^n \approx 1 + n\varepsilon$.

d) Finally, replace the combination qd with the dipole moment p, the unit vector $\hat{\mathbf{z}}$ with $\hat{\mathbf{p}}$ and the fraction z/r with cos θ, where θ is the angle between the dipole vector **p** and the position vector **r**.

If you did everything correctly, you should get the result $\mathbf{E} = \dfrac{p}{4\pi\varepsilon_0 r^3}\left(3\cos\theta\hat{r} + \hat{p}\right)$.

e) Sketch some field vectors for the electric field above. Does it look like what you would expect? ❖❖

E-3. Gauss's Law

$$\Phi_E = \oiint_S \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$$

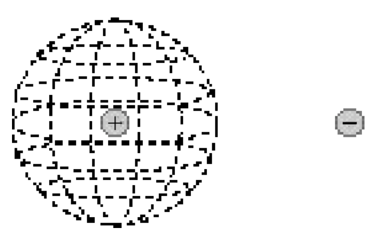
Questions for discussion

1. Using what you know about the electric field, write down some rules for electric field lines.

2. Consider a pair of point charges ±Q, fixed in place near one another as shown.

a) On the diagram above, sketch the field created by these two point charges.

b) Now consider an imaginary spherical surface enclosing the +Q charge:

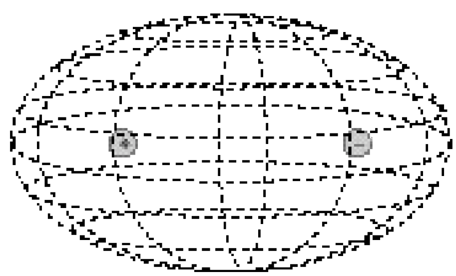


i) Reproduce here your drawing of the electric field lines from part (a), so you can get a sense of how the field lines pierce the imaginary spherical surface.

ii) How much electric flux passes outward through the imaginary spherical surface? You should be able to arrive at the answer very quickly using Gauss's Law.

iii) By examining the field lines and how they pierce the imaginary spherical surface, try to explain *why* the flux turns out to be what Gauss's Law said it was. (For example, try to explain why the net flux through the surface is *outward*.)

c) Next, consider an imaginary *ellipsoidal* surface enclosing both charges:

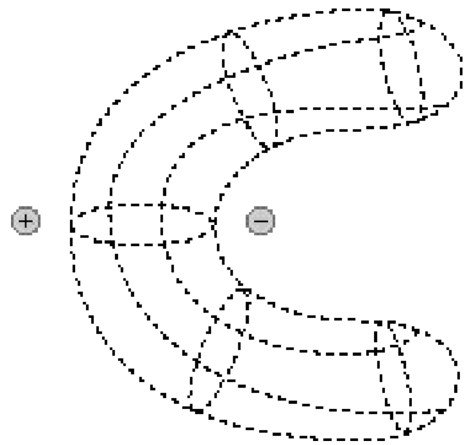


i) Once again, reproduce your drawing of the electric field lines from part (a), so you can get a sense of how the field lines pierce the imaginary ellipsoidal surface.

ii) How much electric flux passes outward through the imaginary ellipsoidal surface? Again, you should be able to arrive at the answer very quickly using Gauss's Law.

iii) By examining the field lines and how they pierce the ellipsoid, try to explain *why* the flux turns out to be what Gauss's Law said it was.

d) Finally, consider an *irregular* imaginary closed surface that winds around between the charges as shown:

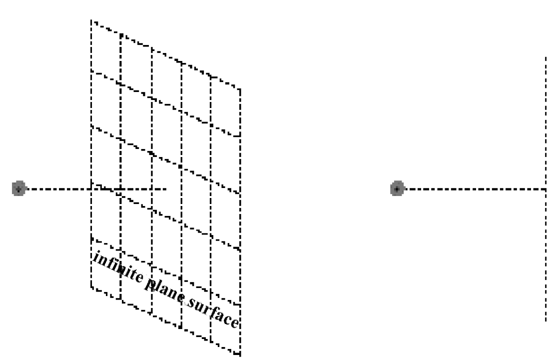


i) Once again, reproduce your drawing of the electric field lines from part (a), so you can get a sense of how the field lines pierce the irregular imaginary closed surface.

ii) How much electric flux passes outward through the irregular imaginary spherical surface? Again, you should be able to arrive at the answer very quickly using Gauss's Law.

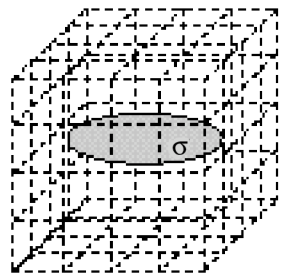
iii) By examining the field lines and how they pierce the irregular surface, try to explain *why* the flux turns out to be what Gauss's Law said it was.

3. The diagram below shows a single point charge Q. Sketch the field created by Q, and find the amount of electric flux passing through the imaginary infinite plane surface.



perspective view **side view**

4. A thin disk of radius R has uniform surface charge density σ. Let us imagine a "cubical" surface enclosing the disk.



a) What is the electric flux passing outward through the imaginary cubical surface?

b) Can you use this result to find the electric field created by the disk? Why or why not? You may want to sketch qualitative field lines on the diagram above.

5. Use symmetry arguments to find the most general form for the electric field vector (magnitude *and* direction) for the following types of charge distributions.

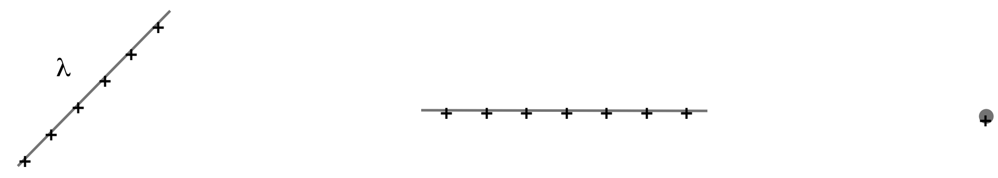
a) Spherically Symmetric - the charge distribution only depends on the radial distance r from the origin (for example, the point charge; a spherical shell of charge with a uniform surface charge density).

b) Cylindrically Symmetric - the charge distribution is infinitely long and only depends on the radial distance r from the axis of symmetry (for example, an infinitely long line of charge with a uniform linear charge density).

c) Planar Symmetry - the charge distribution is infinite in two directions and only depends on the third direction in a cartesian coordinate system (for example, an infinite sheet of charge with uniform surface charge density).

Problems

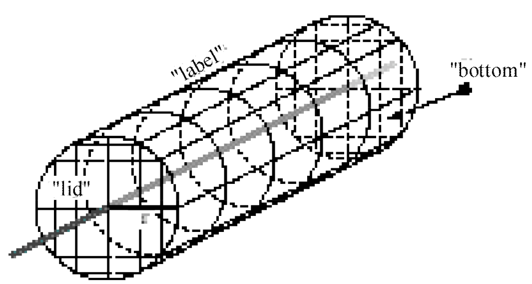
1. Consider a long line of charge, with uniform positive charge per unit length λ.



perspective view side view end view

a) Sketch the electric field created by this charge distribution.

The figure below shows an imaginary surface that can be used with Gauss's Law to determine the strength of the electric field at any distance r from the line charge.



The imaginary surface is kind of like a soup can, with a *label* part, a *lid* part, and a *bottom* part.

b) Is the magnitude of the electric field the same at all points of the label part of the Gaussian surface? Why or why not?

c) What angle do the electric field vectors make with the label, at various points of the label?

d) Is this angle the same at *all* points of the label?

e) In terms of the (unknown) electric field strength E, how much electric flux passes through the label? Compute this *directly* using the flux integral: $\Phi_{\text{label}} = \iint\limits_{\text{label}} \vec{E} \cdot d\vec{A}$

f) Answer parts (c) - (e) for the *lid part* of the Gaussian surface.

g) Answer parts (c) - (e) for the *bottom part* of the Gaussian surface.

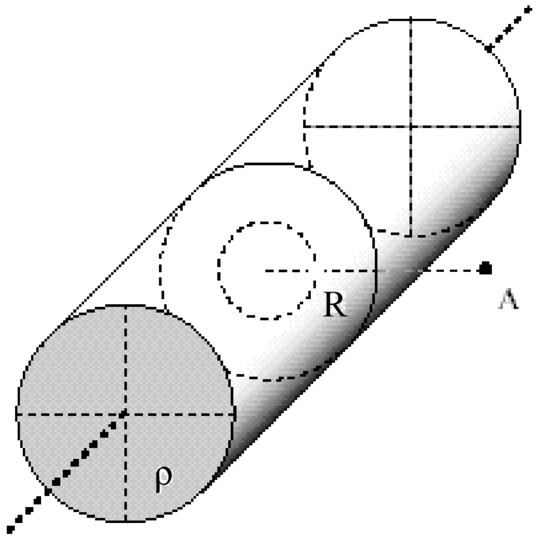
h) What is the total flux passing outward through the closed Gaussian surface?

i) How much charge is enclosed by the Gaussian surface? (use information about the charge distribution)

j) What is the electric field strength E(r) at any distance r from the line charge?

k) For this derivation to work, why is it necessary that the line of charge be infinitely long - or, in practice, very long compared to r? ❖

2. A very long tube of radius R is full of charged stuff with uniform positive charge per unit volume ρ, as shown in the figure below.



a) Sketch the electric field created by this charge distribution.

b) Find the electric field ouside the tube, for $r > R$.

c) Find the electric field inside the tube, for $r < R$.

d) Is the electric field continuous at the surface of the cylinder? That is, are the values of $E_{inside}(r=R)$ and $E_{outside}(r=R)$ equal? ❖

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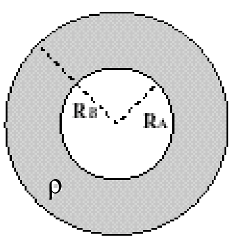
3. A spherical shell with inner radius R_A and outer radius R_B is filled with a material with uniform charge per unit volume ρ_o. The inside of the shell is empty.



cross section

a) What is the electric field inside the shell, for $r < R_A$?

b) Find the electric field ouside the sphere, for $r > R_B$.

c) Find the electric field inside the shell, for $R_A < r < R_B$.

d) Suppose the surface at R_B is coated with a uniform *surface* charge density σ. Which regions of the electric field are affected and what are the new values for the field? ❖

---

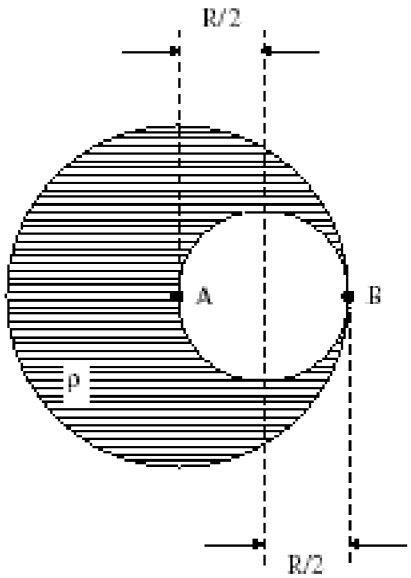
4. A sphere of radius R is filled with <u>nonuniform</u> charge per unit volume $\rho(r) = \rho_o r/R$, where ρ_o is a constant.

a) What is the total charge contained within the sphere?

b) What is the electric field outside the sphere, for $r > R$?

c) What is the electric field inside the sphere, for $r < R$? ❖

5. Initially we have a ball of charge with radius R and uniform positive charge density ρ. Then we scoop out a spherical cavity of radius R/2 so that the cavity is centered halfway from the center of the ball. (See figure below.)



a) What is the direction of the electric field at point A? (Hint: Use superposition. This charge distribution is the sum of a *positive ball* of radius R and a smaller *negative ball* of radius R/2.)

b) Consider the region entirely outside the charge distribution. In this region, what does the field of the large positive ball look like? What does the field of the smaller negative ball look like in this region?

c) Based on your answers to part (b), what does the net electric field outside the charge distribution look like?

d) Find the magnitude and direction of the electric field vector at points A and B.

e) Find the magnitude and direction of the electric field vector at the point located a distance R/2 directly to the left of point A. ❖

---

6. Consider an infinite plane of charge with uniform surface charge density σ.

a) How does the electric field at a point a distance z *below* the plane compare to the field at a point *above* the plane in terms of magnitude an direction?

Construct a cylindrical gaussian surface with cross-sectional area A and height 2z centered on the plane with the axis perpendicular to the plane. Again, this closed surface is made of three other surfaces: a 'label' a 'lid' and a 'bottom.'

b) In terms of the (unknown) electric field strength E, how much electric flux passes through the label? Compute this *directly* using the flux integral: $\Phi_{\text{label}} = \iint_{\text{label}} \vec{E} \cdot d\vec{A}$

c) Repeat part (b) for the lid and bottom. [Be careful of the signs!]

d) Find the electric field (magnitude and direction) at points a distance z from the plane. ❖❖

E-4. Conductors

Questions for discussion

1. In Physics 7B problems, we commonly consider "a ball of charge, with uniform charge per unit volume ρ." How do you know that the ball in such a problem is made of *non-conducting* material, such as plastic?

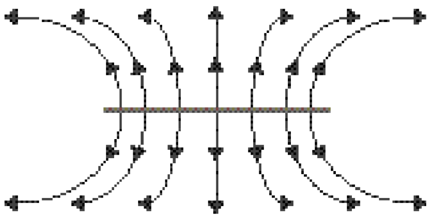
2. Give an intuitive justification for each of the following facts about conductors, all of which hold in equilibrium:

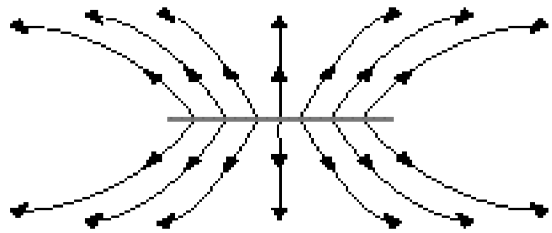
a) *Within a conducting material, the electric field vanishes.*

b) *Any net charge on a conductor must reside on a surface of the conductor.*

c) *The electric field at a surface of a conductor is always perpendicular to the conducting surface.*

3. The figure shows two charged disks. One of them is made of *plastic* (with a uniform positive surface charge sprayed on it), and the other is made of *metal* (with excess positive charge on its surface).



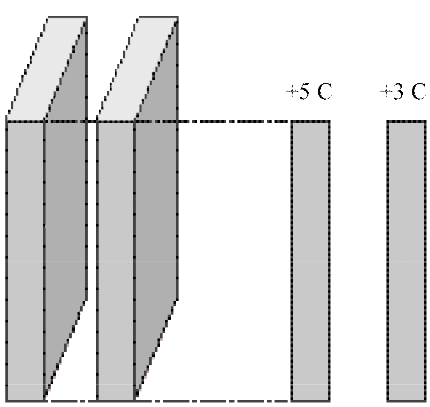


a) Which is which?

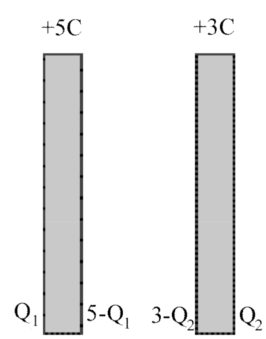
b) Evidently, the surface charges on the metal disk cannot be uniformly distributed (otherwise they would create the same field as that created by the uniformly charged plastic disk). Do you think that the surface charges on the metal are concentrated more near the center of the disk, or more near the edge? Why?

Problems

1. A pair of thick conducting slabs are fixed in place near one other as shown, with their faces parallel. The faces have area A, which we will take to be very large compared to the slabs' separation.



Initially the slabs are neutral, but then a net charge of +5 C is placed on the left slab, and a net charge of +3 C is placed on the right slab. When things have settled down, some amount of charge has migrated to the outer faces of the slabs, and some amount of charge has migrated to the inner faces of the slabs. This situation is diagrammed below.



Notice that the charges on the left slab add up to +5 C as required by charge conservation- and likewise for the right slab. Notice also that, in accordance with the rules for conductors, all charges reside on the surfaces of the conductors. The goal of this question will be to use other rules for conductors to determine the unknown charges Q_1 and Q_2.

a) Treating Q_1 and Q_2 as known quantities, use superposition to find an expression for the electric field within the left slab.

b) Taking into account the fact that the slab is a conductor, use this expression to find Q_1.

c) Repeat parts (a) and (b) for the right slab, in order to find Q_2.

d) What are the charges on the inner surfaces of the slabs?

e) You should have found that the charges on the inner surfaces of the slabs are equal and opposite. This will be true regardless of how much charge is on either slab. Can you prove this by drawing an appropriate Gaussian surface and making use of the fact that the field in either slab vanishes? ❖

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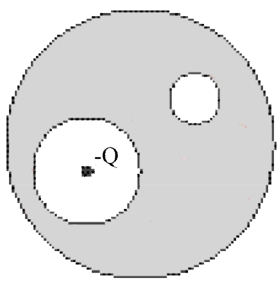
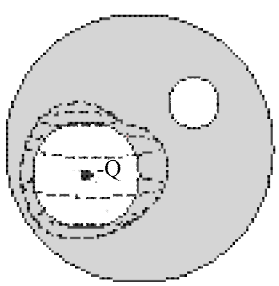
2. A metal ball carrying a net charge of +7Q has two spherical cavities scooped out of it.



net charge on conductor = +7Q

In the center of the first cavity, a point charge -Q floats motionless in midair. The second cavity is empty.

Consider the imaginary surface indicated with dotted lines in the figure below.



net charge on conductor = +7Q

First of all, note that this imaginary surface encloses the first cavity. Second, notice that every point of this imaginary surface lies within the conducting material.

a) What is the electric flux passing outwards through the imaginary surface? Compute this directly using $\Phi_{\text{imaginary surface}} = \iint_{\text{imaginary surface}} \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}$.

b) What can you therefore say about the charge enclosed by the imaginary surface?

c) How much charge lies on the inner surface of the cavity? How is this charge distributed?

d) Now draw another imaginary surface, this one enclosing the *second* cavity, and repeat questions (a) - (c) for this surface.

e) How much charge resides on the *outer* surface of the conductor? How is this charge distributed?

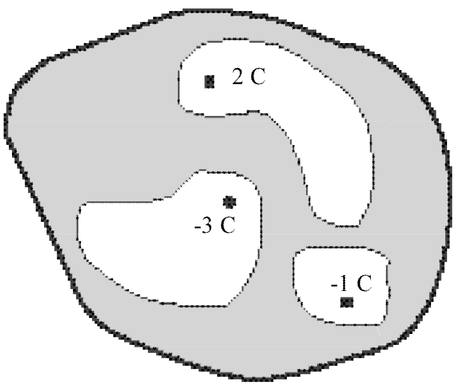
f) Sketch the electric field created by this object, both outside the conductor and within the cavities.

g) What is the strength of the electric field within the first cavity, at a distance r from the -Q point charge?

h) What is the strength of the electric field within the second cavity, at a distance r from the center of the cavity?

i) What is the strength of the electric field outside the ball itself, at a distance r from the center of the ball? ❖

3. The irregularly shaped conductor shown below has three inner cavities. Initially it has a charge of -5 C placed on it. Then point charges of 2 C, -1 C, and -3 C are introduced into the cavities as shown.



net charge on conductor = -5 C

Once things have settled into equilibrium, find the charges residing on each of the conductor's four surfaces. ❖❖

E-5. Electric Potential

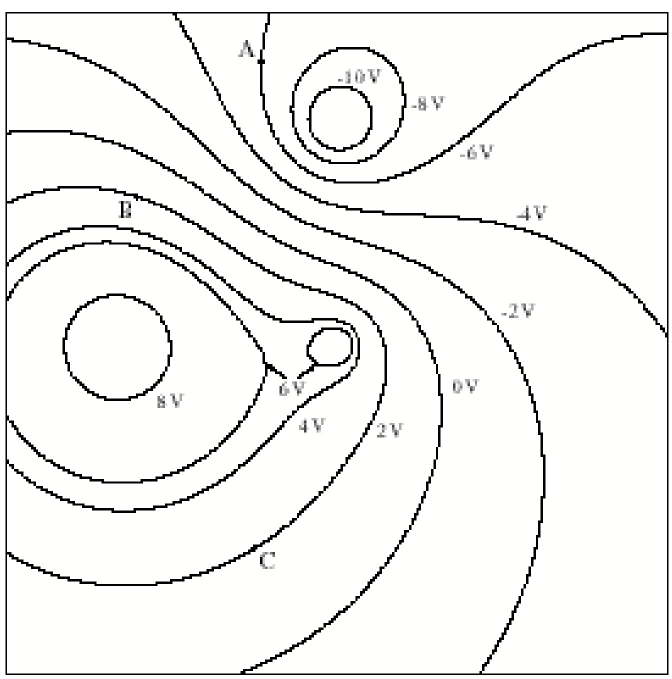
Summary
An electric potential energy requires two charges.
An electric potential is established by a single charge

Questions for discussion

1. True or false: A point charge placed in an external electric field accelerates toward the direction of *decreasing electric potential.*

2. A point mass m = 0.05 kg, with charge Q = +3 C, is sitting in an external potential V_i = 17 V. How much work will it take to move the point charge to a point with higher potential V_f = 22 V?

3. The diagram below shows some equipotential lines associated with a group of three fixed point charges. (The point charges are not shown. The letters A, B, and C in the figure are not point charges. They refer simply to locations in the diagram.)



a) How much work would it take to move a +1 Coulomb test charge from point A to point B?

b) How much work would it take to move a +1 Coulomb test charge from point A to point C?

c) How much work would it take to move a +1 Coulomb test charge from point B to point C? Why?

d) Where in the diagram is the electric field the strongest? How do you know?

e) Draw the electric field lines for this configuration of charges. Explain how you generated your picture.

4. Recall that in a conductor, the electric field within a conducting material must be zero. What does this imply about the electric *potential* within the material? (Hint: How are the electric field and the electric potential related mathematically?

Problems

1. The source of the sun's energy is a process called nuclear fusion. In this process, protons collide and create larger particles. As a result, huge amounts of energy are liberated. The problem, however, is that for fusion to occur, the protons have to "collide"—that is, they have to pass near to each other—say, within a proton diameter, 10^{-15} m.

a) Suppose that one proton is fixed in place. If another proton starts out from very far away, then how fast must it be going initially, if it is to approach to within 10^{-15} m of the fixed proton?

The question now arises, Are protons inside the sun actually moving that fast?

b) Calculate how high the temperature would have to be inside the sun, in order for the speed you found in part (a) to be a typical rms speed for particles in an ideal gas.

c) The actual temperature in the solar interior is about 10^7 K. Is this consistent with the fusion mechanism? ❖

---

2. A thin rod of length L lies along the x-axis. It has a uniform linear charge distribution λ_o.

a) What is the value of the electric potential at a given point x located to the right of the rod? Take V=0 at infinity.

b) What is the strength of the electric field at the point x? (You can find the answer by doing another integral, or you can simply differentiate your answer from part (a): $E_x = -dV/dx$.) ❖

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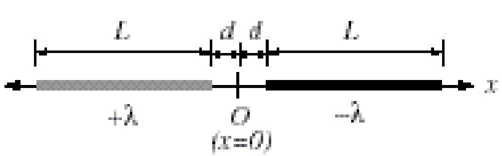
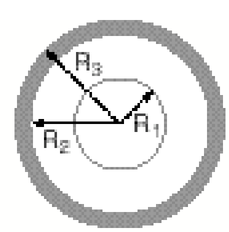
3. A thin rod of length L and uniform positive charge per unit length $-\lambda$ is positioned on the x–axis with one end at x = d. A second rod, also of length L but with uniform charge per unit length $+\lambda$, is positioned with one end at x = –d. Take $V = 0$ to be at infinity.



Find the electric potential due to the two rods at the origin O (x = 0). If the electric potential at O is zero, explain why. Otherwise find both the magnitude and sign of the potential at O. ❖

---

4. A thin conducting shell of radius R_1 is centered inside a thick conducting shell with inner radius R_2 and outer radius R_3. The inner shell has positive charge $2Q_0$ on it, while the outer shell has net charge $-Q_0$ on it.



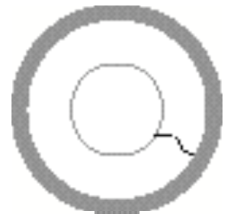
Inner shell has net charge $+2Q_0$.
Outer shell has net charge $-Q_0$.

a) What is the charge residing on the inner surface of the thick shell (at $r = R_2$) ? What is the charge residing on the outer surface of the thick shell (at $r = R_3$) ? Explain your reasoning.

b) Taking the potential $V(r)$ to be zero at infinity, i.e. $V(r=\infty)=0$,

- Find $V(r=R_3)$, the potential at the outer surface of the thick shell.
- Find $V(r=R_2)$, the potential at the inner surface of the thick shell.
- Find $V(r=R_1)$, the potential at the surface of the thin shell.
- Find $V(r=0)$, the potential at the center of the thin shell.

The two spheres are now connected by a thin conducting wire.



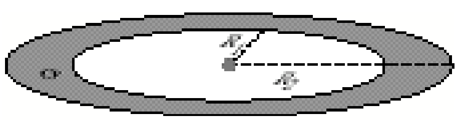
c) After a long time passes, which of the following is true? Explain your reasoning.

(i) All of the charge resides on the inner (thin) shell.
(ii) All of the charge resides on the outer (thick) shell.
(iii) There is charge on both shells, but there is more positive charge on the inner shell than on the outer shell.
(iv) There is charge on both shells, but there is more positive charge on the outer shell than on the inner shell. ❖

---

5. For a *charged disk* with radius R and uniform surface charge density σ, the electric potential at the center is given by $V = \sigma R/2\varepsilon_0$. (Here we have taken the potential to be zero at infinity.)

a) Given this, find the electric potential energy of a gold nucleus (with charge +79e), if it happens to be located at the center of a *charged annulus* as shown.



b) What is the magnitude of the electric field of the annulus at this center point?

c) What force acts on the gold nucleus when it is located there?

d) The gold nucleus is given a tiny nudge out of the plane of the annulus. What will happen to it?

e) Find the speed of the gold nucleus when it gets very far away from the annulus. (Denote the mass of the nucleus by M.) ❖

---

6. Consider again a charged annulus of inner radius R_1 and an outer radius R_2 with a uniform surface charge density σ. Take the electric potential to be zero at infinity, as usual.

a) Find the electric potential along the axis of a thin ring of radius r and width dr of the annulus.

b) Integrate your answer over the radius r to find the electric potential along the axis of the annulus.

c) At the very center of the annulus, what is the potential? Does this agree with your result from part (a) of problem 5? ❖

7. A very long line of charge, with uniform positive charge per unit length ℓ, is held fixed in place. Located a distance D from the line charge, and also fixed in place, is a point mass m with positive charge Q.

a) Use Gauss's Law to calculate the strength of the electric field created by the line charge, at any distance r from the line charge.

b) If the positive point charge Q is suddenly released, it will accelerate away from the line charge. Will the acceleration be constant?

c) How fast will the point charge be going by the time it has reached a distance 5D from the line charge? ❖

---

8. A ball of radius R has total charge +Q distributed uniformly over its volume.

a) Use Gauss's Law to calculate the electric field created by the ball, at points outside the ball ($r \geq R$).

b) Use Gauss's Law to calculate the electric field created by the ball, at points inside the ball ($r \leq R$).

c) Taking the potential to be zero at infinity, i.e. $V(r=\infty)=0$, find the potential $V(r=0)$ at the center of the ball.

d) Suppose that a point charge +q is at rest at the center of the ball. If the point charge is given a slight nudge, what will happen? How fast will the point charge be travelling at infinity? ❖

---

9. Conside a dipole of dipole moment p oriented along the z-axis (see worksheet E-2, Problem 7 and the Challenge Problem for E-2 for further discussions of dipoles).

a) Modeling the dipole as two charges q separated by a distance d and centered on the origin, find the electric potential along the axis of the dipole a height z above the origin. Take the potential to be zero at infinity, i.e. $V(z=\infty)=0$.

b) Take the far-field approximation of this potential. That is, expand your answer to first order in d using the approximation $(1+\varepsilon)^n \approx 1+ n\varepsilon$ and then replacing qd with the dipole moment p.

c) Find the z-component of the electric field along the z-axis. Compare your answer with the result found in E-2.

d) Repeat parts (a) through (c), this time looking at points along the plane perpendicular to the dipole axis and intersecting the dipole, i.e. the z=0 plane. ❖❖

E-5 Challenge Problem

Classical Radius of the Electron.

In this challenge problem, we will calculate a quantity known as the *classical radius of the electron.*

a) Consider a sphere of charge, with radius R and total charge q. What is the charge density, ρ?

b) What charge is enclosed in a sphere of radius $r<R$? Call this $q(r)$.

c) What charge, dq, is contained in a spherical shell of width dr and radius r?

d) Suppose we have already built a sphere of radius r with charge $q(r)$. How much work would it take to bring the charge found in part (c) to the surface of the sphere? (Assume we have an infinitely big sphere of total charge dq that we are contracting down to a radius r. Ignore the effects that charges in one part of the shell have on the other parts of the shell.

e) How much work will it take to assemble the entire sphere of radius R and charge Q?

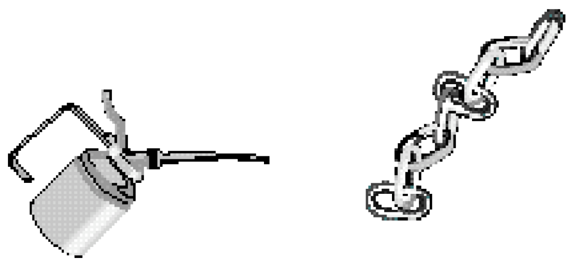
f) Einstein said that mass and energy were equivalent using his famous formula $E=mc^2$, where c is the speed of light and E is the energy content of a particle with mass m. Model an electron as a sphere of the type considered above, with charge *e* and radius r_e. Assume that all of the mass of the electron is from the energy associated with assembling the charge as we have done above. Plugging in the known values for the mass and charge of the electron, solve for r_e, the *classical radius of the electron.*

If we were instead to have considered the electron as a *shell* of charge, then we would have gotten the same answer, just with a different fraction in front. Of course, all of this ignores the considerable effects of quantum mechanics, which says that the electron should be a point particle. The classical radius is basically the radius below which Quantum Electrodynamics becomes important for describing the electron. ❖❖

E-6. Capacitance

Questions for discussion

1. The two metal objects below have net charges of +73 pC and -73 pC, and this results in a potential difference of 19.2 V between them.



a) What is the capacitance of this system?

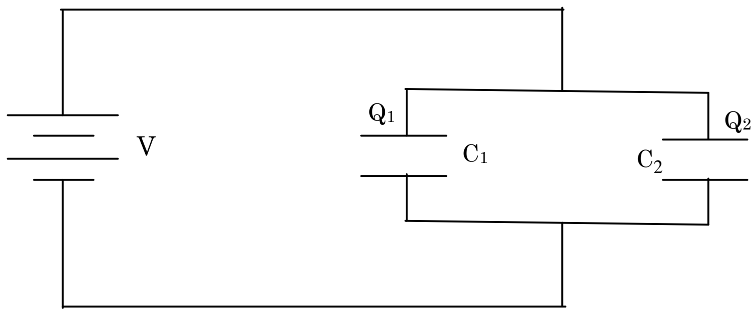
b) Suppose that the charges on the two objects are changed to +210 pC and -210 pC, respectively. What is the capacitance of the system now? How about the potential difference between them?

c) What are some ways in which you might alter the capacitance of this system?

2. Why does the capacitance of a capacitor get larger when the two pieces are closer together?

3. How does the presence of a dielectric affect the capacitance of a parallel-plate capacitor?

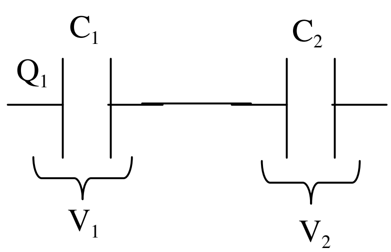
4. Consider two capacitors, C_1 and C_2, connected in parallel, as shown.



a) If a voltage V is put across the parallel capacitors, what is the charge on each capacitor?

b) What is the equivalent capacitance of the two capacitors in parallel?

5. Consider two capacitors, C_1 and C_2, connected in series, as shown.



a) If a charge Q is put on C_1, what is the charge on C_2?

b) If a charge Q is put on C_1, what is the voltage across each capacitor?

c) What is the equivalent capacitance of the two capacitors in series?

Problems

1. In a parallel-plate capacitor, the plates are in the shape of circles of radius R.

a) If the plate separation is d, then what is the capacitance?

Suppose a battery of voltage ε is connected across the plates of the capacitor.

b) How much charge will build up on the plates?

c) Once this charge has built up, how much energy will be stored in the capacitor? ❖

---

2. A conducting plate of area A has a charge +Q on it. (See Figure 1 at right.)

a) What is the electric field created by this plate? (Ignore edge effects.) Sketch this field at right.

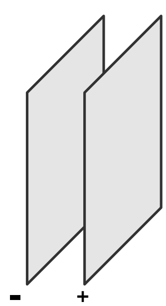
Figure 1

Now consider another conducting plate of area A, but with charge –Q on it. (Figure 2 at right.)

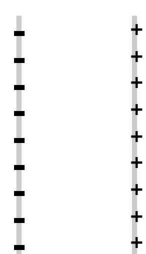
b) What is the electric field created by this second plate? Sketch this field at right.

Figure 2

Now consider that the plates are separated by a distance d, as shown in the two views below.



perspective view



side view

c) What is the field outside of the plates? Explain. (Again, ignore edge effects.)

d) What is the field between the plates? Draw the net field on the figure above.

e) Find the energy stored in the electric field (Again, ignore edge effects.)

f) What is the force exerted on one plate by the other?

g) What is the capacitance of this system?

h) Show that the energy stored in the capacitor, , is the same as we found in part (e).❖

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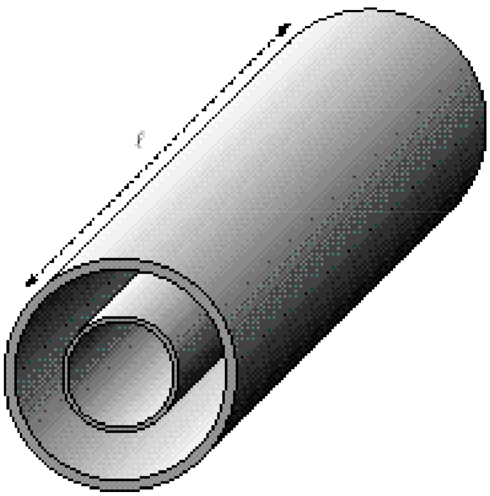
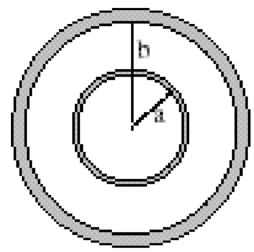
3. In this problem we will calculate the capacitance of a pair of concentric metal cylinders.



perspective view



end view

a) Imagine placing charges ±q on the plates, and sketch the resulting electric field in the region between the cylinders.

b) Calculate the strength of the electric field at points between the cylinders.

c) Find the difference in potential between the plates, by integrating the field you found in part (b) from one plate to the other.

d) What is the capacitance of the pair of cylinders? ❖

---

4. In a simple parallel-plate capacitor, the plates are circular, with radius R, and are separated by a distance D.

Suppose you connect a battery of voltage V_0 across the plates of this capacitor.

a) How much charge will accumulate on the plates?

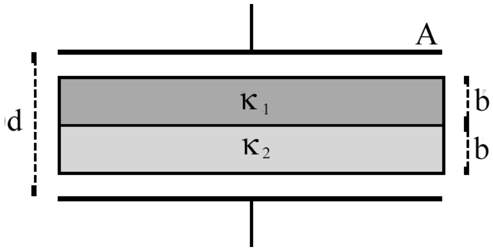
b) How strong will the electric field between the plates be?

c) How much electrostatic energy will be stored in each cubic centimeter of volume between the plates?

d) Considering your answer from part (c), how much energy will be stored in the capacitor as a whole?

e) Is your answer from part (d) consistent with the standard formulas, $U = \frac{1}{2}CV^2$ and $U = Q^2/2C$, for the electrostatic energy stored in a capacitor? ❖

5. A parallel-plate capacitor is filled with two different dielectrics as shown.



a) Find the capacitance by regarding this arrangement as four capacitors in series.

Now we will find the capacitance by a more systematic method:

b) Assume that the charge on the plates is $\pm q$, and find the electric field in the four regions.

c) Assuming $\kappa_1 > \kappa_2$, sketch the (total) electric field in each of the four regions.

d) Find the difference in potential V between the plates, by integrating the field you found in part (b) from one plate to the other.

e) Verify that the capacitance is given correctly by q/V. ❖

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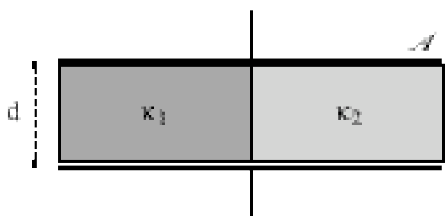
6. A parallel-plate capacitor is filled with two different dielectrics as shown.



a) Find the capacitance. (Hint: Regard this as two capacitors in parallel.)

b) If the charge on the plates is $\pm q$, then find the electric field in the two regions.

c) Assuming $\kappa_1 > \kappa_2$, sketch the (total) electric field in each of the four regions. ❖

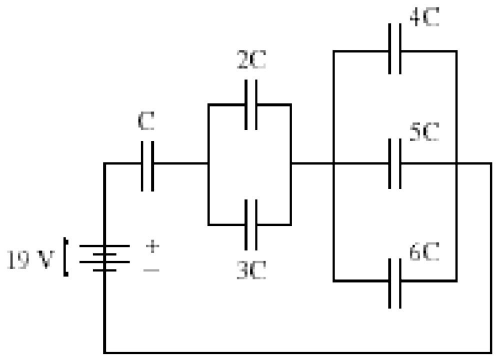
---

7. A parallel plate capacitor has plates of area A and spacing d between the plates. It is initially charged with charge Q, and is then disconnected from the battery. A <u>conducting</u> metal slab of width b is then positioned between the plates. The slab touches neither plate.

a) Find the capacitance of the system after the slab has been introduced. Answer in terms of A, d, b, and ε_0. (You should be able to come up with more than one way of obtaining the answer.)

b) How much work was done by the person who moved the conducting metal slab from very far away to its final position between the two plates? ❖

8. Consider the following circuit.



If C = 1 microfarad (μF), then how much charge resides on each of the capacitors in this circuit? ❖❖

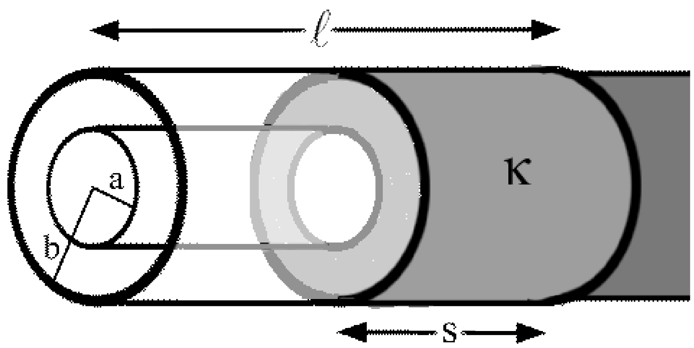
E-6 Challenge Problem

Force on a Dielectric in a Cylindrical Capacitor

Consider a cylindrical capacitor with inner radius a, outer radius b, and length ℓ (ignore any fringing effects).

a) What is the capacitance of this capacitor?

Now suppose we have a long length dielectric material shaped like a thick cylindrical shell, also of inner radius a and outer radius b so that it can potentially fill the capacitor. Call the length of dielectric filling the capacitor s, as shown below.



b) What is the capacitance of this conductor as a function of the amount of dielectric filling the capacitor, s.

Suppose the capacitor is connected to a voltage source and allowed to charge up. The voltage source is disconnected when the charge on the capacitor is Q.

c) What is the total energy stored in the capacitor, U, as a function of Q and s?

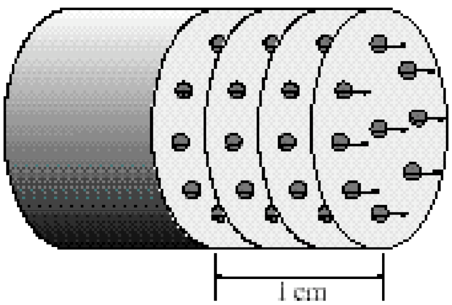
d) Recall from 7A that the force on an object in the direction of increasing a variable x is given by $F_x = -\frac{\partial U}{\partial x}$. Using this and your result from part (c), find the force on the dielectric that is filling the capacitor. Is the force going to pull the dielectric farther into the capacitor or push it out?

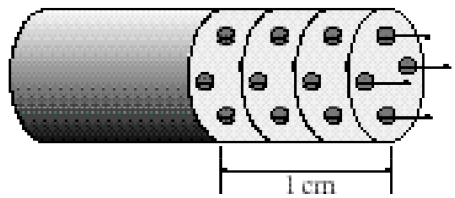
e) Find the force on the dielectric if, instead of fixing the charge, we fix the voltage across the capacitor (by, say, hooking it up to a battery) ❖❖

E-7. DC Circuits

Questions for discussion

1. The figure below shows a cutaway view of two current-carrying wires, a thick one and a thin one.





In the thick wire, electrons are moving with a speed of 3 cm/sec. In the thin wire, electrons are moving faster, with a speed of 5 cm/sec. Let both wires have the same electron density (the number of electrons per unit volume) be ρ.

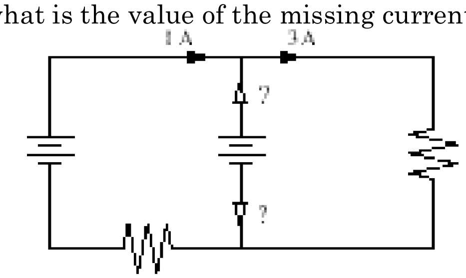
a) First consider the thick wire. In one second, how many electrons will pass a given point in the wire?

b) Next consider the thin wire. In one second, how many electrons will pass a given point in this wire?

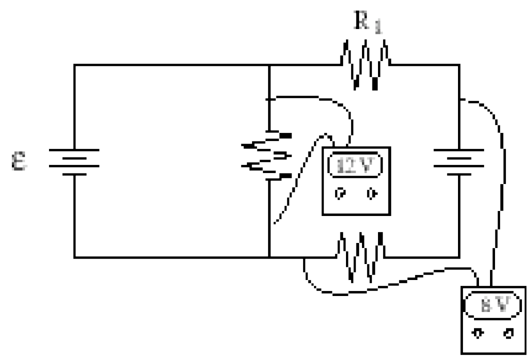
c) Which wire carries the greater current?

d) What are the *numerical* values of the two currents, in amps?

2. In the circuit shown below, what is the value of the missing current? In which direction does it flow?



3. Voltmeters have been attached to the following circuit, in order to ascertain voltages between various points.

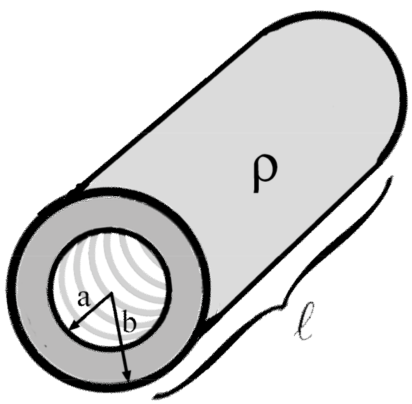


a) What is the voltage ε of the battery on the left?

b) What is the voltage drop across the resistor R_1?

Problems

1. A resistor is made out of a material with resistivity ρ. The resistor is in the shape of a thick cylindrical shell of inner radius a, outer radius b, and length ℓ. The resistor is attached to the circuit at the ends of the cylinders, so that current flows along the length.



a) What is the resistance of this resistor?

b) Suppose we change the length to 2ℓ. How does the resistance change? Does this make sense given what you know about adding resistors in series?

c) Suppose we change the outer radius to c. What is the new resistance? Show that this is the same resistance that we would get if we added a resistor with inner radius a and outer radius b in parallel to another resistor with inner radius b and outer radius c.

d) The original resistor is hooked up to a voltage source of voltage V. What current runs through the resistor?

e) What power does the resistor emit when it is hooked up to this circuit?

***f)** Find the resistance of the resistor if, instead of running the current along the length, we attach one end of the circuit to the inner radius and the other end to the outer radius, so that current now flows radially out through the resistor? ❖

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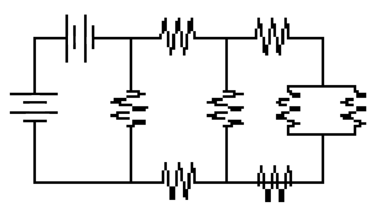
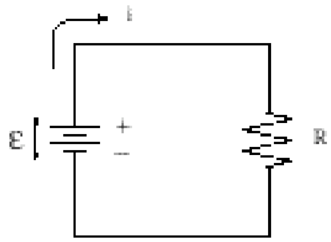
2. In the following circuit, all batteries are 1 V, and all resistors are 1 Ω.



Reduce the circuit to an equivalent voltage and resistance in series. ❖

3. Consider the following simple circuit.



a) What is the potential difference between one terminal of the battery and the other?

b) Considering your answer from part (a), how much work does it take to bring an element of charge dq from the "bottom" terminal to the "top" terminal?

c) Considering your answer from part (b), at what *rate* is the battery doing work on the charges in the circuit? Answer in terms of the symbols on the diagram.

d) The current flowing in this circuit is constant in time. What does this imply about the *speed* with which the charge carriers move around the circuit?

e) Recall that according to the work-kinetic energy theorem, the net work done on the charge carriers must equal the change in their kinetic energy. Considering your answer from part (d), what is the net work done on the charge carriers as they go around the circuit?

f) We found in part (c) that the battery does *positive work* on the charge carriers as they move around the circuit. So something else must be doing *negative work* on the charges as the move around, in order to satisfy the work-kinetic energy theorem. What element in the circuit is doing negative work on the charges?

g) In order to satisfy the work-kinetic energy theorem, what must be the rate at which heat is lost through the resistor? Answer in terms of the symbols on the diagram. ❖

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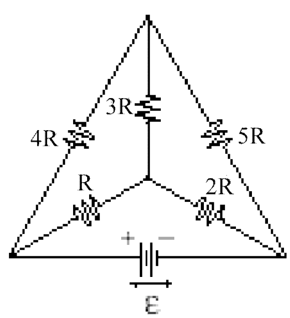
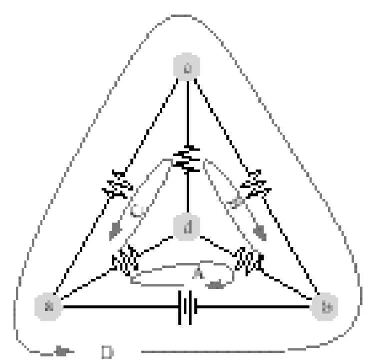
4. Consider the following circuit.



a) How many distinct currents flow in this circuit?

b) If we had to solve for these currents, how many independent equations would we need?

This diagram shows several different loops and junctions in the circuit.



The four loops shown are labeled A - D, and the junctions are labeled a - d.

c) Which loops and/or junctions would you choose, if you had to solve for the currents? (Many answers are possible here.)

d) Set up a system of equations that would allow you to solve for the currents. (Do not bother actually solving this system.) ❖❖

M-1. Introduction to Magnetism

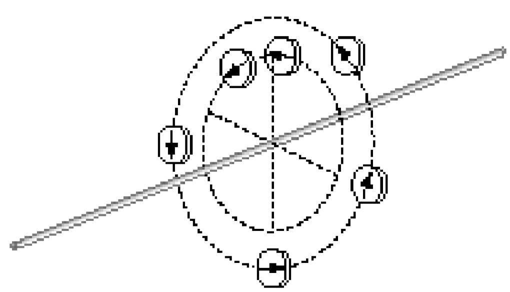
Questions for discussion

1. Suppose you have a compass that points North when you are standing in Paris. If you take that compass to Capetown, will it point South?

2. A compass often comes in handy when you are hiking. But in order to find geographic North using the compass, you have to correct its reading. Why is this? Is there anywhere in the United States you could be hiking, where you would *not* have to correct the compass reading?

3. It was considered a major discovery in 1820 when Oersted noticed that compasses deflect near current–carrying wires. What would you say was the main implication of this discovery?

4. The diagram below shows the readings a compass would give when placed in various locations near a current-carrying wire.



a) Based on this diagram, try sketching some *magnetic field lines* for the magnetic field produced by the wire.

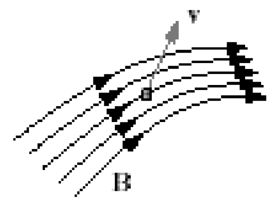
b) Which way is the current flowing in this wire?

M-2. The Lorentz Force Law

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$$

Questions for discussion

1. The diagram below shows a proton moving through an external magnetic field at a particular instant of time. (Here, the velocity vector is coming out of the plane of the page.)



a) At the instant shown, would the magnetic field be exerting any force on the proton? If so, in which direction? Sketch the force vector on the diagram above.

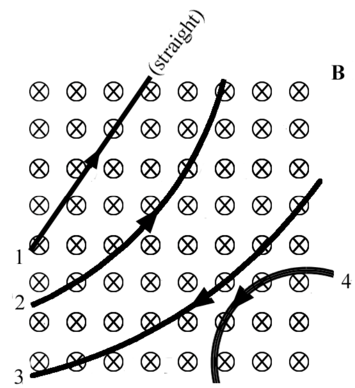
b) Would your answers change if the particle were an electron?

2. A positively charged particle is moving with the indicated velocity through an external magnetic field, which is *not shown*. This external magnetic field exerts a force on the moving charge, which *is* shown.

a) What is the direction of the external magnetic field at the location of the particle?

b) Answer part (a) again, this time assuming the particle is *negatively* charged.

3. In the figure below, a uniform magnetic field points *into the page*.[1] (The magnetic field vectors are indicated by ⊗'s.) Four particles with the same mass follow the paths shown as they pass through this magnetic field with identical, constant speeds.



What can you conclude about the charge on each particle?

4. An electron moves horizontally from the left with speed v and enters a uniform vertical electric field of magnitude E_0 pointing upwards. In the absence of any other forces, the electron would be deflected vertically by the $F_{on\ q} = qE_{ext}$ force. Sketch the direction of a magnetic field that could cancel this force and allow the electron to maintain its horizontal path.

5. When a particle moves under the influence of a magnetic field, the *speed* of the particle remains constant. How does this come about?

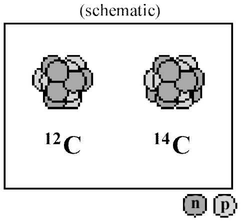
[1] For a summary and review of cross products, right hand rules, and drawing vectors that leave the plane of the page, see "Vectors and Right Hand Rules in Magnetism," in the Supplementary Material at the end of the workbook.

Problems

1. *The mass spectrometer.*

In a sample of ordinary carbon, most of the carbon atoms consist of six electrons bound to a nucleus of six protons and six neutrons. But a small fraction of the carbon atoms in the sample will be slightly heavier, consisting of six electrons, six protons, and *eight* neutrons.

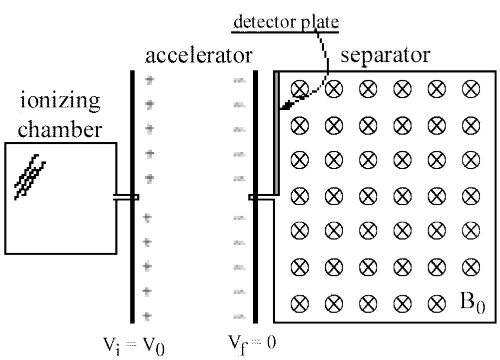
Nuclei for two isotopes of carbon (schematic)



As you can see, the masses of these *isotopes* are very nearly $M_{12} = 12m_p$ and $M_{14} = 14\ m_p$, respectively, where $m_p = 1.67\times10^{-27}$ kg is the mass of a proton.

Because these isotopes are the same as far as their *charged* constituents, their chemical properties are for all practical purposes identical. So we cannot separate these isotopes from one another via chemical procedures.

A mass spectrometer uses a magnetic field to separate the isotopes as follows.



In the *ionizing chamber*, an electric discharge runs through the atoms in the carbon gas, causing them to become singly ionized (which means that they have lost one of their outer electrons). When the ions emerge from the ionizing chamber, they are accelerated through a potential difference V_0, at which point they enter a uniform magnetic field B_0 as shown. The magnetic field deflects the ions into a circular path, and they eventually collide with a detector plate.

The reason this whole scheme works is that ions of different isotopes will end up in different places on the detector plate. This allows identification of the relative abundances of the isotopes.

In this problem we will determine the separation of ^{12}C and ^{14}C on the detector plate.

a) How fast is a ^{12}C ion going when it enters the magnetic field? Answer in terms of V_0, m_p, and e. (For simplicity, assume that the ions emerging from the ionizing chamber are at rest.)

b) Draw a force diagram for a ^{12}C ion when it is traveling inside the magnetic field in a circular path with the speed calculated in part (a). (Neglect gravity.)

c) Use Newton's Second Law $\mathbf{F}_{net} = m\mathbf{a}$ to determine the radius of the ^{12}C ion's circular path. Answer in terms of V_0, B_0, m_p, and e.

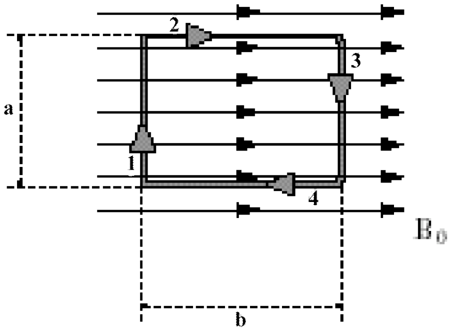
d) Looking at your answer for part (c), write down an expression for the radius of a ^{14}C ion's circular path within the magnetic field.

e) What is the separation of the ions on the detector plate? Answer in terms of V_0, B_0, m_p, and e.

f) If we want the separation to be at least 1 cm, and the magnetic field we are using is 0.1 Tesla, then how large is the accelerator voltage V_0 going to have to be?

g) With the voltage set at this level (and with B_0 = 0.1 T), where will the ^{12}C ions strike the detector plate? ❖

2. In the figure below, a rectangular loop of wire is immersed in an external magnetic field B_0 pointing to the right.



The loop carries current i_0 in the clockwise direction, and is held fixed in place.

a) How large is the force exerted on segment 1 of the loop by the external magnetic field? In which direction does it point?

b) Answer the same questions for segments 2, 3, and 4.

c) If the loop were not held fixed in place, what would be the net effect of the forces on each segment of the loop?

d) What torque does segment 1 of the loop experience? (Consider the magnetic force on the segment to act at the midpoint of the segment, and take the torque about the center of the loop.)

e) Answer the same question for segments 2, 3, and 4. In each case, consider the magnetic force on the segment to act at the midpoint of the segment, and take torques about the center of the loop.

f) What is the net torque on the loop about its center? Give the magnitude and direction.

g) Now compute the torque on the loop using the standard formula $\boldsymbol{\tau}_{\text{on loop}} = \boldsymbol{\mu}\times\mathbf{B}$, where $\mu = i\mathbf{A}$ is the magnetic moment of the loop. Does your answer from part (f) agree, in both magnitude and direction? ❖

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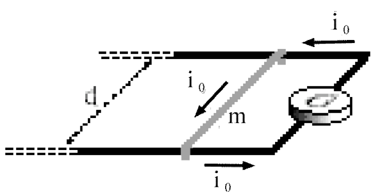
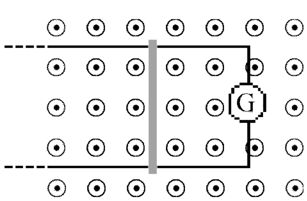
3. A metal wire of mass m slides without friction on two horizontal rails spaced a distance d apart, as shown.



perspective view



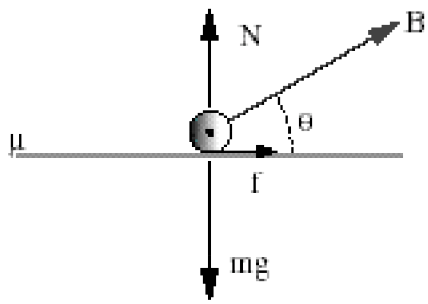
top view
showing B out of page

The track lies in a uniform external magnetic field B_0, pointing perpendicular to the plane of the rails. A constant current i_0 flows from the generator G along one rail, across the wire, and back along the other rail.

a) At any given time, what is the force exerted on the sliding wire by the external magnetic field? Give both the magnitude and direction.

b) Find the velocity of the sliding wire as a function of time, assuming it to be at rest at time t=0. ❖

4. A copper rod of mass m rests on two horizontal rails a distance d apart, and carries a current i_0 from one rail to the other. The coefficient of static friction is μ. A uniform magnetic field of unknown magnitude B points upward at angle θ. Here is a side view:



In this diagram, the current i_0 in the wire is coming out of the page, and **f** denotes the friction force. Notice that the magnetic field vector **B** is *not* a force vector.

a) The magnetic force on the wire has not been included in the above diagram. What is the magnitude of this force? Sketch the force vector on the diagram, and break it into horizontal and vertical components.

b) If the wire is just on the verge of sliding sideways, then how large is the magnetic field B? Your answer will involve θ.

c) By minimizing your expression for B with respect to θ, find the smallest magnetic field that could cause the wire to slide sideways.

d) If the magnetic field were *vertical*, which direction would the force on the wire point? Why is the optimal magnetic field found in part (c) *not* vertical? ❖❖

M-3. Magnetic Fields

Questions for discussion

1. Sketch the magnetic fields created by these currents:

a) A long straight wire.

b) A circular current loop.

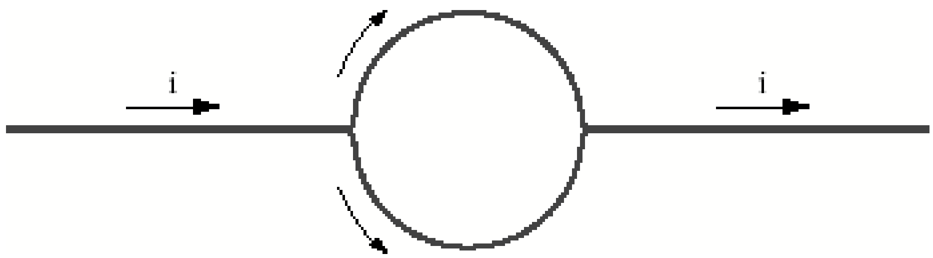
2. Suppose we have two long, straight current-carrying wires running parallel to one another. If the two currents flow in the same direction, will the wires attract each other or repel each other?

3. A long straight wire has a net charge per unit length λ.

a) If the wire is fixed in place, what is the strength of its *electric* field?

b) If the wire moves along its length with speed v, what is the strength of its *magnetic* field? Does the wire still have an electric field when it moves this way?

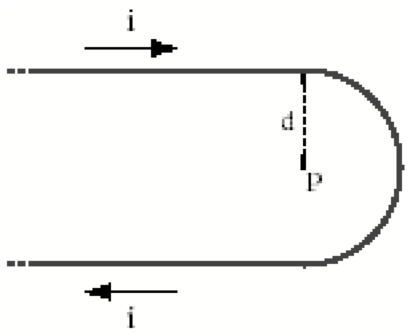
4. A straight conductor carrying a current i is split into identical semicircular turns as shown.



What is the magnetic field strength at the center? Carefully explain your reasoning.

Problems

1. A long wire is bent into the hairpin-like shape shown in the figure.



a) What is the *direction* of the magnetic field at the indicated point P, which lies at the center of the half-circle?

b) What is the *magnitude* of the magnetic field at that point? (Hint: At the center of a circular current loop of radius R, the loop's magnetic field has magnitude $B_{\text{loop, at center}} = \mu_0 i / 2R$.)

c) Suppose an electron is at rest at point P. What force does the magnetic field exert on it? ❖

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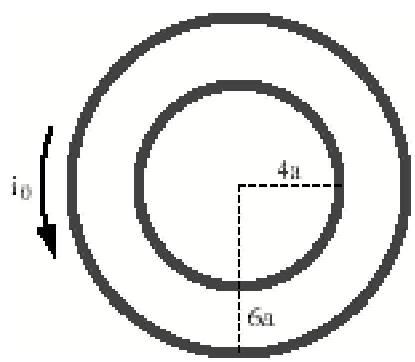
2. Two circular wire loops are fixed in place as shown. The large loop carries current i_0 counter-clockwise as viewed from above.



a) Suppose we want the magnetic field to vanish at the common center of the loops. Should we generate a *clockwise* current in the small loop, or a *counter-clockwise* current?

b) How strong a current will we need to generate? (Hint: At the center of a circular current loop of radius R, the loop's magnetic field has magnitude $B_{\text{loop, at center}} = \mu_0 i / 2R$.) ❖

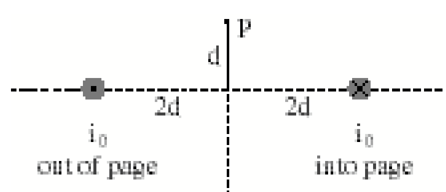
---

3. Two straight wires are fixed in place near one another, and carry equal currents i_0 in opposite directions.

a) Sketch the two contributions to the net magnetic field at point P.

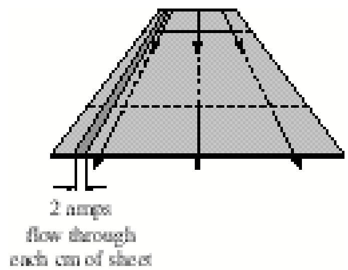
b) Add these vectors graphically to obtain a sketch of the net magnetic field vector at P.

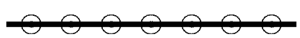
c) Calculate the magnitude and direction of the magnetic field at P. ❖



end view

4. An infinite metal sheet has current flowing through it as shown in the figure below.

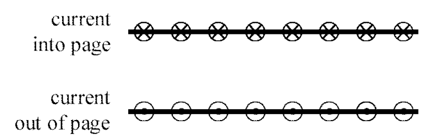




end view

a) Sketch the magnetic field created by this current distribution. (Hint: Think of the plane as consisting of infinitely many current-carrying wires. Also, use symmetry reasoning.)

Now imagine that two such current sheets are fixed in place near one another.



end view

b) Sketch the magnetic field in all three regions (above, in between, and below the two current sheets).

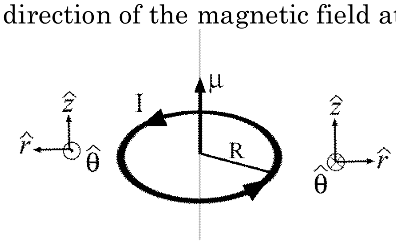
c) What would the field look like if *both* current distributions were flowing into the page? ❖❖

M-3 Challenge Problem

The Biot-Savart Law and Magnetic Dipoles

In this challenge problem we will use the Biot-Savart Law to show that, far away from a magnetic dipole, the magnetic field has the same form as the electric field for a dipole. The most common type of magnetic dipole we will encounter is the current loop.

Consider a loop of radius R in the z=0 plane with a counterclockwise current I flowing through it. The magnetic dipole moment, **μ** (not to be confused with μ_0!), has a magnitude of IA, where A is the area of the loop, and points in the direction of the magnetic field at its center.



The Biot-Savart Law says that, for a given current distribution, $\vec{B} = \int_{current} \frac{\mu_0 I}{4\pi\rho^2}\left(d\vec{\ell} \times \hat{\rho}\right)$, where I is the current and $d\ell$ is a little piece of line element, which is somewhere on the current and points in the direction of the current. ρ is the vector that points from the little piece of current we are looking at to the point where the magnetic field is to be calculated. The integral runs over the entire current.

a) Parameterize the current using the variable θ, which will run from 0 to 2π. Let θ=0 when we are looking at the piece of current that is at coordinates x=R, y=0, z=0. What is $d\vec{\ell}$? Again, work with the cartesian unit vectors.

b) Consider a point at coordinates (x,y,z). What is $\vec{\rho}$? $\hat{\rho}$? ρ? Write these out in terms of the cartesian unit vectors ($\hat{x}, \hat{y}, \hat{z}$) and the variables r (the distance from the origin to the point (x,y,z)), θ (definied in part (a) above), x and y. Calculate $d\vec{\ell} \times \hat{\rho}$.

c) If you've done part (b) correctly, when you write everything out you should have the following factor in your integrand: $\left[r^2 + R^2 - 2R(x\cos\theta + y\sin\theta)\right]^{-3/2}$. This integral is too messy for us to solve as is, and we are only interested in the far-field anyway. Assume that r>>R (we are very far away from the dipole). Expand this term to first order in R (which means throw out the R^2 and use $(1+\varepsilon)^n \approx 1 + n\varepsilon$).

d) Write out the full integral again, plugging in your results from part (b) and (c). Throw out any terms that have an R^3 or higher in them. Replace IA with μ. We have now taken the far-field limit!

e) Integrate! Remember that the integral of sine or cosine over a full period is 0 and the integral of $\sin^2\theta$ and $\cos^2\theta$ is π.

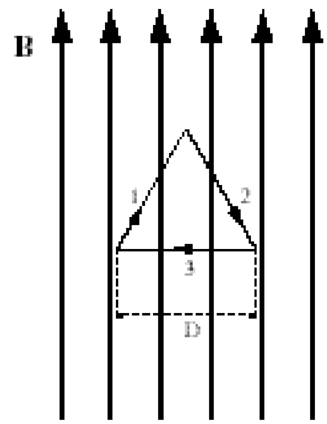
f) Show that your answer is equivalent to $\vec{B} = \frac{\mu_0\mu}{4\pi r^3}(3\cos\varphi\,\hat{r} - \hat{\mu})$. Compare the form of this to the form of the field of an electric dipole. ❖❖

M-4. Ampère's Law

$$\oint_C \vec{B} \cdot d\vec{s} = \mu_0 I_{\text{enc}}$$

Questions for discussion

1. A uniform magnetic field **B** points upward, as shown. (The source of this field could be a pair of current sheets. But in any case, these current sheets are far enough away not to be shown in this diagram.)



The figure also shows an imaginary closed path, in the shape of a planar equilateral triangle of side D.

a) What is the value of the line integral of the magnetic field around this imaginary closed path? (You should be able to answer very quickly using Ampère's Law.)

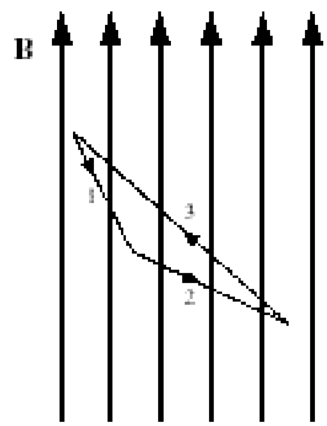
b) Calculate the line integral of the magnetic field along segment 1 of the imaginary triangle. Do this directly using the definition

$$\text{line integral of } \mathbf{B} \text{ along segment 1} \equiv \int_{\text{segment 1}} \mathbf{B} \cdot d\mathbf{s}.$$

c) Do the same for segments 2 and 3 of the imaginary triangle.

d) Add your results, and see if your overall value of $\oint_{\substack{\text{triangular}\\\text{path}}} \mathbf{B} \cdot d\mathbf{s}$ matches the result from part (a).

2. The situation is the same as for Discussion Question 1, but this time the triangle is *scalene.*



a) What is the value of the line integral of the magnetic field around this imaginary closed path? (Again, you should be able to answer very quickly using Ampère's Law.)

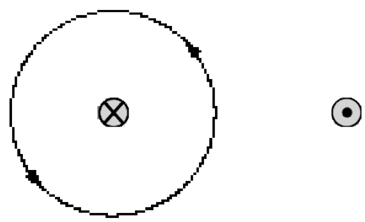
b) Identify those legs of the triangle, if any, along which $\int_{\text{leg}} \mathbf{B} \cdot d\mathbf{s}$ is *positive.* Next, identify those legs, if any, along which $\int_{\text{leg}} \mathbf{B} \cdot d\mathbf{s}$ is *negative.* Finally, identify those legs, if any, along which $\int_{\text{leg}} \mathbf{B} \cdot d\mathbf{s}$ is *zero.*

c) Considering your results from part (b), explain how it comes about that the line integral around the imaginary path has the value you found in part (a).

3. Two parallel wires fixed in place near one another carry equal and opposite currents i. The left current is into the page, the right current is out of the page.

a) On the diagram above, sketch the magnetic field created by these two currents.

b) Now consider an imaginary *circular path* encircling the left wire:



NOTE: The line indicates an imaginary *path*, and does <u>not</u> represent an imaginary *surface* of any kind. The entire diagram is planar, and no perspective is intended.

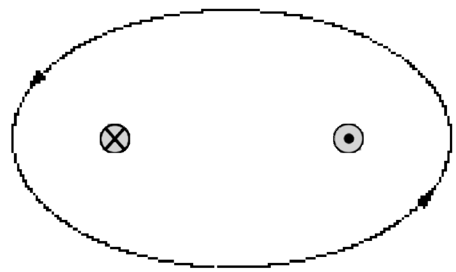
Again, the dashed line is simply a *circle*, and is <u>no</u>t a 2-D representation of any kind of surface.

i) Reproduce your drawing of the magnetic field lines from part (a) on the just figure above, so you can get a sense of how the imaginary circular path runs along the magnetic field lines.

ii) What is the value of the line integral of the magnetic field around the imaginary circular path? You should be able to arrive at the answer very quickly using Ampère's Law.

iii) By examining the field lines and how they run along the imaginary circular path, try to explain *why* the line integral turns out to be what Ampère's Law said it was. (For example, try to explain why the line integral around the path comes out to be *negative*.)

c) Next, consider an imaginary *elliptical* path encircling both wires:



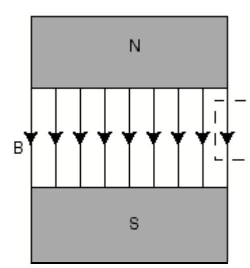
NOTE: Once again, the dashed line indicates an imaginary *path*, and does not represent an imaginary *surface* of any kind. The entire diagram is planar, and no perspective is intended.

i) Reproduce here your drawing of the magnetic field lines from part (a), so you can get a sense of how the imaginary elliptical path runs along the magnetic field lines.

ii) What is the value of the line integral of the magnetic field around the imaginary elliptical path? You should be able to arrive at the answer very quickly using Ampère's Law.

iii) By examining the field lines and how they run along the imaginary elliptical path, try to explain *why* the line integral turns out to be what Ampère's Law said it was.

3. Show that a uniform magnetic field *cannot* drop abruptly to zero at an edge as shown in the figure below. (Hint: consider the amperian loop shown as a dashed rectangle.) In actual magnets, there is always a "fringing field." Modify the figure to indicate a more realistic situation.



Problems

1. A thick wire of radius R carries total current i distributed *uniformly* across its cross-section.

a) Sketch the magnetic field created by the wire, at points both inside and outside the wire.

b) Find the strength of the magnetic field created by the wire, at any point inside the wire ($r < R$).

c) Find the strength of the magnetic field created by the wire, at any point outside the wire ($r > R$). ❖

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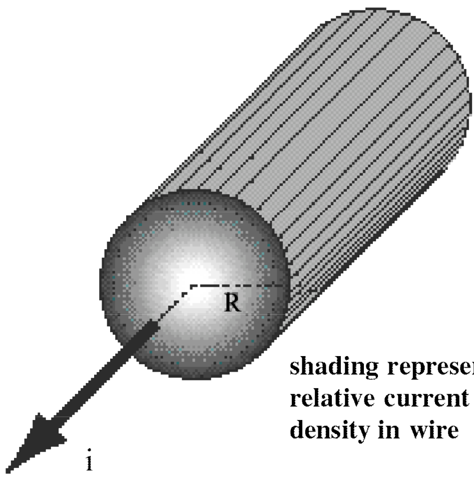
2. A thick wire of radius R carries total current i distributed *non-uniformly* across its cross-section. The current density within the wire (in amps per square meter) is given by $j(r) = j_0(r/R)^4$.



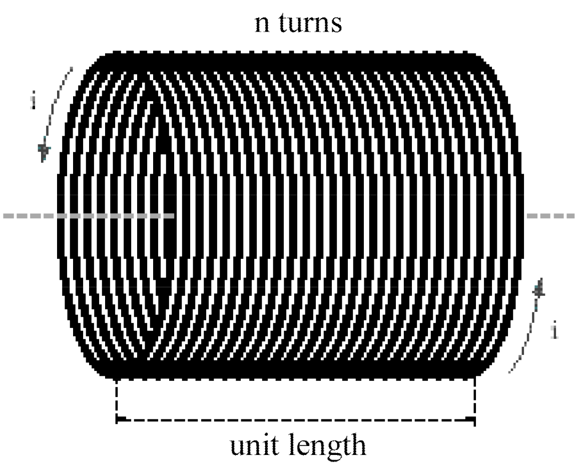
shading represents relative current density in wire

a) Sketch the magnetic field created by the wire, at points both inside and outside the wire.

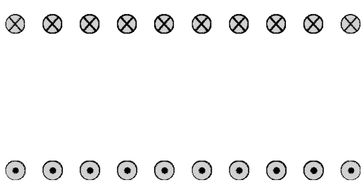
b) Find the strength of the magnetic field created by the wire, at any point inside the wire ($r < R$).

c) Find the strength of the magnetic field created by the wire, at any point outside the wire ($r > R$). ❖

3. An infinitely long solenoid with n turns of wire per unit length carries a current i.



perspective view



schedmatic cutaway view

(The solenoid's length is compressed in these diagrams.)

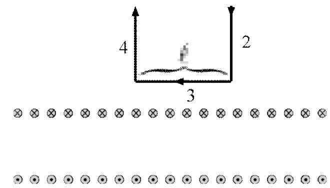
We will use Ampère's Law to calculate the strength of the magnetic field within the solenoid.

a) Put in a cylindrical coordinate system (coordinates r, θ, z). Approximating the solenoid as infinitely long, which of the three coordinates will the magnetic field strength depend on?

b) Sketch the magnetic field of the solenoid, at points both inside and outside the solenoid. (Hint: Use a right hand rule to get the direction of the field inside the solenoid.)

c) Use your results from (a) and (b) to write down the general form for the **B**-field in this type of setup.

Next, consider the imaginary path shown below:



schedmatic cutaway view

Segment 1 is placed at r = ∞ and has a length ℓ. Segment 3 is placed at some radius r > R and also has a length ℓ, so that the entire rectangular path lies outside of the solenoid.

d) What do you expect the magnitude of the magnetic field will be infinitely far away from the solenoid? What is the value of $\int_{\text{segment}} \mathbf{B} \cdot d\mathbf{s}$ for segment 1 of the path?

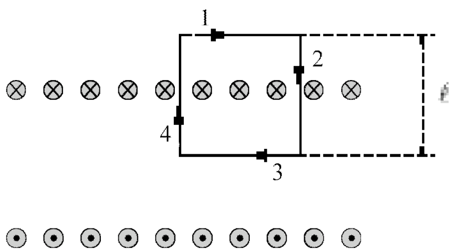
e) Find the value of $\int_{\text{segment}} \mathbf{B} \cdot d\mathbf{s}$ for the segments 2 and 4 of the path. Explain your reasoning.

f) Find the value of $\int_{\text{segment}} \mathbf{B} \cdot d\mathbf{s}$ for segment 3 of the path. Express your answer in terms of the unknown magnetic field strength B.

g) What is the value of $\oint_{loop} \mathbf{B} \cdot d\mathbf{s}$ for the entire square loop? Again, you may express this value in terms of the unknown magnetic field strength B.

h) How much current is encircled by the square loop? Using Ampere's Law, what does this tell you about the magnetic field strength outside the solenoid?

Next, consider another imaginary path shown below:



schedmatic cutaway view

Each side of this square path has length ℓ, and this time, segment 3 is *inside* the solenoid.

i) What is the value of $\int_{\text{segment}} \mathbf{B} \cdot d\mathbf{s}$ for segment 1 of the square path? Why?

j) Find the value of $\int_{\text{segment}} \mathbf{B} \cdot d\mathbf{s}$ for the other three segments of the square path. Where necessary, express these values in terms of the unknown magnetic field strength B. Explain your reasoning in each case.

k) What is the value of $\oint_{loop} \mathbf{B} \cdot d\mathbf{s}$ for the entire square loop? Again, you may express this value in terms of the unknown magnetic field strength B.

ℓ) How much current is encircled by the square loop? (Note: The diagram is schematic, so it is not acceptable simply to count the number of ⊗'s encircled in the diagram. To determine the current encircled by the loop, you must consider the number of turns per unit length in the solenoid, and the length of the square loop.)

m) What is the strength of the magnetic field within the solenoid? ❖

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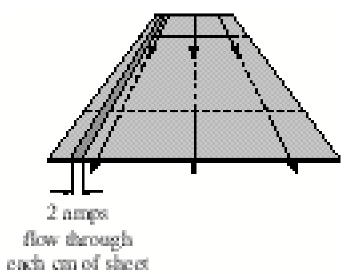
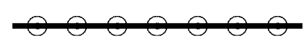
4. Use Ampère's Law to calculate the strength of the magnetic field created by an infinite current sheet.





end view

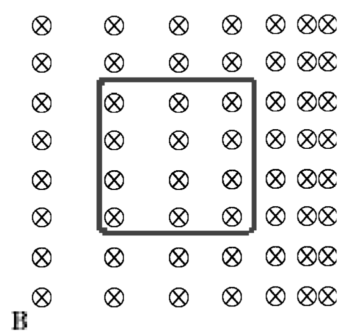
(See Problem 4 of Worksheet M-3, *Magnetic fields*.) The sheet carries current per unit length K = 200 amps/meter. ❖❖

M-5. Faraday's Law

$$\oint_C \vec{E} \cdot d\vec{s} = \mathcal{E}_{\text{induced}} = -\frac{d\Phi_B}{dt}$$

Questions for discussion

1. In the figure below, there is a non-uniform magnetic field pointing into the page.



a) If you move the metal loop to the right, will a current be induced in it?

b) If so, will the induced current be clockwise or counter-clockwise?

c) Suppose you want to move the loop to the left at constant speed. Will you have to exert any force to do this? Why or why not?

d) Suppose instead that you want to move the loop upwards. Will a current be induced in it? Why or why not?

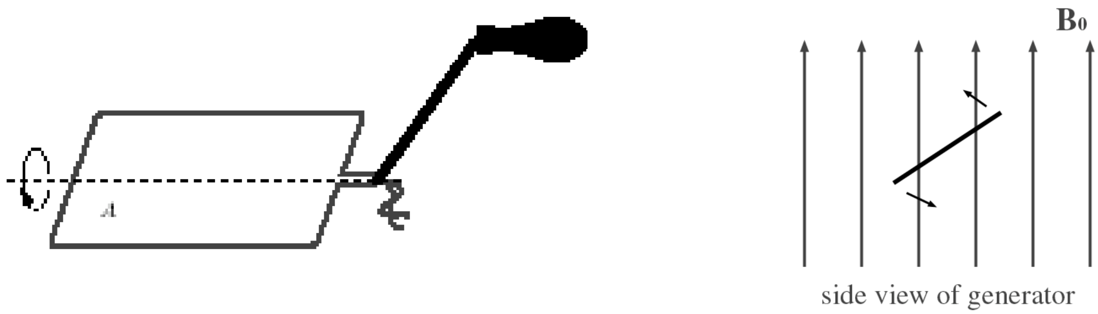
2. The diagram below shows an infinitely long current-carrying wire, with two metal loops nearby.

By means of a current generator (not shown), we cause the current in the wire to increase with time.

a) Will a current be induced in loop 1? If so, will it be clockwise or counter-clockwise?

b) Will a current be induced in loop 2? If so, will it be clockwise or counter-clockwise?

3. A crude current generator consists of a loop of wire with area A and resistance R. The loop is connected to a handle, so that someone can cause it to rotate within a uniform magnetic field B_0 that points upwards.



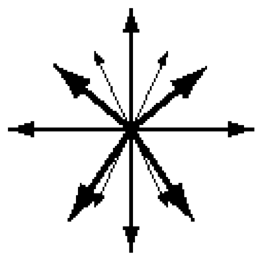
a) How does this device generate current?

b) Why will you have to do work in order to turn the handle?

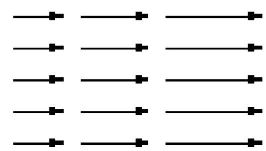
4. ***Gauss's Law for magnetism.***

$$\Phi_B = \oiint_S \vec{B} \cdot d\vec{A} = 0$$

Consider the two vector fields shown below.



vectors radiating outwards from a point



vectors increasing along their own direction (field uniform in direction normal to page)

a) Why can't these be pictures of magnetic fields?

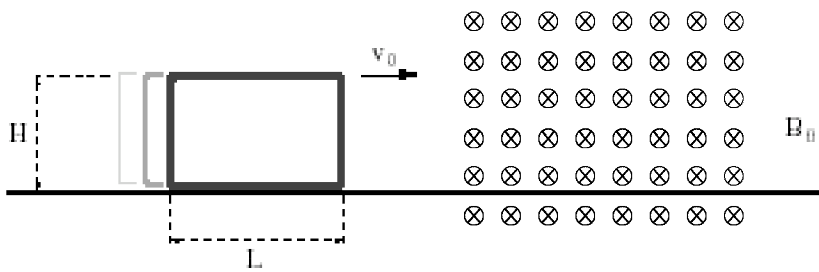
b) For the field on the left, draw an imaginary closed surface for which $\Phi_B \neq 0$.

c) Do the same for the field on the right.

5. The left hand side of Faraday's Law involves the induced emf, which is the line integral of the electric field around a closed loop. The right hand side involves the flux through a surface. The loop used on the left is the *boundary* of the surface from the right. However, the same loop can be the boundary for many different surfaces (Think of the film of soap on a bubble wand). Why does Gauss' law for magnetic fields imply that we will get the same emf around a given loop no matter what surface we use to compute the changing flux through, as long as the loop is the boundary of the surface?

Problems

1. A rectangular loop of wire has length L, height H, and resistance R. The loop balances on one edge and slides frictionlessly with speed v_0. It encounters a region of uniform magnetic field B_0 pointing into the page.



There are five time intervals of importance here:

i. The time before the loop enters the field;
ii. The time while the loop is partly within the field;
iii. The time while the loop is entirely within the field;
iv. The time while the loop is partly out of the field;
v. The time while the loop is entirely outside the field.

a) For each of the five time intervals, determine the sense of the induced current (clockwise, counter-clockwise, or no current).

b) Taking clockwise currents to be positive, sketch a graph of the induced current versus time.

c) Do you think that the final speed of the loop will be less than v_0, equal to v_0, or greater than v_0? Why?

d) Ignoring the effect alluded to in part (c), determine the magnitude of the induced current for each of the five time intervals. ❖

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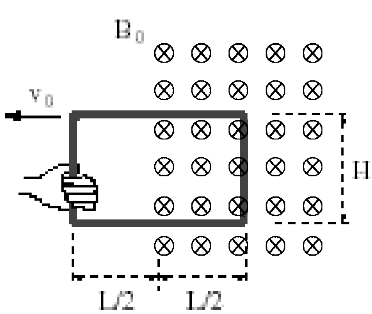
2. The rectangular metal loop has height H, width L, and resistance R. Initially the loop is situated halfway within a uniform magnetic field B_0. You want to pull the loop out of the field at a constant speed v_0.



In this problem, we will calculate in two different ways the amount of *work* you will have to do in order to extract the loop.

a) Explain why you *are*, in fact, going to have to do work in order to extract the loop. (Try to give more than one explanation.)

b) During the time you are pulling the loop out of the field with speed v_0, what is the magnitude of the induced current in the loop? Is this clockwise or counter-clockwise?

First method: Forces

c) What force does the magnetic field exert on each segment of the loop during this time?

d) What is the net magnetic force on the loop? In which direction does it point?

e) In order to pull the loop to the left with constant speed v_0, what force will you have to exert?

f) Over what distance will you have to exert this force?

g) How much work will you have to do in order to extract the loop?

Second method: Energy

h) During the time you are pulling the loop out of the field with speed v_0, what is the rate at which heat is being dissipated by the loop's resistance?

i) For how long a time will heat energy be dissipated at this rate?

j) How much total heat energy will be dissipated during the time you pull out the loop?

k) How much work will you have to do in order to extract the loop?

ℓ) Do your answers for part (g) and part (k) agree? ❖

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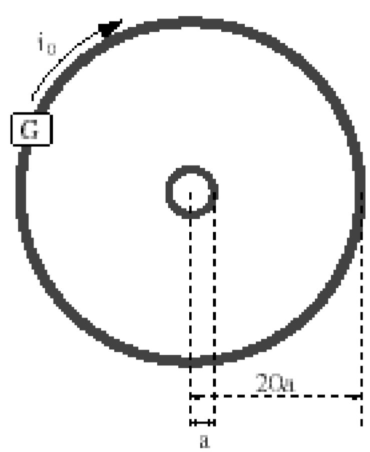
3. The metal loops shown below are concentric circles lying in the same plane.



The large loop of radius 20a is connected to a current generator. The small loop of radius a is not connected to anything.

a) Suppose that a current i_0 is flowing clockwise around the large loop. Sketch the magnetic field created by the large loop.

b) How strong is this magnetic field at the center point?

c) Assume that the small loop is *so* small that the magnetic field of the large loop is approximately *uniform* over its extent. Under this assumption, what is the flux of the magnetic field through the small loop?

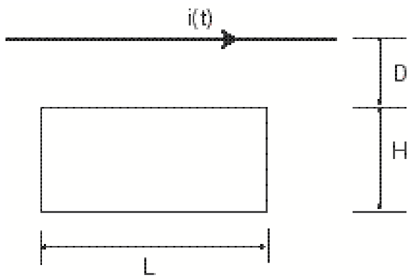
Now imagine that, starting at t=0, the current generator causes the current in the large loop to increase at a steady rate, until it reaches $4i_0$ at time t=T. After time t=T, the current in the large loop remains steady at $4i_0$.

d) Sketch the current in the large loop as a function of time.

e) During the time $0 < t < T$ when the current in the large loop is increasing, will a current be induced in the small loop? Why or why not? If so, then will the induced current be clockwise or counter-clockwise?

f) If the resistance of the small loop is R, then find the magnitude of the current induced in the small loop. ❖

4. A rectangular metal loop has height H, length L, and resistance R. It sits a distance D from a long straight wire.



The current in the long straight wire varies in time according to

$$i(t) = i_o\left(1 - \frac{t}{T}\right) \qquad \text{for } t \le T$$
$$i(t) = 0 \qquad \text{for } t > T.$$

a) What is the magnetic flux, F_B, through the loop as a function of time?

b) What is the emf induced around the loop as a function of time?

c) What is the current induced in the loop? Which way does this current flow? ❖❖

M-5 Challenge Problem

Torque on a Rotating, Charged Cylinder

Consider a cylindrical shell of length ℓ and radius R (ℓ>>R). Let the cylinder have a total charge Q uniformly distributed on its surface. The cylinder is an insulator, so the charges are fixed in place. In this challenge problem, we will see the effect that the charge has on the moment of inertia of the cylinder.

a) The cylinder is rotating about its axis with angular velocity ω. What is the current per unit length on the cylinder due to the rotating charges?

b) What is the magnetic field at points inside the solenoid? (Hint: Our system is similar to a solenoid)

c) Consider an imaginary loop of radius R centered on the axis. What is the magnetic flux through this loop?

d) If we give the cylinder an angular acceleration α, what voltage will be induced around the loop?

e) What is the magnitude and direction of the *induced* electric field at the surface of the cylinder? (Hint: Remember the relationship between voltage drop and electric field)

f) What net torque, τ_F, does the cylinder experience due to the effects of Faraday's law? How will the direction of the torque compare to the direction of the angular acceleration?

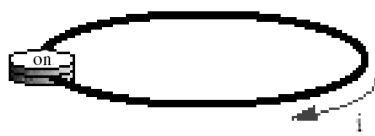
g) Let an external torque be put on the cylinder. Recall that $\tau_{net}=I\alpha$, where I is the moment of inertia (which, for a hollow cylinder, is mR^2). What is the effective moment of inertia for our cylinder, I_{eff} (defined as $\tau_E=I_{eff}\alpha$)? ❖❖

M-6. Inductance

Questions for discussion

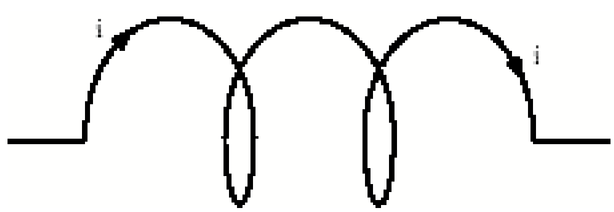
1. Consider a circular loop made of wire. The loop has a current generator, but initially the generator is turned off, so there is no current flowing in the loop.

Now suppose we gradually turn up the generator, until there is a current i flowing in the loop.



During the time when we are increasing the current, an *induced voltage* in the loop will tend to oppose the increase in current. Where does this "back-EMF" come from?

2. Consider the coil of wire shown below.



By means of a current generator, the current flowing through the coil is gradually increased at a rate of 3 mA/sec. As a result, a back-EMF of 12 V is induced in the coil.

a) What is the inductance of the coil?

Next, the current generator is set to increase the current flowing through the coil at the slower rate of 1 mA/sec.

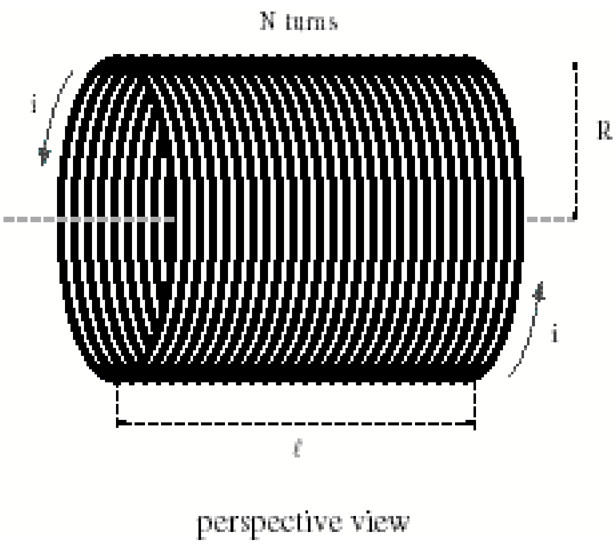
b) What is the inductance of the coil now?

c) What will be the back-EMF induced in the coil?

d) What are some ways in which you could change the inductance of this coil?

Problems

1. In Problem 3 of Worksheet M-4, *Ampère's Law*, you calculated the magnetic field created by a solenoid of length ℓ and N turns, when the solenoid carries current i.



(The solenoid's length is compressed in this diagram.)

a) If you can't remember how to do this, now would be a good time to try it again.

b) Knowing the expression for the solenoid's field, calculate the *flux* of the solenoid's field through itself, when the current flowing in the solenoid is i. (Hint: First calculate the flux through just one of the loops.) (Note that the coils in this solenoid have radius R.)

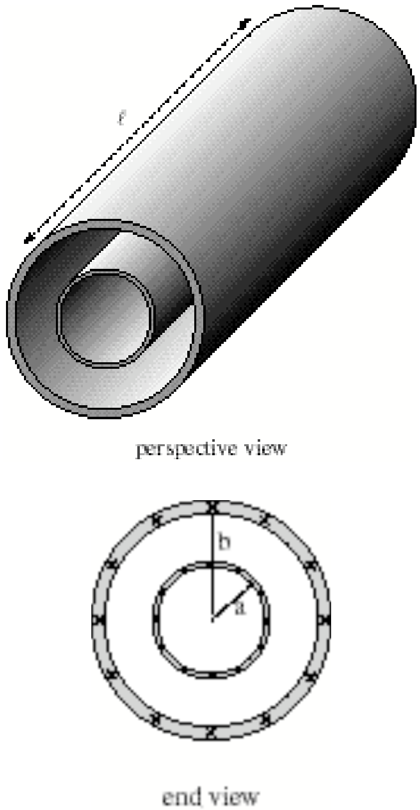
c) If the current through the solenoid changes, a back-EMF will be induced in the wires that make up the solenoid. How strong will this back-EMF be?

d) What is the *inductance* of the solenoid? Express this in terms of the number of turns per unit length, $n \equiv N/\ell$.

e) When a current i_0 flows through the solenoid, how much energy is stored in the magnetic field of the solenoid? Find this both using the formula for the energy stored in an inductor and by integrating the energy density of the magnetic field over the volume of the solenoid. Do the two results agree?

f) Suppose we have two solenoids A and B with the same length and cross-sectional area. Solenoid A has 400 turns per meter, and solenoid B has 700 turns per meter. Which solenoid has the greater inductance? Can you explain why? ❖

2. In a *coaxial cable*, concentric metal cylinders carry currents ±i.

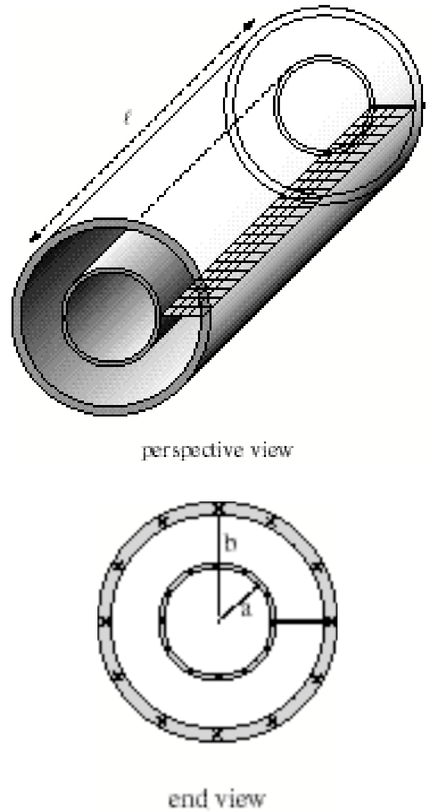


a) What is the magnetic field strength at points outside the large cylinder? ($r > b$)

b) What is the magnetic field strength at points inside the small cyclinder? ($r < a$)

c) What is the magnetic field strength at points between the cylinders? ($a < r < b$)

Consider an imaginary rectangular surface linking the two cylinders, oriented perpendicular to the magnetic field.



d) What is the flux of the cable's magnetic field through this surface? Remember: The magnetic field depends on the radius, so in calculating the flux you will have to integrate over the coordinate r.

e) Use this flux to calculate the inductance of a length ℓ of the coaxial cable. ❖

---

3. A long straight wire of length ℓ and radius R carries a current i distributed uniformly over its cross section.

a) How strong is the magnetic field at points inside the wire? ($r < R$)

b) How much energy is stored in each cubic centimeter of this magnetic field? In other words, what is the *energy density* associated with this magnetic field? ❖❖

M-6 Challenge Problem

Nested Solenoids and Mutual Inductance

Consider two solenoids, each of length ℓ, centered on the same axis and nested within eachother. The larger of the solenoids has radius r_1 and has n_1 turns per unit length. The smaller solenoid has radius $r_2<r_1$ and has n_2 turns per unit length. Ignore fringing effects in this problem.

a) What are the inductances, L_1 and L_2 of the two solenoids?

b) A current I_1 is put through solenoid 1. What is the magnetic flux through solenoid 2?

If we put a *changing* current I_1 through solenoid 1, then a voltage will be induced in solenoid. The relationship between the changing current and induced voltage is given by $\mathcal{E}_2 = M_{21}\frac{dI_1}{dt}$, where M_{21} is the *mutual inductance.*

c) What is the mutual inductance M_{21}?

d) Show that the mutual inductance M_{12} is equal to M_{21}.

Suppose the two solenoids are linked together such that the current through solenoid 1 equals the current through solenoid 2. Let the connection be such that if current is flowing clockwise through solenoid 1, then the current will be flowing counterclockwise through solenoid 2.

e) In the limiting case where the radii of the two solenoids are equal and $n_1=n_2$ what do you expect the total inductance of the system to be?

f) Find the magnetic field everywhere when the solenoids are linked together and a current I is flowing through them.

g) Find the magnetic flux through one loop of the inner solenoid. Do the same for one loop of the outer solenoid. How do these fluxes compare in the limiting case considered in part (e)? Remember that the relative signs of the two fluxes will be important!

h) What is the total magnetic flux through the nested solenoid system?

i) What is the self-inductance of the nested solenoid system?

j) Verify your result from part (i) by considering the following limiting cases:

- $n_1 = n_2$ and $r_1=r_2$.
- $r_1=r_2$
- $n_1=0$
- $n_2=0$

k) Using your results from parts (a), (c), and (i), find the total inductance in terms of the two self-inductances of the individual solenoids and the mutual inductance between the two solenoids. ❖❖

M-7. Displacement Current and Maxwell's Equations

Part 1: Displacement Current

$$I_D = \varepsilon_0 \frac{d}{dt} \iint_{surface} \vec{E} \cdot d\vec{A}$$

Questions for discussion (Part 1)

1. In the version of Ampere's Law that we have been working with up to now, we had to find the current enclosed by a loop. How did you determine whether a current was 'enclosed' or not? Try to relate the calculation of 'enclosed current' to the calculation of a flux through a surface.

2. In Faraday's Law, why did it not matter *which* surface you are calculating the flux through, as long as the surface is bounded by a given closed loop?

3. Consider a closed surface. For *steady state* configurations (which is what we were working with in magnetostatics), what is the net amount of current piercing the surface? (Consider current *entering* the surface to be negative and current *leaving* the surface to be positive.)

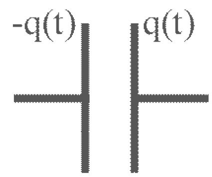
4. Why do we need to add another term to Ampere's Law to make it consistent?

5. How is the net displacement current entering a closed surface related to the net *actual*current entering that same surface?

6. What are the units of electric flux? Show that the displacement current indeed has the units of a current.

7. A single point charge q is located on the z-axis a little bit above the xy-plane. If the charge moves upwards with speed v, how much displacement current flows through the xy plane?

8. Two circular plates have equal but opposite charges q that vary with time.

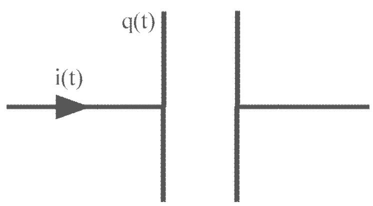


a) If the plates are *charging up*, draw the magnetic field created between the plates as a result of the changing electric field. (Think of the changing electric field as a current flowing in the direction of the change.)

b) Draw the magnetic field created between the plates when the plates are charging down.

Problems (Part 1)

1. Consider an infinitely long current-carrying wire connected to a circular place capacitor of radius R and separation d, with R>>d. Ignore all fringing effects in this problem and assume that the charge is uniformly distributed over the capacitor plate.



a) At a certain instant in time, the charge on the capacitor is q(t) and the current in the wire is i(t). Find the electric and magnetic fields everywhere, ignoring the effects of Faraday's Law and the displacement current.

b) At the instant in time considered above, what is the rate of change of the electric field between the capacitor plates?

c) Consider an Amperean loop of radius $r<R$ between the capacitor plates. What is the displacement current enclosed by this loop?

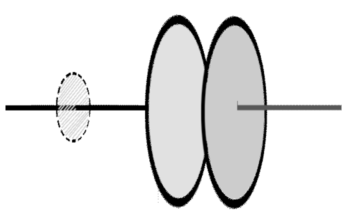
d) Find the displacement current density everywhere between the plates, $j_D(r,t)$.

e) What is the magnetic field created by this displacement current density?

f) Show that the total displacement current passing through a plane between the two capacitor plates is equal to the current entering the capacitor. ❖

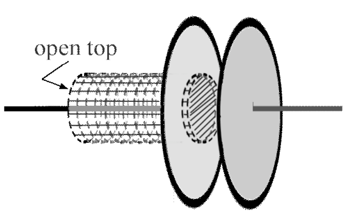
---

2. Consider again the wire and capacitor from problem 1. An Amperian loop of radius $r<R$ is placed around the wire. Consider two different surfaces that have the Amperean loop as their boundary. The first is just the disk that is in the plane of the loop, as shown below.



a) What current pierces this surface?

Now consider a different surface, which is the 'label' and 'end' part of a cylinder. The Amperean loop forms the boundary of the open 'lid' part, and the 'end' part is now between the capacitor plates, as shown below.



b) What *displacement* current pierces this surface?

c) Does the *wire* ever pierce the surface?

d) For Ampere's law to be consistent, we need the total enclosed current (regular and displacement) to be the same for both surfaces. What missing current is there in the second surface above?

e) The missing current is from current flowing radially out along the capacitor plate from the point of contact between the plate and the wire. Find the current density in the plate as a function of radius.

f) Show that the current entering the plate minus the current flowing radially out at a radius r is equal to the rate of change of charge on the portion of the capacitor plate with $r<R$.

Note that in an real system, since the capacitor plates are finite we won't be able to exploid symmetry as much. Namely, the charge *won't* be uniformly distributed over the plate and there will be fringing fields, all of which will complicate the analysis. ❖❖

Part 2: Maxwell's Equations

Gauss' Law $\oiint_S \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$

Gauss' Law (Magnetism) $\oiint_S \vec{B} \cdot d\vec{A} = 0$

Faraday's Law $\oint_C \vec{E} \cdot d\vec{s} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{A}$

Ampere's Law $\oint_C \vec{B} \cdot d\vec{s} = \mu_0 I_{\mathrm{enc}} + \mu_0 \varepsilon_0 \frac{d}{dt} \iint_S \vec{E} \cdot d\vec{A}$

Lorentz Force Law $\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$

Questions for Discussion and Problems for Part 2 will be available on the course website.

C-1. RC Circuits

Questions for discussion

1. Why does the current vary with time in an RC circuit[1]?

2. For the simplest type of RC circuit (consisting only of a battery $\boldsymbol{\varepsilon}$, a resistance R, and a capacitance C), the *time constant* is τ = RC. What is the significance of this time constant? In other words, what does the time constant for an RC circuit tell you?

3. Consider again the simple type of RC circuit discussed in question 2.

a) The capacitor is initially uncharged and a switch closes and completes the circuit at time t=0. What, if any, is the capacitor's effect on the circuit at the instant the switch is thrown? That is, does it act like a battery, an open switch, or a closed switch?

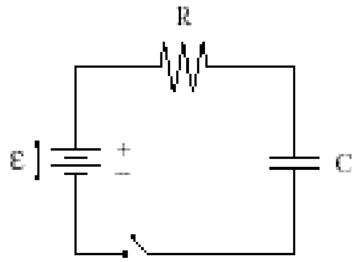
b) After a very long time (a few time constants, in practice), what, if any, is the capacitor's effect on the circuit? That is, does it act like a battery, an open switch, or a closed switch?

c) Now consider the case when the capacitor is initially charged to some charge q and determine the behavior of the capacitor at the instant the switch is thrown and a very long time after the switch is thrown.

[1] Some problems in RC circuits require solutions to differential equations. For the results you will need, and derivations of those results, see "Differential Equations for Circuit Problems" in the Supplementary Material at the end of the workbook, in the section "RC and LR Equations."

Problems

1. In the RC circuit shown below, the capacitor is initially uncharged, and the switch is initially open.



a) Immediately after the switch is closed, what is the voltage across the capacitor?

b) Immediately after the switch is closed, what current flows in the circuit? (Hint: You may want to re-draw an equivalent circuit, based on your answer from part (a). Then use the Loop Rule.)

c) After a very long time, what current flows in the circuit? Why?

d) After a very long time, what is the voltage across the capacitor? (Hint: Loop Rule, taking into account your answer from part (c).)

e) After a long time, how much charge has accumulated on the capacitor plates?

f) Sketch a graph showing the charge on the capacitor plates as a function of time. Let t=0 be the instant the switch was closed.

g) Sketch a graph showing the current in the circuit as a function of time. Again, let t=0 be the instant the switch was closed.

h) How are these two graphs related? Can you explain why the graphs are related in this way? ❖

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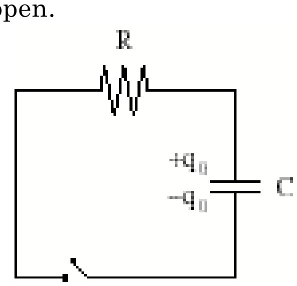
2. In the RC circuit shown below, the capacitor carries an initial charge q_0. The switch is initially open.



a) What will happen when the switch is closed?

b) Immediately after the switch is closed, how much current flows in the circuit? (Hint: Loop Rule.)

c) After a long time, how much current flows in the circuit? Why?

d) After a long time, what is the voltage across the capacitor? How about the charge on the capacitor?

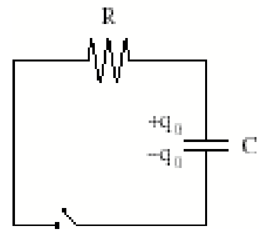
e) Sketch graphs showing the charge on the capacitor and the current flowing in the circuit as functions of time.

f) How much energy was stored in the electric field of the capacitor initially?

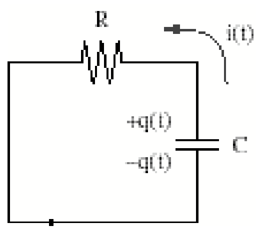
g) How much energy is stored in the electric field of the capacitor after a long time?

h) What happened to this energy? ❖

3. Consider once again the RC circuit from Problem 2. Initially the capacitor carries a charge q_0, and the switch is open.



When the switch is closed at t=0, charge will begin to leak off of the capacitor plates, resulting in current flow around the circuit. At some arbitrary time t, the circuit therefore looks like this:



a) Write down the Loop Rule for this circuit.

b) What is the relationship between the current i(t) and the charge q(t) contained on the plates? Use this to express your Loop Rule entirely in terms of q, dq/dt, and constants.

c) Verify that the function $q(t) = q_0 e^{-t/RC}$ satisfies the Loop Rule at all times. Hence, this function gives the charge on the capacitor at any given time after the switch is closed.

d) Sketch a graph of the charge on the capacitor as a function of time.

e) From the expression $q(t) = q_0 e^{-t/RC}$, find an expression for i(t), the current flowing in the circuit at any time after the switch is closed. Sketch a graph of this function.

f) Verify that these graphs agree with the ones you drew for part (e) of Problem 2.

g) From your expression for i(t), find an expression for P(t), the rate of heat loss through the resistor at any given time after the switch is closed.

h) By integrating P(t) over time, from t=0 to t=∞, show that the total amount of heat dissipated by the resistor is none other than the initial energy stored in the capacitor, as required by energy conservation. (This justifies your answer to part (h) of Problem 2.) ❖

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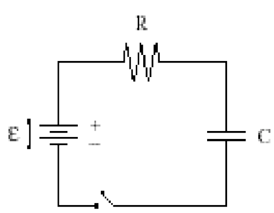
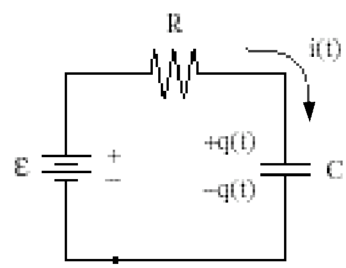
4. Consider once again the RC circuit from Problem 1. Initially the capacitor uncharged and the switch is open.



When the switch is closed at t=0, charge will begin to accumulate on the capacitor plates, resulting in current flow around the circuit. At some arbitrary time t, the circuit therefore looks like this:



a) Write down the Loop Rule for this circuit.

b) What is the relationship between the current i(t) and the charge q(t) contained on the plates? Use this to express your Loop Rule entirely in terms of q, dq/dt, and constants.

c) Plug the fuction $q(t) = \alpha\left(1 - e^{-\beta t}\right) + \gamma e^{-\beta t}$ into your differential equation and solve for the constants α and β.

d) Use the fact that q=0 when t=0 to solve for γ.

e) Sketch a graph of the charge on the capacitor as a function of time. ❖

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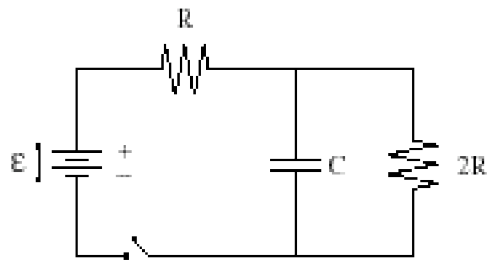
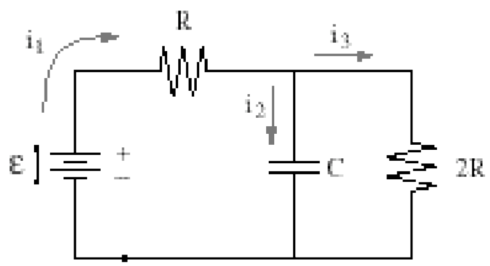
5. In the RC circuit shown below, the capacitor is initially uncharged, and the switch is open.



Then, at t=0, the switch is closed:



a) Immediately after the switch is closed, what are the currents i_1, i_2, and i_3?

b) After a long time, what are the currents i_1, i_2, and i_3?

c) After a long time, how much charge is on the capacitor plates?

d) Write down two loop rules and one junction rule so that we have three equations with three unknowns.

e) Using the relation $i_2 = dq/dt$, where q is the charge on the capacitor, combine your equations from part (d) to get a first order differential equation for q.

f) Refering to the general solution to that differential equation, solve for q(t) and fine $i_2(t)$. Does your answer for $i_2(t)$ match with your expectations from parts (a) and (b)? ❖❖

C-2. LR Circuits

Questions for discussion

1. Why does the current vary with time in an LR circuit[1]?

2. For the simplest type of LR circuit (consisting only of a battery $\mathcal{E}$, a resistance R, and an inductance L), the *time constant* is $\tau = L/R$. What is the significance of this time constant? In other words, what does the time constant for an LR circuit tell you?

3. Consider again the simple type of LR circuit discussed in question 2.

a) The inductor initially has no current flowing through it, and a switch closes and completes the circuit at time t=0. What, if any, is the inductor's effect on the circuit at the instant the switch is thrown? That is, does it act like a battery, an open switch, or a closed switch?

b) After a very long time (a few time constants, in practice), what, if any, is the inductor's effect on the circuit? That is, does it act like a battery, an open switch, or a closed switch?

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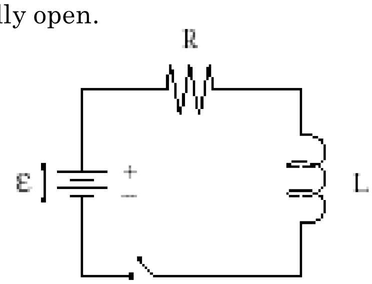
[1] Some problems in LR circuits require solutions to differential equations. For the results you will need, and derivations of those results, see "Differential Equations for Circuit Problems" in the Supplementary Material at the end of the workbook, in the section "RC and LR Equations."

Problems

1. In the LR circuit shown below, the switch is initially open.



a) Immediately after the switch is closed, what current flows in the circuit? Why?

b) After a very long time, what current flows in the circuit? Why?

c) Sketch a graph showing the current in the circuit as a function of time. Let t=0 be the instant the switch was closed.

d) Sketch a graph showing the energy stored in the magnetic field of the inductor as a function of time. Again, let t=0 be the instant the switch was closed. ❖

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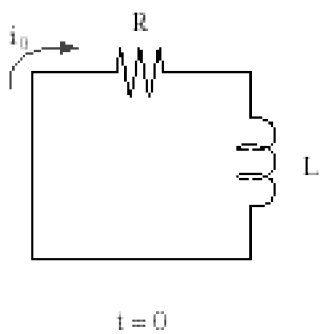
2. Consider the LR circuit shown below. At the particular instant shown (which we can call t=0), the current flowing in the circuit is i_0.



a) After a long time, how much current flows in the circuit? Why?

b) How much energy was stored in the magnetic field of the inductor initially?

c) How much energy is stored in the magnetic field of the inductor after a long time?

d) What happened to this energy? ❖

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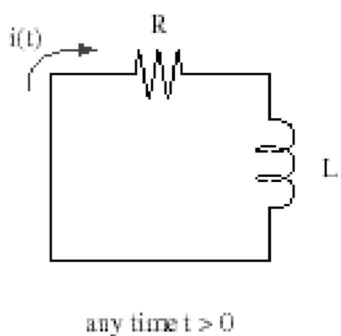
3. Consider once again the LR circuit from Problem 2. At t=0, the current flowing in the circuit is i_0. At some later time t, the current in the circuit will have some new value i(t):



a) Write down the Loop Rule for this circuit at the time t shown above.

b) Verify that the function $i(t) = i_0 e^{-t/RC}$ satisfies the Loop Rule at all times. Hence, this function gives the current in the circuit at any given time after the switch is closed.

c) Sketch a graph of the current in the circuit as a function of time.

d) From the expression for i(t), find an expression for P(t), the rate of heat loss through the resistor at any given time after the switch is closed.

e) By integrating P(t) over time, from t=0 to t=∞, show that the total amount of heat dissipated by the resistor is none other than the initial energy stored in the inductor, as required by energy conservation. (This justifies your answer to part (d) of Problem 2.) ❖

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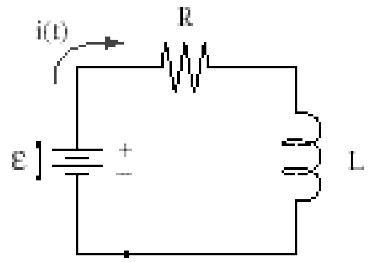
4. Consider once again the LR circuit from Problem 1. Initially the switch is open. At time t=0 the switch closes. At some arbitrary time t, the circuit looks like this:



a) Write down the Loop Rule for this circuit.

b) Plug the fuction $i(t) = \alpha\left(1 - e^{-\beta t}\right) + \gamma e^{-\beta t}$ into the loop rule and solve for the constants α and β.

d) Use the fact that i=0 when t=0 to solve for γ.

e) Sketch a graph of the current on the capacitor as a function of time. ❖

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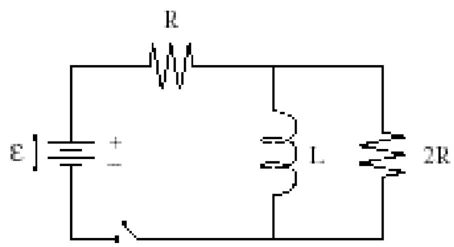
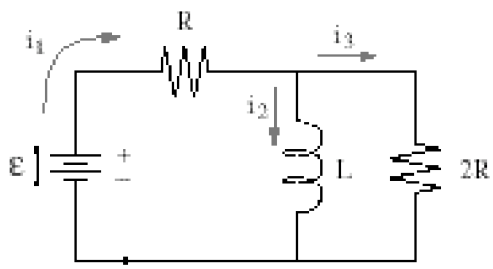
5. In the LR circuit shown below, the switch is open and no currents are flowing.



Then, at t=0, the switch is closed:



a) Immediately after the switch is closed, what are the currents i_1, i_2, and i_3?

b) After a long time, what are the currents i_1, i_2, and i_3?

c) Write down two loop rules and one junction rule and rearrange so we get a differential equation for the current i_2.

d) Solve explicitly for $i_2(t)$.

When the currents have finally reached the values calculated in part (b), the switch is again opened.

e) Find the currents i_1, i_2, and i_3 immediately afterwards.

f) Find the currents i_1, i_2, and i_3 a very long time later.

g) Again solve explicitly for $i_2(t)$. You can reset the time to t=0 when the switch is opened. ❖❖

C-3. LRC and AC Circuits

Questions for discussion

1. In an LC circuit[1], why does the current oscillate?

2. In an LC circuit, the inductor stores energy in its magnetic field. Likewise, the capacitor stores energy in its electric field. How do these stored energies vary with time as the circuit oscillates? How does their *sum* vary with time?

3. Adding a small resistor to an LC circuit will not change the fact that the current oscillates back and forth. But the resistor *will* have an important effect on things - what is it?

4. How do you expect the sum of the stored energies in the capacitor and the inductor to behave in an LRC circuit? Does this violate conservation of energy?

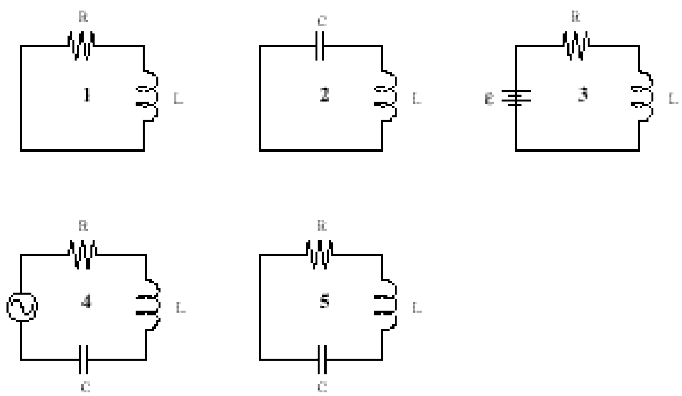
5. Consider a simple AC circuit containing a resistor, a capacitor, an inductor, and a sinusoidal voltage source, all in series.

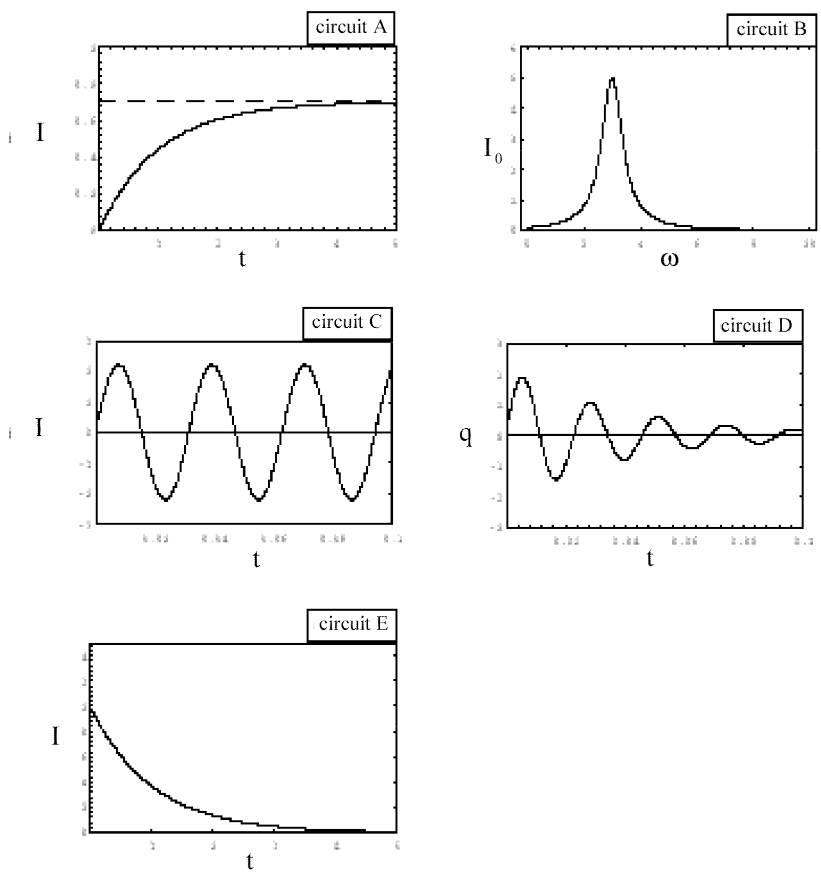
a) Why does the inductor cause the amplitude of the current going through this circuit to be small when the AC frequency is very high?

[1] Some problems in LC and LRC circuits require solutions to differential equations. For the results you will need, and derivations of those results, see "Differential Equations for Circuit Problems" in the Supplementary Material at the end of the workbook, in the section "LRC Equations."

b) Why does the capacitor cause the amplitude of the current going through this circuit to be small when the AC frequency is very low?

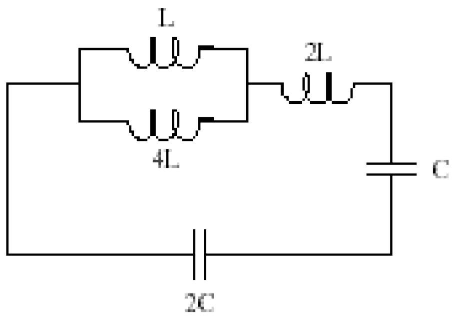
6. Match the circuits shown below to their corresponding graphs. (Each circuit corresponds to one and only one graph.)





Problems

1. The figure below shows a circuit with several inductors and capacitors, with both series and parallel combinations.



What is the natural oscillation frequency of this circuit? ❖

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2. In an LC circuit with inductance L and capacitance C, the charge on the capacitor is initially q_0 and the current is initially i_0.

a) How much energy is initially in the capacitor?

b) How much energy is initially in the inductor?

c) What is the total energy initially contained in the circuit?

d) What is the peak value of the current as it oscillates?

e) What is the peak value of the charge on the capacitor as it oscillates?

f) What is the natural oscillation frequency of the circuit?

g) Sketch the charge on the capacitor and the current in the circuit as functions of time. ❖

---

3. Consider again the LC circuit from problem 2.

a) Write down the loop rule for this circuit.

b) Use the relation between the charge on the capacitor and the current in the circuit to rewrite the loop rule as a second order differential equation in the variable q.

c) Plug the trial solution $q(t) = \alpha\cos(\omega t + \varphi)$ into your result from part (b) and solve for the natural frequency of the oscillation, ω.

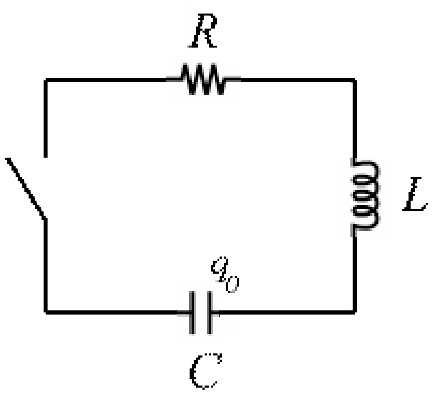
d) Use the initial conditions to solve for the constants α and φ.

e) What is the energy stored in the capacitor as a function of time?

f) What is the energy stored in the inductor as a function of time?

g) Verify that the total energy stored in the circuit is constant. ❖

4. Consider a series LRC circuit, with an inductor L with no current initially flowing through it, a capacitor C which is initially charged to charge q_0, and a resistor R that is small enough so that the circuit is *underdamped*.



a) At the instant the switch is closed, what current flows through the circuit?

b) At some time t after the switch is closed, the charge on the capacitor is q(t) and the current going through the circuit is i(t). Write the loop rule for this circuit. Rewrite this as a second order differential equation for q(t).

c) Plug in the trial solution $q(t) = \alpha e^{-t/\tau}\cos(\omega t + \varphi)$ into the loop rule. (Since e^x is never 0, we can cancel out the common exponential in all of the terms.)

d) Rewrite your answer from part (c) so that it has the form: $A\cos(\omega t + \varphi) + B\sin(\omega t + \varphi) = 0$. Since this must hold for all times, the coefficients A and B must be zero.

e) Set the coefficient of the sin term equal to zero and solve for τ. This is the *decay time* for the circuit.

f) Set the coefficient of the cos term equal to zero, plug in your result for τ, and solve for ω. This is the *natural frequency* of the circuit.

g) Plug in the initial conditions to finish solving for q(t).

h) Show that, in the limit as the resistance of the circuit goes to zero, the solution reduces to the solution for an LC circuit.

i) What is the maximum value of resistance that allows underdamped behavior?

j) What is the *quality factor*, Q (defined as 2π times the number of cycles needed for the energy stored in the circuit to decay be a factor of 1/e) of the circuit?

k) Show that when the resistance is such that the circuit is critically damped, the quality factor reduces to 0. ❖❖

C-4. AC Circuits - Impedance

$$V_0 \sin(\omega t) = Z I_0 \sin(\omega t - \varphi)$$

Questions for discussion

1. An AC source of frequency ω is placed across a resistor with resistance R. Use the loop rule to find the impedance, Z_R, and phase angle, φ_R of a resistor.

2. An AC source of frequency ω is placed across a capacitor with capacitance C. Use the loop rule to write a differential equation for the charge on the capacitor and solve it. Use your solution to find the impedance, Z_C, and phase angle, φ_C of a capacitor. (Hint: $\cos(\theta) = \sin(\theta + \frac{\pi}{2})$)

3. Qualitatively,why does the impedance of a capacitor increase when the driving frequency of the source is decreased?

4. An AC source of frequency ω is placed across an inductor with inductance L. Use the loop rule to write a differential equation for the current through the inductor and solve it. Use your solution to find the impedance, Z_L, and phase angle, φ_L of an inductor. (Hint: $-\cos(\theta) = \sin(\theta - \frac{\pi}{2})$)

5. Qualitatively,why does the impedance of an inductor increase when the driving frequency of the source is increased?

6. In a series LRC circuit with an AC source of frequency ω the impedance is given by $Z = \sqrt{Z_R^2 + (Z_L - Z_C)^2}$. What is the *resonant frequency* ω_0? That is, with what frequency should the AC source drive the circuit such that the amplitude of the current is maximal?

7. Sketch a graph showing the amplitude of the current as a function of the driving frequency in a series LRC circuit.

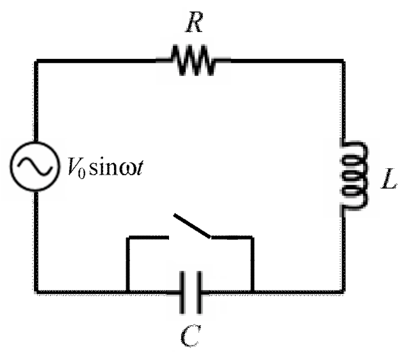
8. The time-evolution of a voltage or a current in an AC circuit is given by two numbers: the *amplitude* of the oscillation (a *magnitude*) and the argument of the sine function (an *angle*). Given this, explain why phasors (and/or complex numbers) are useful in analyzing AC circuits.

9. (If you are using *phasors* to analyze circuits) Draw a phasor diagram of a series LRC circuit with an AC source (note that all three circuit elements share the same current phasor). Vectorially add the three voltage phasors and show that the impedance of the circuit is, indeed, $Z = \sqrt{Z_R^2 + \left(Z_L - Z_C\right)^2}$ and that the phase angle is given by $\tan\varphi = \dfrac{Z_L - Z_C}{Z_R}$.

10. (If you are using *complex numbers* to analyze circuits) Use the loop rule and complex notation to get an equation relating the voltage source and the current. Find the amplitude of the current ($|I|^2 = I^*I$) and show that $Z = \sqrt{Z_R^2 + \left(Z_L - Z_C\right)^2}$. Also, show that the phase angle is given by $\tan\varphi = \dfrac{Z_L - Z_C}{Z_R}$ using the fact that, for a complex number a+b*i*, the phase is given by $\tan^{-1}(b/a)$.

Problems

1. Consider the series LRC circuit shown below, with L = 100 mH, C = 1000 μF, and R = 50 Ω. The AC source produces a voltage $V(t) = V_0 \sin(\omega t)$, with V_0 = 200 V and ω = 90 hertz. A switch is placed across the capacitor, but left open for this problem.



a) What is the total impedance of this circuit?

b) Find the amplitude and phase angle of the current coming out of the AC source.

c) What is the maximum voltage drop across each circuit element, V_{R0}, V_{L0}, and V_{C0}?

d) What is the earliest time, t, for which q(t), the charge on the capacitor, is zero?

e) What is the resonant frequency of this circuit?

f) If we tune the driving frequency to resonance, what will be the new current and phase angle for the circuit?

g) What average power is dissipated by the resistor?

h) What average power is delivered to the circuit? ❖

---

2. Consider again the circuit from problem 1, but suppose now that the switch is closed.

a) What is the new impedance of this circuit?

b) What is the new current flowing out of the voltage source?

c) What is the voltage drop across the inductor as a function of time?

d) How much time does it take for the inductor to go from having no current flowing thorough it to having the maximum amound of current flowing through it? ❖

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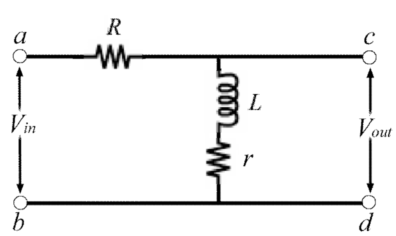
3. A high-pass filter is a circuit that lets high-frequency signals pass with ease but virtually eliminates low-frequency signals. Consider the RL high-pass filter shown below.



a) If a voltage $V_{in} = V_0 \sin\omega t$ is put across the gap *ab*, what current will run from *a* through the two resistors and inductor to *b*?

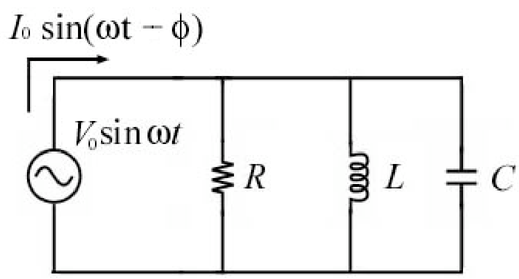
b) What voltage will be read across leads *cd* (V_{out})?

c) Find the ratio of the amplitudes of the two voltages $V_{0,out}/V_{0,in}$.

d) At what frequency will the strength of the outgoing signal be only half of the incoming signal?

e) Replace the inductor with a capacitor C and repeat parts (a) through (d). What type of filter is this? ❖

4. Consider a *parallel* LRC circuit driven by an AC source.



a) What is the impedance of this circuit?

b) What is the current coming out of the AC source as a function of time?

c) What is the maximum current going through each circuit element, I_{R0}, I_{L0}, and I_{C0}?

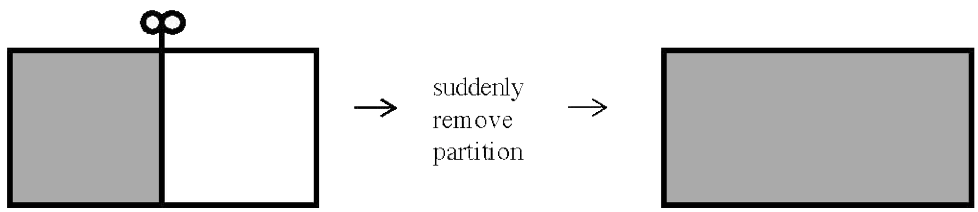
d) What is the voltage across the source when the current through the source is a maximum?

e) What is the current through the source when the voltage across the source is a maximum?

❖❖

Free Expansion of an Ideal Gas

A "free expansion" occurs when a gas is allowed to expand freely into a vacuum. In the usual setup, there is an insulated box with a divider in the middle.



Initially there is some ideal gas is on the left with volume V_i, pressure p_i, and temperature T_i. The other half of the box is empty.

But then someone rapidly removes the divider-- or perhaps someone quickly pokes a little hole in the divider. However the problem may be phrased, the point of a free expansion is that the other half of the box is suddenly made available to the gas.

Many students insist that because the gas is expanding to fill the other half of the box, it should cool off. But in a free expansion of an ideal gas, the temperature remains the same! How can we understand this?

In one common answer, we point out that since the box is insulated, Q = 0 for the transformation (no heat is added or taken away) and since there are no walls for the expanding gas to push against, the gas does no work, so W = 0. Hence from the First Law, $\Delta E_{int} = Q - W = 0$. And if the internal energy doesn't change, then from $E_{int} = (3/2)NkT$, we see that the temperature doesn't change either.[1]

This explanation works fine, but there is a more insightful answer. To understand why the temperature remains constant in a free expansion, we should remember what temperature really *means*. As Boltzmann showed, temperature really has to do with the amount of kinetic energy carried by a typical particle. This is the intuitive meaning of Boltzmann's formula $\langle KE \rangle_{particle} = (3/2)kT$.

With this in mind, let's pretend that you are trying to argue with Ludwig Boltzmann about free expansions. You believe that the gas *cools* when it undergoes a free expansion, and Ludwig is trying to convince you that the temperature remains *constant* in a free expansion.

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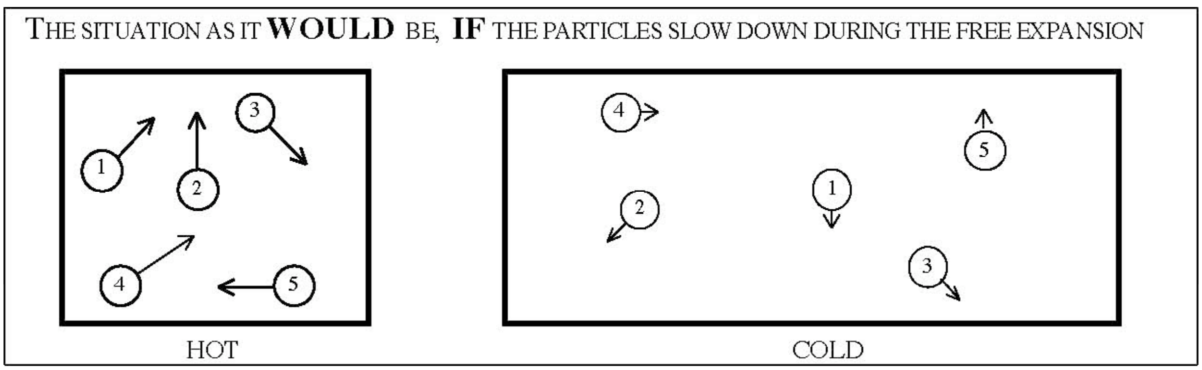
[1] More generally, recall that the internal energy of an ideal gas is given by E = (d/2)NkT. Here d represents the number of degrees of freedom of the ideal gas particles: d = 3 for monatomic particles, d = 5 for diatomic particles at intermediate temperatures, etc. For definiteness, we will suppose that the particles in this discussion are *monatomic*.

YOU: With all due respect Herr Boltzmann, the gas is expanding, right? So it seems like it should cool off.

LB: Well, if you say so, then let us agree that the gas cools off! But as I have discovered, when we say “temperature,” we *really* mean something like “the kinetic energy of a typical particle.” So if we are going to propose that the gas cools off during the free expansion, then you must convince me that the kinetic energy of a typical particle decreases. Or, in other words, you must convince me that a typical particle slows down when the gas freely expands.

YOU: Fine, so the particles slow down. You say it your way, I’ll say it mine!

LB: But let’s take a closer look at this. According to you, the situation is like so:



Before the partition is removed, the particles are moving quickly. After the partition is removed, the particles have slowed down. Is this what you are saying?

YOU: Yep.

LB: Well, look at the particle labeled with the number 1. It’s moving slowly now. It used to be moving quickly. What do you suppose could have happened to it in order to slow it down?

YOU: Ummm.....

LB: Particle 1 was way over by the left wall of the box when I removed the partition ... I never touched particle 1 ... How can it know that I removed the partition?

YOU: Well, lots of other particles were close to the partition when you removed it. It’s not really fair to pick particle 1 just because it helps your case!

LB: But if I remove the partition fast enough (which is what we're assuming), then practically *all* the particles are in the same situation as particle 1. I never touched any of them. What can possibly have happened to them to slow them down?

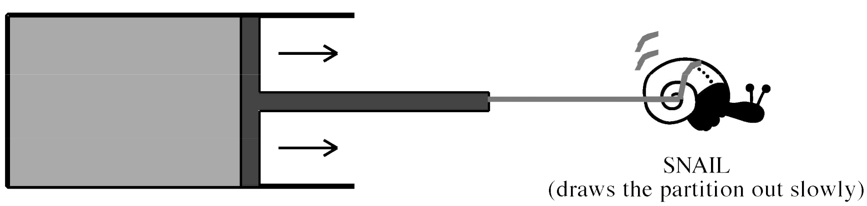
YOU: I see what you mean, it doesn't seem like anything could slow them down. But don't they slow down anyway, just because they have more room?

LB: It's like this. I'm driving my car in Iowa, listening to the news on the radio. The reporter announces that the US has just annexed Canada. This doesn't magically cause my car to slow down, does it?

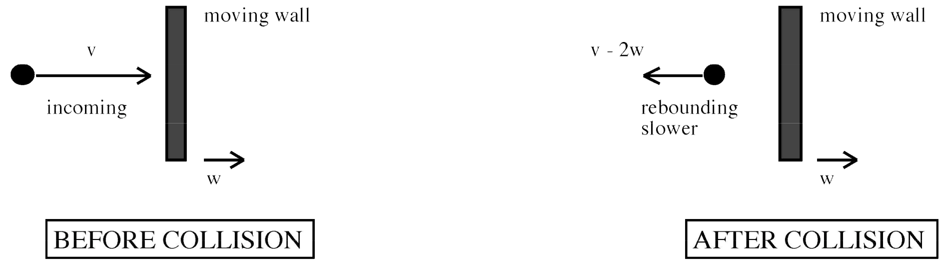
YOU: I guess not. But then isn't there a problem with the ideal gas law? After all, the volume has doubled...

LB: Correct, so the pressure must drop. Using the ideal gas law, piVi = NkTi becomes $p_f V_f = NkT_f$. And since $T_f = T_i$ (as I hope you agree now), we must have $p_i V_i = p_f V_f$. So if the volume doubles, the pressure must decrease by half.

YOU: OK, I see how that works. But now I'm confused about something else. What difference does it make that in a free expansion we remove the partition rapidly? According to your arguments, it seems like the gas should *never* cool in *any* expansion, even one in which we slide the divider *slowly* to the right.



LB: That would certainly be a disaster! Because we know that in slow expansions like this, the gas does cool off - at least, provided we don't add any heat to the gas. And I can explain why. According to my own rules, if I am claiming that the gas cools off, then I must tell you how it happens that the particles slow down. And here it is: If a particle collides with a wall that is moving away from it, then the particle will rebound from the wall *slower* than it struck the wall. This is a simple fact about elastic collisions:



And when you slide the divider *slowly* to the right, you give all the particles time to collide with the moving wall. So you allow all the particles to lose some speed when they rebound. So you cool the gas.

YOU: Gee Ludwig, I guess I never looked at things close up like that. I really learned something today!

Entropy of the Ideal Gas

In this supplement, we will show that the change in entropy between any two states of an ideal gas is given by the following formula:

$$\Delta S = \frac{d}{2} N k_B \ln\frac{T_2}{T_1} + N k_B \ln\frac{V_2}{V_1}$$

The fact that entropy is a *state variable* allows us to define the entropy difference between any two states of an ideal gas. The only way that we know of to compute entropy in this class is using $\Delta S = \int \frac{dQ}{T}$. However, this formula only holds for *reversible* processes. Since S is a state variable, the entropy change does not depend on the path connecting two points, which allows us to draw in any path we like to find the entropy change, and we will be ensured that the formula will hold for any process, reversible or not. The path we will work with is going to be an *isobar* that takes the gas reversibly from the volume V_1 to V_2 at the constant pressure P_1, followed by an *isochor* that takes the gas reversibly from the pressure P_1 to P_2 at the constant volume V_2. The problem will be to find the entropy change between (P_1, V_1) and (P_2, V_2).

Isobar

In order to use the entropy formula above, we need to express the integrand in terms of a single variable, since both T and Q will, in general, vary along the path. We are at a constant pressure, and going from V_1 to V_2, so V seems like it should be an appropriate variable. First, use the ideal gas law to reexpress T in terms of V : $T = PV/Nk_B$. Next, we re-express dQ. The first law says that $\Delta U = W - Q$. The work we can express as $W = P_1\Delta V$. The equipartion theory tells us $\Delta U = \frac{d}{2} Nk_B\Delta T$. Substituting in the ideal gas law gives $\Delta U = \frac{d}{2}(P_fV_f - P_iV_i)$, so $\Delta U = \frac{d}{2}P_1\Delta V$ and $Q = \left(\frac{d}{2}+1\right) P_1\Delta V$. To change this into a differential, we just replace 'Δ' with 'd', giving

$$\begin{aligned}
\Delta S_{\text{isobar}} &= \int_{V_1}^{V_2} \frac{\left(\frac{d}{2}+1\right)P_1 dV}{\frac{P_1 V}{Nk_B}} \\
&= \left(\tfrac{d}{2}+1\right)Nk_B \int_{V_1}^{V_2}\frac{dV}{V} \\
&= \left(\tfrac{d}{2}+1\right)Nk_B \ln\frac{V_2}{V_1}
\end{aligned}$$

Isochor

The isochor follows in exactly the same was as the isobar, with the Ps and Vs switching places in every place except for the work, which is just 0. Thus, we can immediately jump to the result

$$\Delta S_{\text{isochor}} = \frac{d}{2} N k_B \ln \frac{P_2}{P_1}$$

Result

Adding the two results together gives

$$\Delta S = \left(\frac{d}{2} + 1\right) N k_B \ln \frac{V_2}{V_1} + \frac{d}{2} N k_B \ln \frac{P_2}{P_1}$$

We can simplify this a little bit. First, separate the d/2 part in the first term and collect common coefficients:

$$\Delta S = \frac{d}{2} N k_B \left(\ln \frac{V_2}{V_1} + \ln \frac{P_2}{P_1}\right) + N k_B \ln \frac{V_2}{V_1}$$

The sum of two logarithms is just the logarithm of the products:

$$\Delta S = \frac{d}{2} N k_B \ln \frac{V_2 P_2}{V_1 P_1} + N k_B \ln \frac{V_2}{V_1}$$

Finally, by the ideal gas law, $\frac{V_2 P_2}{V_1 P_1} = \frac{T_2}{T_1}$, giving the promised result!

$$\boxed{\Delta S = \frac{d}{2} N k_B \ln \frac{T_2}{T_1} + N k_B \ln \frac{V_2}{V_1}}$$

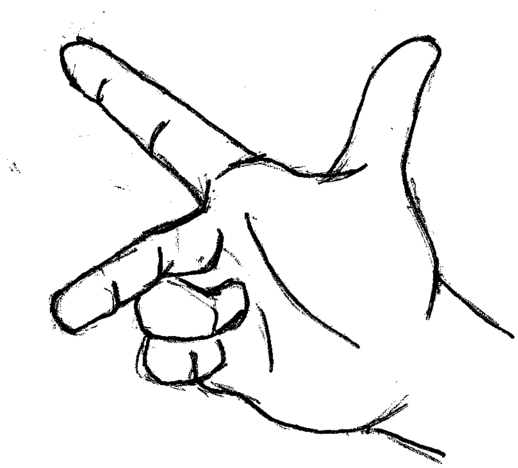
Vectors and Right Hand Rules in Magnetism

Drawing 3D Vectors on a 2D Page

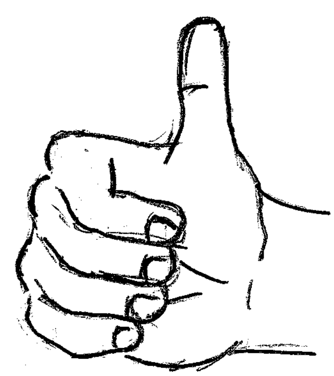
When working with magnetism, we will often have to deal with different vectors pointing in all three directions in space, since we are using the cross product extensively. Since not everyone is an artist and perspective views tend to mask, rather than illuminate, the important features of a diagram, there is a convention for expressing vectors that leave the plane of the page. Vectors lying within the plane of the page are drawn normally. For a vector that is to point *out* of the plane of the page towards the viewer, we use a circle with a dot in the center (as if you were looking head on into an arrow). For vectors going *into* the page, the vector is represented as a circle with an X through it (as though you were looking head on at the feathered tail of an arrow.

"out of page" "into page"

Configurations of the Right Hand



Configuration 1



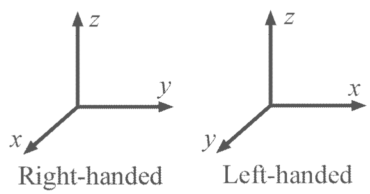
Configuration 2

In configuration 1, the vectors will point in the thumb, index finger, and middle finger directions. To simplify writing, we will refer to the triplet of quantities (a,b,c) to mean "a points in the direction of the thumb, b points in the direction of the index finger, and c points in the direction of the middle finger."

In configuration 2, the vectors or currents will point in the direction of the thumb or curl around in the direction of the fingers.

Right Handed Coordinate Systems

In magnetism, we will want to use *right-handed coordinate systems*. A right-handed coordinate system is one in which the coordinates have a definite order. For cartesian coordinates the order is (x,y,z) while for cylindrical the order is (r, θ, z). The ordering is such that, in configuration 1 above, the first coordinate is the thumb, the second is the index finger, and the third is the middle finger. If we switch the order of any two coordinates, we are left with a left-handed system. If we cyclically permute the coordinates (for instance, to (y,z,x) or (z,x,y)), then the system is still right-handed.



Applications

The Cross Product

Use configuration 1. If **A** × **B** = **C**, then the direction of **C** is determines from the directions of **A** and **B** by using the tripled (**A**, **B**, **C**) in configuration 1.

Force on a Moving Charge or Current

Use configuration 1. The force on a charge in a magnetic field is **F**=q(**v** × **B**). For a *positively* charged particle, then the triplet used in configuration 1 will be (v, B, F). For a *negatively* charged particle, then the triplet used in configuration 1 will be (v, B, -F). If we are looking at the force on a current, then we use the tripled (direction of current, B, F).

Magnetic Field from a Straight Wire

Use configuration 2. Your thumb is the direction of the current and your curled fingers point in the direction of the magnetic field circling the current.

Magnetic Field from a Loop of Wire

Use configuration 2. Your curled fingers will curl in the direction of the current and your thumb will point in the direction of the magnetic field at the center of the current.

Area Vector for an Open Surface

An open surface has a loop of some sort as its boundary. Give this boundary a direction (clockwise or counterclockwise). Use configuration 2. Your fingers curl around in the direction of this orientation, and your thumb will point in the direction of the area vector.

Direction of Induced Current

Use configuration 2. Your thumb points in the direction of *changing* magnetic field and your fingers curl in the direction that the induced current will follow.

Differential Equations for Circuit Problems

When dealing with circuits with capacitors, inductors, or AC sources, differential equations occur when applying the loop rules. There are a few simple cases that pop up frequently. In this supplement, three common differential equations are presented, with solutions.

LR and RC Equations

In LR or RC circuits, the differential equations are first-order in derivatives. The variable we are solving for is going to be the charge on the capacitor or the current through the inductor. The general form of the equations will be:

$$\frac{dy}{dt} = -Ay + B$$

Here, y is the variable we are solving for (either q(t) or i(t)) and A and B are constants that depend on the circuit that we are considering. The general solution to this equation is:

$$y(t) = \frac{B}{A}\left(1 - e^{-At}\right) + y_0 e^{-At}$$

y_0 is the *initial condition*: the value the variable y at time t=0. 1/A is known as the *time constant* - it gives the time it takes for the value of y to decay by a factor of e.

LRC Equations

In LRC circuits, the differential equations are second-order in derivatives, which means that we will need two initial conditions (typically the initial values of the charge and current, or two values of the charge or current at different times). The variable that we are solving for will most commonly be the charge on the capacitor. The general form of the equations will be:

$$L\frac{d^2y}{dt^2} + R\frac{dy}{dt} + \frac{y}{C} = 0$$

This is the same type of equation that shows up when analyzing the damped harmonic oscillator, with A acting as the mass (it will be an inductance in our case), B acting as the damping term (the resistance) and C^{-1} acting as the spring constant (the capacitance). Note that L, R, and C need not be actual inductors, resistors, or capacitors - they are just suggestive constants in the above differential equation (like A and B were in the LR and RC case). Recall that for the damped harmonic oscillator, there were three separate types of motion: underdamped, overdamped, and critically damped. For LRC circuits, we will typically only be concerned with the underdamped case.

For an underdamped system, we require that $R < 2\sqrt{\frac{L}{C}}$. If this is the case, then the general solution is

$$y(t) = \alpha e^{-\gamma t}\cos(\omega' t + \beta)$$

Here, α and β are the two constants that will be determined by the initial conditions. The main property of circuits with both capacitors and inductors is oscillation. For a pure LC circuit, we define the *undamped frequency of oscillation*:

$$\omega_0 = \sqrt{\frac{1}{LC}}$$

The actual frequency of oscillation when we have a damping term is:

$$\omega' = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} = \sqrt{\omega_0^2 - \gamma^2}$$

Finally, the decay constant (the inverse of the time constant) is given by

$$\gamma = \frac{R}{2L}.$$

AC Equations

In AC circuits, we have a sinusoidally varying voltage source, so we are considering *driven* oscillators. The general equation that will pop up when we analyze AC circuits is

$$L\frac{d^2y}{dt^2} + R\frac{dy}{dt} + \frac{y}{C} = V_0\sin(\omega t)$$

This is the *inhomogenous* version of the simple LRC equation, and the general solution will have two parts: A *transient part*, which will die out after a few time constants, and a *steady state* part. The transient part will just be the solution to the homogenous equation: that is, a solution when the right hand side is 0. The steady state part of the solution is easy, since there will be no decay (it's steady state after all) and no initial conditions (those were taken care of in the transient part of the solution). We typically don't care about the transient part when analyzing AC circuits, so what we will need for the above differential equation is:

$$y(t) = y_{\text{transient}}(t) + y_{\text{steady state}}(t)$$
$$y_{\text{steady state}}(t) = \frac{-A}{\omega}\cos(\omega t + \varphi)$$

The solution is written this way so that the derivative of y, which is the current, has a very simple form. Note that in the differential equation and the steady state solution the frequency ω is the *driving frequency* which is completely independent of the frequencies ω_0 and ω' from the LRC circuit.

A is known as the *amplitude* of the solution and is usually written as $A = V_0/Z$, where Z is known as the *impedance* of the circuit. For the equation written above, the impedance is

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

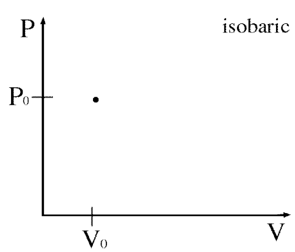
φ is known as the *phase angle* and offsets the sinusoidal variation so that the current is not necessarily at a maximum when the voltage is.

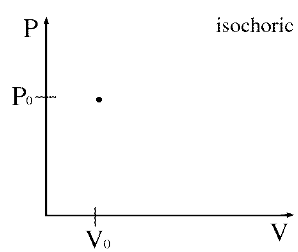
$$\tan\varphi = \frac{\omega L - \frac{1}{\omega C}}{R}.$$

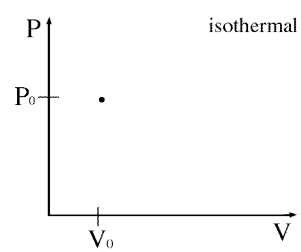
T-S1. Ideal Gas Transformation

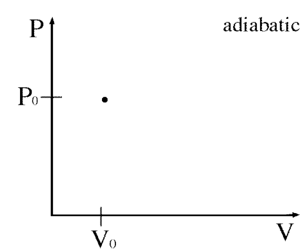
In this supplement, you will derive all of the important properties of the four ideal gas transformations that are most commonly used. The gas has d degrees of freedom and N particles. For each transformation, the gas starts at a pressure P_0, a volume V_0, and a temperature $T_0 = P_0V_0/Nk_B$. Assume that all transformations are reversible. For each of the transformations below, fill in the table based on the final states indicated (along with variables d and γ, if necessary) and draw a PV diagram.

Transformation	$\mathbf{P_f}$	$\mathbf{V_f}$	$\mathbf{T_f}$	$\mathbf{\Delta U}$	**Q**	**W**	$\mathbf{\Delta S}$
Isobaric		V_f					
Isochoric	P_f						
Isothermal		V_f					
Adiabatic			T_f				







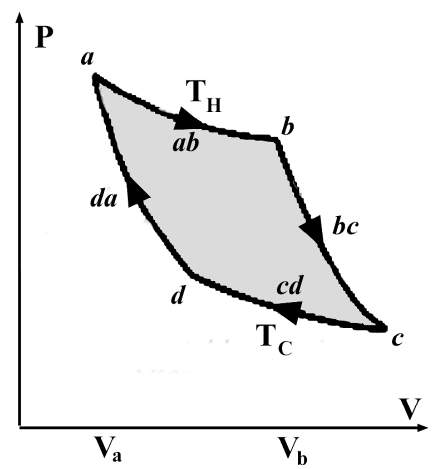


T-S2. Efficiency of the Carnot Engine

In this supplement, you will derive the efficiency of a Carnot Engine in two ways. The first way will be long and tedious, and only be applicable to a Carnot Engine whose working substance is an ideal gas. The second way will be short and simple and will be applicable to any type of Carnot Engine.

Part 1: The Hard Way

Consider a Carnot Engine operating between two heat reservoirs of constant temperature T_H and T_C whose working substance is an ideal gas with d degrees of freedom per particle. Recall that a Carnot Engine consists of an isotherm that expands the gas at constant temperature T_H, followed by an adiabat that expands the gas until the temperature is T_C, followed by an isotherm that compresses the gas at constant temperature T_C, followed lastly by another adiabat taking the gas back to the starting point. Suppose that in the first isothermal expansion the gas goes from volume V_a to V_b.



The Carnot Engine

i) For each 'corner' of the path, labeled *a*, *b*, *c*, and *d* in the PV diagram above, find the missing pressures, volumes, and internal energies in the table below.

Point	T	U	V	P
a	T_H		V_a	
b	T_H		V_b	
c	T_C			
d	T_C			

ii) Find the change in internal energy, the work, and the heat associated with each of the four legs of the cycle, labeled *ab*, *bc*, *cd*, and *da* in the PV diagram above. Fill in your answers in the table below.

Leg	ΔU	W	Q
ab			
bc			
cd			
da			

iii) What *net work*, W_{net}, does one complete cycle of our Carnot engine output?

iv) Which legs have a positive heat transfer? That is, in which steps do we put heat *into* the gas? What is the total heat input, Q_{in}, of one complete cycle?

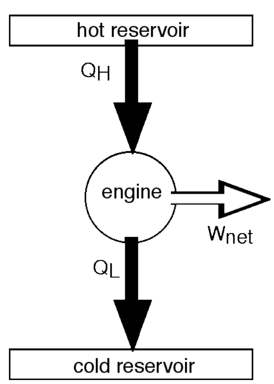
v) What is the efficiency of this Carnot engine?

Part 2: The Easy Way

To find the efficiency of the Carnot engine the short way, all we need to know about the engine are the following facts:

- By definition, the Carnot cycle is *reversible*
- The Second Law of Thermodynamics, which states that $\Delta S_{universe} \geq 0$, with $\Delta S_{universe} = 0$ only in reversible transformations.
- For a reversible process, the change in entropy can be found by $\Delta S = \int \frac{dQ}{T}$.
- Entropy is a *state variable.*

We will use the following schematic for the Carnot engine:



i) What pieces make up the 'universe' shown?

ii) What is the total entropy change of the working substance of the engine after one full cycle?

iii) What is the entropy change for each of the other elements of the universe found in part (i)?

iv) What is the *total* entropy change of the universe for one cycle?

v) Given that the Carnot engine is reversible, what is the relation between the heats and the temperatures?

vi) Use the definition of efficiency and your result from (v) to find the efficiency of the Carnot engine.

Physics 7B

Labs

NAME: ____________________ SECTION DAY/TIME: ____________________________
GSI: ___________________________ LAB PARTNER: ____________________________

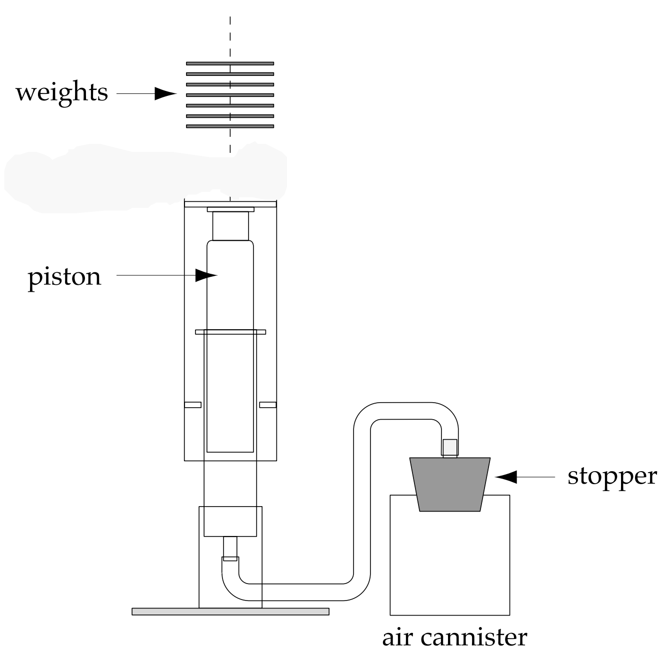
Lab 1: Thermodynamic cycles and engines

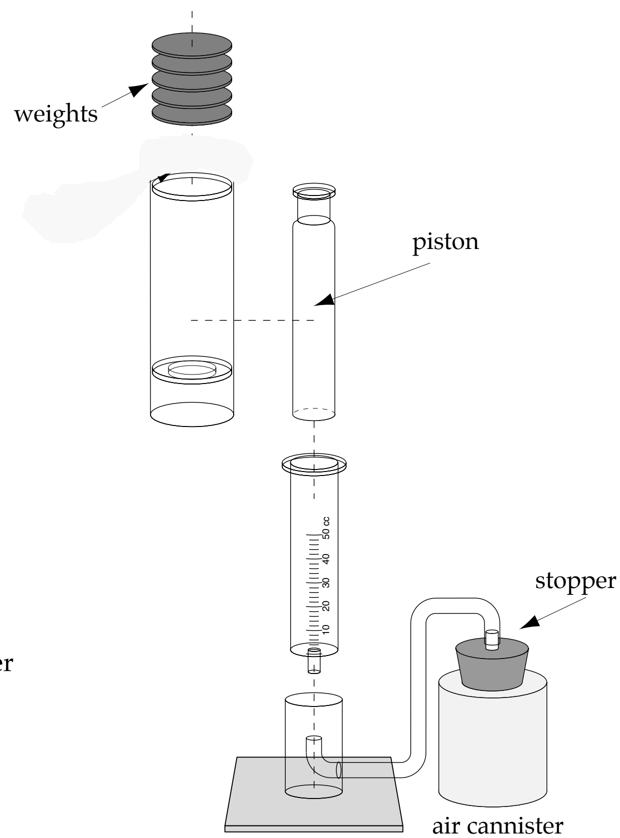
Introduction

As presented in textbooks, heat engines and refrigerators can seem very theoretical. The point of this lab is thus to help you draw connections between abstract p–V diagrams and real life. We want p–V diagrams and cycles to make sense, both mathematically and physically.

Equipment and useful information

◊ Cylindrical piston (Radius = 0.014 meters. Mass of piston & outer sleeve = 0.100 kg.)
◊ Containers of hot and cold water.
◊ Ten 10-gram masses. Never put more than 100 grams onto the piston, or else air may leak out.
◊ Air pressure = 1.00×10^5 N/m^2.





Pre–lab Questions

[Do 1a and 1b before coming to lab. Your GSI will initial these pre–lab questions when you arrive in lab.]

1. This question gives you a sense of the pressure differences we'll see in this experiment.

(a) When no masses are placed on the piston, what is the pressure of the air inside the piston? Show your work here.

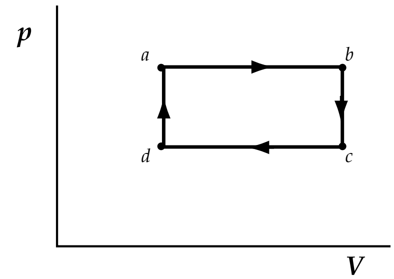
(b) When 100 grams are placed on the piston, what is the pressure of the air inside the piston?

GSI Initials: ___________

The lab starts on the next page

2.

- With everything at room temperature, place 100 grams on the piston.
- If necessary, remove the stopper from the air canister, let the piston slide down almost as far as it will go, and then replace the stopper (making it airtight!). We want the piston to start near the bottom.
- If the temperature of your hot water is below 50° C, scoop some hot water from one of the hot plates. If the water is boiling, dilute it with some cooler water in your beaker.



NOTE: As suggested by question 1, the pressure and volume differences in this experiment are actually very small.

The gas in the piston now corresponds to point *a* on this *p*–*V* diagram. Here, you'll figure out how to make the gas inside the piston undergo the *cycle* shown. Then, you'll actually do it, and answer questions about each step.

In these experiments, you may add or remove masses from the piston, and you may place the air canister in the water beakers; but you may not push or pull on the piston, because that tends to make it leak.

(a) How will you make the gas go along path *ab*? Along path *bc*? *cd*? *da?* Write down your plan for each of the four steps.

- Now do it. If something doesn't work as predicted, see if you can correct the problem. Next to your original plan, jot down any modifications you used.
- When you are finished, call your GSI over and explain the necessary steps. Your GSI will initial here when your explanation is correct.

GSI Initials: __________

(b) In all these questions "gas" refers to the gas trapped inside the piston. Along path ab, is the work done *by* the gas *on* the piston positive, negative, or zero? Answer this question, and others like it, both in terms of the abstract p–V diagram and in terms of your actual experiment. (Ask your GSI for help with this, if it's not clear what is meant.) Most important, make sure you understand how the theoretical ideas connect to what you're doing.

(c) Along path bc, is the work done by the gas on the piston positive, negative, or zero? How do you know?

(d) Along ab, is the heat added to the gas positive, negative, or zero? How could you have figured this out based on theory, even if you hadn't done the experiment? Hint: think about the First law of thermodynamics.

(e) Along bc, is the heat added to the gas positive, negative, or zero? Explain.

(f) Along path *abcda* (the whole cycle), is the net change in the internal energy of the gas ($\Delta E_{internal}$) positive, negative, or zero? How do you know?

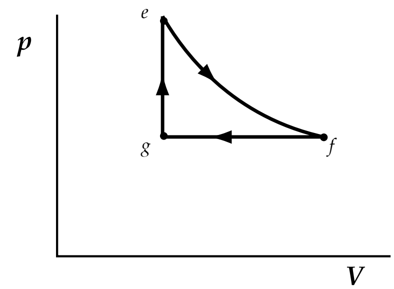
(g) Along path *ab*, the gas does positive work on the piston. Said another way, the piston does negative work on the gas. Along path *cd*, the piston does positive work on the gas. Which of these two W's is bigger in magnitude: the work done by the gas in step *ab*, or the work done on the gas in step *cd*? Explain your answer in terms of the p–V diagram, and *also* in terms of your actual experiment. (Hint: Think about the mass on the piston during *ab* versus the mass on the piston during *cd*.)

(h) Along paths *ab* and *da*, the gas absorbs heat. Along paths *bc* and *cd*, the gas loses heat (i.e., it "absorbs" negative heat). Is the net heat absorbed by the gas zero? Explain how you know. Hint: Your part (f) and (g) answers might be helpful.

(i) In a *heat engine*, net heat is added to a system, and the system uses some of that energy to do work. In the experiment you just performed, did the piston function as a heat engine? Explain.

Question 2 parts (a) through (i) are typical exam problems. Everyone needs to understand this material well. After most students have finished these questions, your GSI will go over them. If you finish early and feel reasonably confident, go on to the next experiment.

3.



NOTE: pV is constant along path *ef*

- Place 80 grams on the piston, and place the air chamber in hot water. If the piston is in danger of reaching the top, use cooler water.
- Let the piston settle.

The gas in the piston now corresponds to point *e* on this new p–V diagram. Once again, you'll figure out how to make the gas inside the piston undergo the *cycle*. But since you already did paths *fg* and *ge* in question 2, let's focus on *ef*. As before, you cannot push or pull the piston. All you may do is add and remove masses, and use the beakers of water.

(a) How will you make the gas go along path *ef*? Since pV is constant along that path, and since $pV = NkT$, the temperature of the gas stays constant along that path.

- Now do it. Make any necessary corrections to your process above.

(b) Discussing this experiment, Jason makes the following comment: "Since the temperature stays constant along *ef*, the gas neither absorbs nor loses heat along that path. Hence, the internal energy of the gas stays constant." Evaluate Jason's argument. What is he right about? What is he wrong about (if anything)? Are there flaws in his reasoning?

(c) Along path *fg*, the gas loses internal energy. Along path *ge*, it gains internal energy. Which of those two $\Delta E_{internal}$'s (if either) is bigger in magnitude: The internal energy lost during step *fg*, or the internal energy gained during step *ge*? Explain.

(d) When the piston goes through this whole cycle (*efge*), does it function as a heat engine? Explain.

Answer the following questions only if you have time. In a related worksheet, you'll cover this material more fully.

4. A heat engine converts heat into work. Needless to say, we want a heat engine to be as efficient as possible. Suppose that, during a cycle, the engine absorbs 10 joules of heat. If the engine does 10 joules of work, then it's 100% efficient (efficiency = 1.0). If it does 9 joules of work, it's 90% efficient (efficiency = 0.9). And so on.

"Work" here refers to the net work. For instance, if the gas inside the piston does 8 joules of work during one leg of cycle, but we do 6 joules of work *on* that gas during another leg of the cycle, then we get only 2 joules of work out of the cycle overall.

(a) Based on the above passage, write a formula for the efficiency of a heat engine.

(b) Your part (a) answer probably contains a Q somewhere. But is that Q the (positive) heat absorbed, or the *net* heat absorbed? To consider the difference, think about the heat engine from question 2: Suppose the engine absorbs 15 joules of heat during dab, and loses 5 joules of heat during bcd. When calculating the efficiency, should you use $Q = 15$ joules or $Q = 10$ joules? Explain your reasoning.

(c) Consider your heat engine from question 2. Is the efficiency 100%? Nearly 100%? Significantly less than 100%? How did you figure it out? (Answer this *without* performing detailed calculations.)

NAME: ____________________ SECTION DAY/TIME: ____________________________
GSI: ___________________________ LAB PARTNERS: ____________________________

Equipotential lines and electric fields

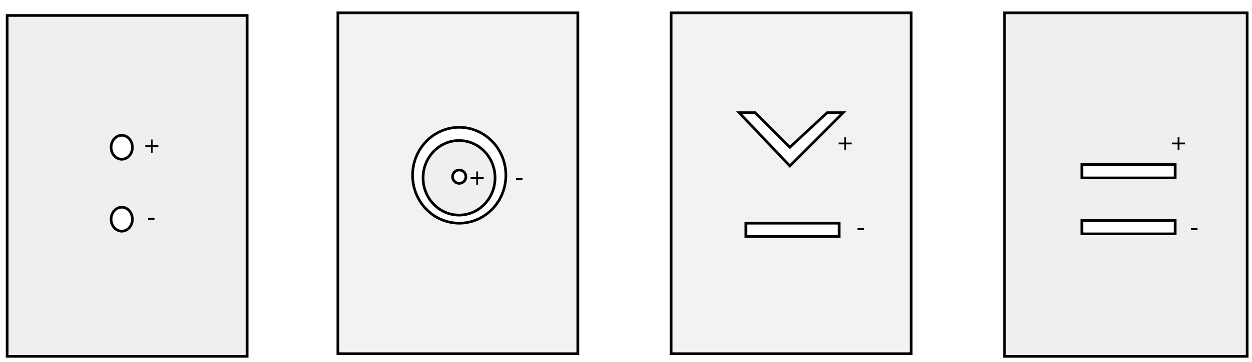
Introduction

This lab gets at one of the most difficult concepts in the course: electric potential, and its relationship to electric fields and potential energy. The lab activities and associated questions can help you get a real-world feel for these concepts and their conceptual underpinnings. The hardest thing about potential, however, is to see how all this fits together.

The activities below are designed to take only a portion of this period. We will take the rest of the time to continue with discussion section activities.

Questions

1. In these pictures, the two conducting regions (marked in white) carry equal and opposite charges. What will the *equipotential lines* look like in each case? Remember, two points have the same potential if a charge would have the same potential energy at either point. Sketch your predictions here using dotted lines.



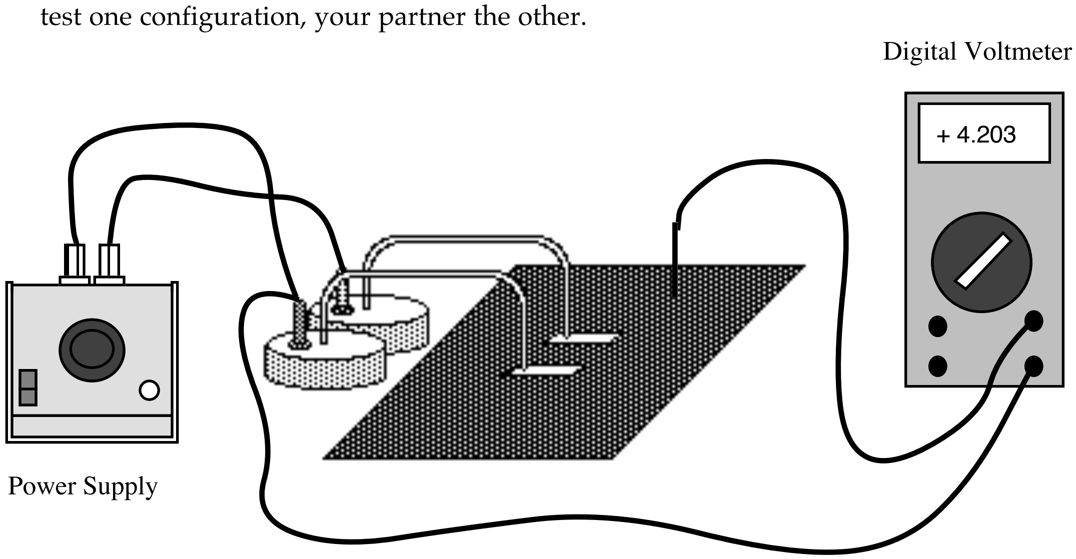
Now pick *two* of the four configurations, and experimentally sketch the equipotential lines, using the procedure on the next page. You should do one; your partner the other. To check your other two predictions, you can look at the work done by other lab tables.

Before starting the experiment, make sure the equipment is working properly, and answer a brief question.

- ♦ Layer a piece of paper on the bottom, then a piece of carbon paper (dark side down), then a piece of the teledeltos paper with conducting paint regions on top. Tape two corners down so you can keep them aligned but still lift up to write on the bottom paper.
- ♦ The equipment should already be set up as drawn on the next page, with the power supply set to 5 volts. So, the power supply enforces a 5 volt potential difference between the two metallic regions on your paper. Touch the voltage probe to one metallic region, and then to the other. It should register 0 V and then plus or minus 5 V (or vice versa).

2. Prediction: *Within* a metallic region, is the potential the same everywhere, or does it vary point by point? *Explain* why; don't just quote a result. (Note, within the metallic region means within the actual metal, not inside a cavity or region surrounded by metal.)

- ♦ Now, test *two* of the predictions you made earlier with the voltage probe (the "free" wire sticking out of the digital voltmeter). Use the following procedure. Remember: you should test one configuration, your partner the other.



Procedure for "sketching" the equipotential lines.

- ♦ Using the voltage probe, find a place on the paper where the potential is 1.0 volt. If it's 1.03 V or 0.98 V, that's fine; just get close. Mark that point by lightly rubbing the probe on the top layer. Check the bottom layer to make sure the mark was transferred by the carbon paper. Then, find another 1-volt point, about a centimeter from the first one. Mark it. And so on. By using symmetry and your above prediction, you may be able to save yourself some work.

- On the **bottom sheet of paper**, connect the dots. This curve is an equipotential line; every point along the curve has potential 1 volt.

- Now make the equipotential lines for 2 V, 3 V, and 4 V, again using light pressure to mark the locations and connecting the dots on the **bottom sheet of paper**. Work pretty fast; it's more important to think about what these lines mean than it is to draw them perfectly.

- If you've made any major errors in your predictions, please correct them now.

3. Using solid lines, add sketches of the electric field for each configuration on p. 1. Explain here how you know how to draw the fields.

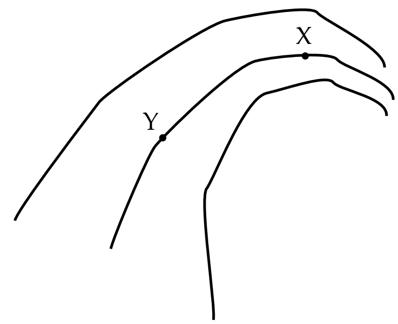
4.

- Estimate the electric field at a point you select between the two conductors on your plot. Record your data here, and explain your measurement. Hint: $E_x = -dV/dx$.

Please lightly erase any stray marks on your conducting paper, so that the next lab group gets a fresh start.

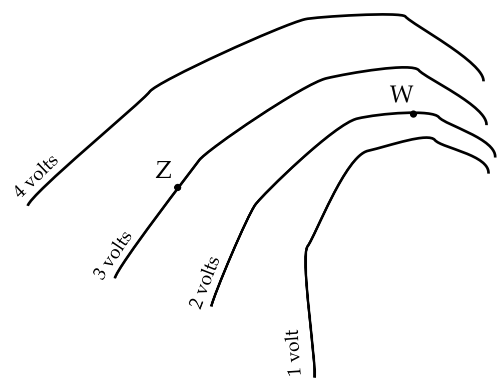
5. (*Just a question, not an experiment.)* In figure 5, at which of these two points, X or Y, is the electric field stronger? How do you know?

FIGURE 5
Segments of equipotential lines



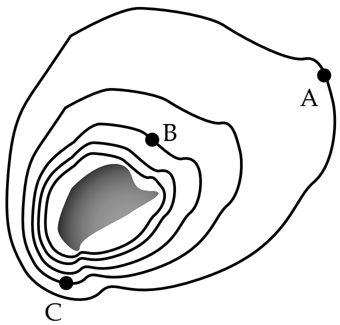
6. **(a)** In question 5, you compared the electric field at two points on the same equipotential line. Now consider two points on different equipotential lines. In figure 6, where is the field stronger—point W or point Z? How do you know?

FIGURE 6
Segments of equipotential lines



(b) Sketch on figure 6 the direction in which a positive charge placed at point *Z* would move. Sketch the direction in which a negative charge placed at point *W* would move.

7.



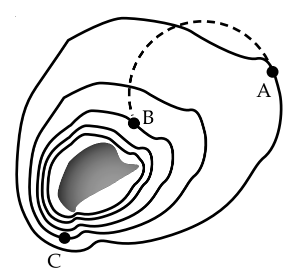
The gray charge distribution shown generates an electric field corresponding to the following equipotential surfaces. The potentials at points A and B are $V_A = 3.0$ V and $V_B = 1.0$ V.

(a) On this diagram, sketch some of the electric field lines resulting from the charge distribution. Is the charge distribution positive or negative? (Yes, you have enough information to tell.)

(b) Where is the electric field strongest? Explain.

(c) How much work would it take to move a $Q = 0.50$ C point charge along a straight line from B to A?

(d) Now consider a semicircular path from B to A. To move the $Q = 0.50$ C charge along this path, would it take more work, less work, or the same work, as compared to part (c)? Explain.



(e) Which takes more work: Moving charge Q from point C to point A, or moving it from point B to point A? Justify your answer.

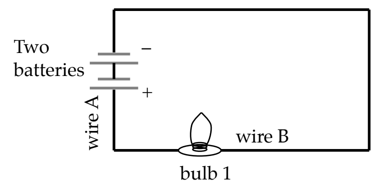
NAME: ____________________ SECTION DAY/TIME: ____________________________
GSI: ___________________________ LAB PARTNER: ____________________________

Lab 3: Introduction to DC circuits

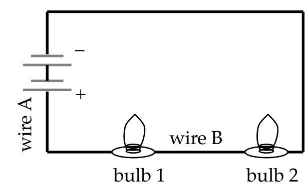
Introduction

This lab introduces direct-current (DC) circuits, focusing on *conceptual understanding*. On a later worksheet, you'll integrate this qualitative understanding with mathematical problem-solving. Although everybody should do questions 1 through 4, people who already know a lot about circuits will be able to get to the challenge problems at the end.

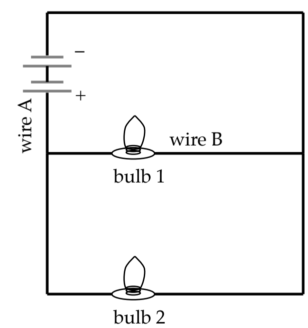
CIRCUIT 1



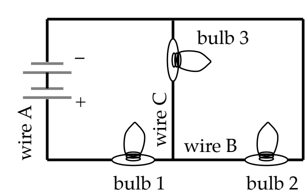
CIRCUIT 2



CIRCUIT 3



CIRCUIT 4



Technical hints

- When your two batteries are hooked up in series; you can think of them as a single, double-strength battery.
- Your GSI will show you how to "transform" one circuit into another. Ask for help if you're having trouble achieving a clean transformation.

Questions

IMPORTANT NOTE: For each question, first write your answer (prediction), then do the experiment. Finally, amend your original answer, if necessary. But don't erase your original prediction—it's helpful to have a record of what mistakes you're liable to make in the future.

1. In circuit 1, which (if either) is bigger: The current through wire A or the current through wire B? What gets "used up" when current flows through a light bulb?

2. When circuit 1 is transformed into circuit 2 (by hooking up the 2nd light bulb), what happens to

(a) The brightness of light bulb 1?

(b) The current through wire A? Why?

3. When circuit 1 is transformed into circuit 3, what happens to

(a) The brightness of bulb 1?

(b) The current through wire A?

(c) The current through wire B? Explain all your answers. If the experiment comes out different from your prediction, you can amend your answer by trying to explain the discrepancy.

> Because the rest of the lab builds on questions 1 - 3, your GSI will go over those three questions with the whole class. If you try question 4 before this discussion, please look over your answers *after* the discussion, to take your GSI's ideas into account.

4. (*This one is hard, but very important.*) When circuit 2 is transformed into circuit 4 (by hooking up the third bulb), what happens to

(a) The current through wire A?

(b) The brightness of bulb 1?

(c) The brightness of bulb 2? Explain your answers intuitive (*not* just with formulas). Check your answers with your GSI.

5. Let I_1 denote the current through wire A in circuit 1. In terms of I_1, what is the current through wire A in. . .

(a) circuit 2? Is it $2I_1$, *or* $I_1/2$, or what? Explain conceptually, even if you know a formula.

(b) circuit 3? Explain.

(c) *(harder)* circuit 4? Explain.

6. With your battery, your three light bulbs, and all the wires you want, build a circuit that produces as much light as possible. Diagram the circuit here, and explain why it's the brightest.

7. With that same equipment, build a circuit that produces as *little* light as possible. Should the circuit use all three bulbs? Be sure to test this issue experimentally. Diagram your circuit, and explain why it's the dimmest.

8. In this lab, you've built a total of six circuits: the four on page 1, the "brightest" circuit from question 6, and the "dimmest" circuit from question 7.

(a) Of those six circuits, which one has the *most* current flowing through the battery? Explain.

(b) Which has the *least* current flowing through the battery? Explain.

9. Give at least *two* separate reasons why it's advantageous to wire holiday lights in parallel. (Ask your GSI if you don't know what we mean by "holiday lights.")

10. Are the electrical outlets in your house/room wired in series or in parallel? Explain.

11. In electrostatic systems, a potential difference (i.e., a voltage) always corresponds to an electric field. Is this also true about circuits? Specifically, does the potential difference between the two terminals of the battery correspond to an electric field anywhere? Or do circuits allow us to have "voltages without fields?"

NAME: ______________________________ DL SECTION NUMBER: __________________

GSI: _________________________________ LAB PARTNERS: ______________________

MAGNETISM LAB:

The Charge-to-Mass Ratio of the Electron

Introduction

In this lab you will explore the motion of a charged particle in a uniform magnetic field, and determine the charge-to-mass ratio (e/m) of the electron. We hope that you will also begin to develop an intuitive feel for magnetism.

There are more prelab exercises for this experiment than has been normal in Physics 7B. Be sure to complete these before arriving at lab—they will count for half of your final lab score, and your GSI will initial page 2 at the start of lab to indicate that you have completed them. We suggest reading through the entire lab before attempting to complete the Prelab questions, so that they will make more sense.

Prelab Questions

1. Using your Physics 7B knowledge about the force on a charged particle moving in a magnetic field, and your Physics 7A knowledge of circular (centripetal) motion, derive an equation for the radius r of the circular path that the electrons follow in terms of the magnetic field B, the electrons' velocity v, charge e, and mass m. You may assume that the electrons move at right angles to the magnetic field.

2. Recall from electrostatics, earlier in the course, that an electron obtains kinetic energy when accelerated across a potential difference V. Since we can directly measure the accelerating voltage V in this experiment, but not the electrons' velocity v, replace velocity in your previous equation with an expression containing voltage. The electron starts at rest. (Don't get capital V, voltage, confused with lowercase v, velocity.) Now solve this equation for e/m. You should obtain

$$\frac{e}{m} = \frac{2V}{B^2 r^2} \qquad \textbf{Eq. 1}$$

3.

The magnetic field on the axis of a circular current loop a distance z away is given by

$$B = \frac{\mu_0 I R^2}{2\left(R^2 + z^2\right)^{\frac{3}{2}}},\qquad \textbf{Eq. 2}$$

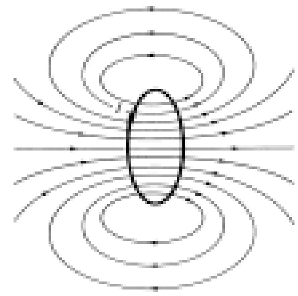


FIGURE 30–15 Magnetic field due to a circular loop of wire.

(Fig. from D. Giancoli's *Physics*)

where R is the radius of the loop and I is the current. (See example in text for a derivation and discussion of this result.) Using this result, calculate the magnetic field at the midpoint along the axis between the centers of the two current loops that make up the Helmholtz coils, in terms of their number of turns N, current I, and radius R—see Fig. 2 on page 5. [Hint: magnetic fields add as any vector fields do.] Helmholtz coils are separated by a distance equal to their radius R.
You should obtain

$$|B| = \left(\frac{4}{5}\right)^{3/2} \mu_0 \frac{NI}{R} = 9.0 \times 10^{-7} \frac{NI}{R} \qquad \textbf{Eq. 2}$$

where B is the magnetic field in tesla, I is the current in amps, N is the number of turns in each coil, and R is the radius of the coils in meters.

GSI Initials: ________

Important Background Information About Atoms

All normal matter is made up of atoms. Atoms have a "size" of roughly 1 Å ("Ångstrom", 10^{-10} meters), and range in mass from about 10^{-27} kg to 10^{-25} kg. Atoms are in turn made up of smaller particles: positively-charged **protons**, uncharged **neutrons**, and negatively-charged **electrons**. Protons and neutrons have almost the same mass (about 10^{-27} kg) while electrons are about 10^{-30} kg. Protons and electrons have equal and opposite charge, $|e| = 1.6 \times 10^{-19}$ C.

An atom's protons and neutrons are contained in the atom's **nucleus**, which is about 1 fm ("femtometer", or "fermi", 10^{-15} meters) across—a miniscule fraction of the atom's total size. The number of protons in an atom's nucleus determines what kind of atom it is, where it sits on the periodic table, and its chemical properties. For instance, any atom with six protons is carbon, whereas any atom with seven protons is nitrogen. The number of neutrons in an atom determines which **isotope** of that atom it is. Helium-4 (2 protons + 2 neutrons = 4) has 1 more neutron than helium-3 (2 protons + 1 neutron) and is therefore a different isotope, but both isotopes are still helium atoms because they both have two protons. The study of nuclei is known as **nuclear physics**.

The nucleus is surrounded by the lighter electrons, which take up most of the volume of the atom. A normal atom has the same number of electrons as protons, and so has zero net charge. If the atom has a different number of electrons than protons it is called an **ion**; ions with more electrons than protons have a net negative charge and are said to be **negatively ionized**, while ions with fewer electrons than protons have a net positive charge and are said to be **positively ionized**. The study of atoms in general, and their electrons in particular, is known as **atomic physics**.

Experiment description

Understanding the electron is essential for understanding atoms and matter in general. Two important properties of the electron are its charge e and its mass m. In this experiment we will measure the ratio of the two (e/m) with the method first used by J.J. Thomson in 1897. The experiment is based on the fact that a charged particle moving in a magnetic field feels a force at right angles to its velocity: $\mathbf{F_B} = q\mathbf{v}\mathbf{x}\mathbf{B}$. If we send a beam of electrons into a magnetic field uniform in strength and direction, then the trajectory of the electrons is a circle whose radius depends on e/m. We measure the radius of the circle for different values of B, and deduce e/m.

The Electron Beam

To produce a beam of electrons, we heat a metal plate called a **cathode** and boil electrons off of its surface. (We won't worry about the details of this boiling off here.) The cathode is held at a low voltage, and the boiled-off electrons accelerate towards a high-voltage plate a few centimeters away called an **anode**. Some electrons pass through a small hole in the anode and are collimated into a narrow beam (see figure 1). The electrons are not accelerated further once they pass through the anode. Since human eyes can't see electrons, the whole experiment is encased in an evacuated glass bulb with a small amount of helium gas inside.

When the gas molecules are struck by electrons they radiate a blue color, making the path of the electron beam—though not the electrons themselves—visible.

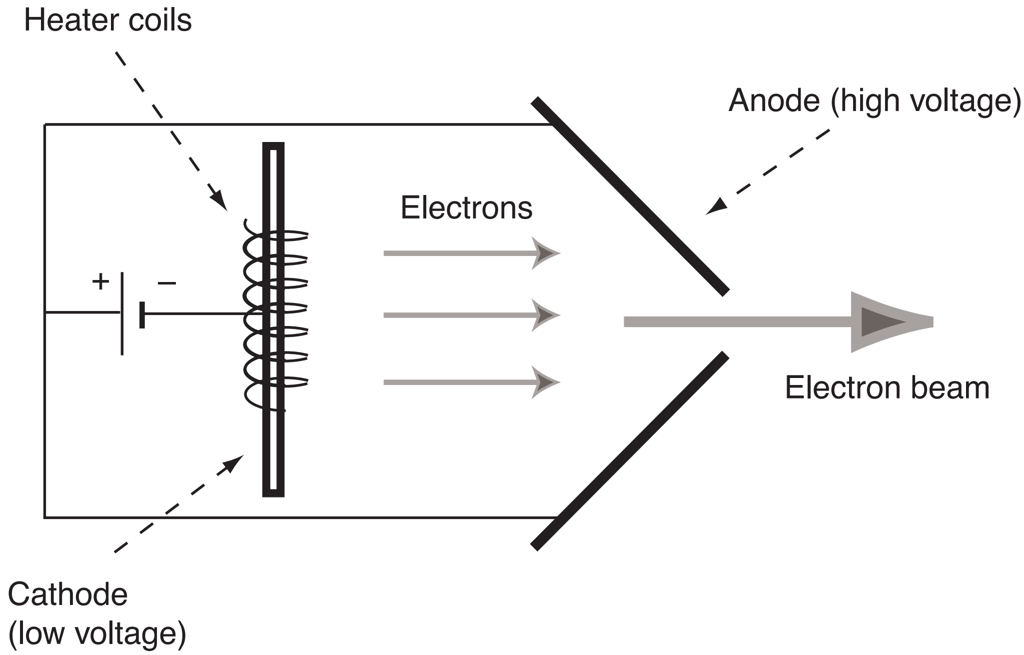


Figure 1: A schematic drawing of the cathode-anode assembly, showing how the electron beam is generated from electrons boiled off of the cathode and accelerated towards the anode.

The Magnetic Field

To produce a uniform magnetic field, we place two large circular coils of wire known as **Helmholtz coils** around the tube, one on either side (see figure 2). The two coils have the same radius and the same number of turns ($R = 0.15$ meters and $N = 130$), and are placed exactly one radius R apart. When a current is passed through both coils in the same direction, the fields add to produce a very uniform magnetic field B_{tot} in the center region between them. The field B_{tot} is pointed along the line joining the centers of the two coils, and its magnitude at the center is related to the current in each coil by Eq. 2 above.

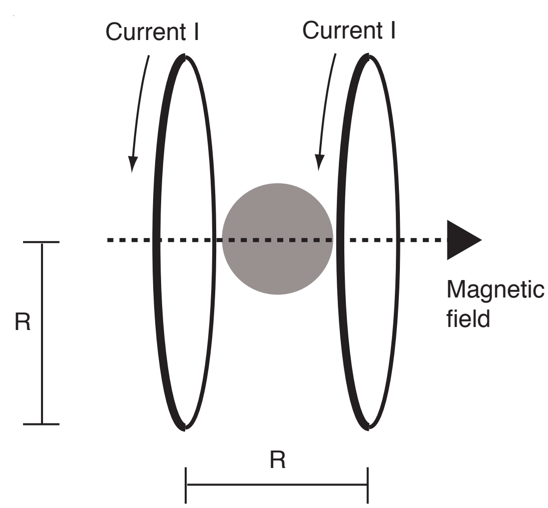


Figure 2: Helmholtz coils. The same current running in the same direction through both coils produces a uniform magnetic field in the shaded center region between the two coils.

We won't ask you to show it here, but you should know that (1) the first derivative, dB_{tot}/dz evaluated at the midpoint between the coils is zero by symmetry; (2) the second derivative, d^2B_{tot}/dz^2, is also zero *if* the coils are separated by a distance equal to their radius R. Because we want as uniform a field as possible, Helmholtz coils are separated by just this distance R.

Parallax Errors

Close one eye and hold up a ruler between you and a far wall. Now move the ruler towards or away from your eye without moving your head, so that the ruler just covers the wall from end to end. If you didn't know better, you'd think that you had just measured the length of the wall to be the same as that of the ruler. This is a **parallax** error, which can occur when a measuring stick is not placed directly against the object it is measuring. (If you had put the ruler right up against the far wall, you'd immediately see your mistake.) Since the circling electron beam is encased in a glass bulb, we can't put a measuring stick directly up against it and so we are susceptible to parallax errors. But there is some additional equipment on the apparatus that will help you avoid these errors—we'll ask you to figure out how rather than describe the procedure here.

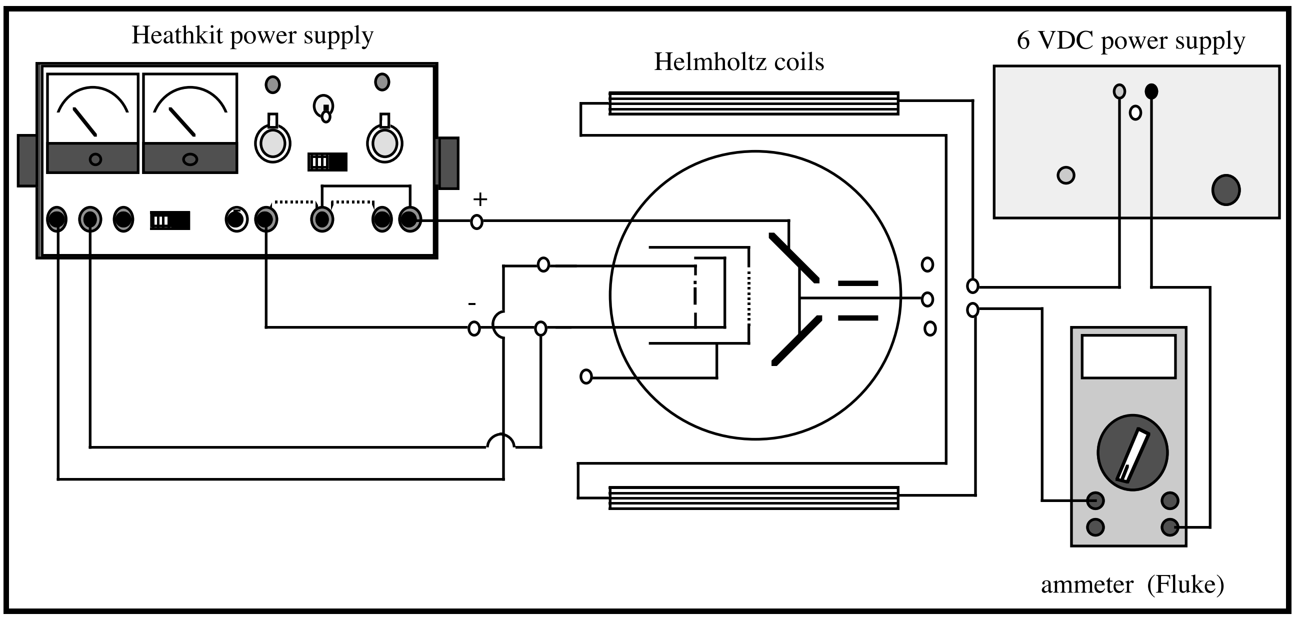


Figure 3: Connections in the e/m experiment.

Procedure

- Connect the power supplies to the baseboard as shown in Figure 3. Switch the Heathkit power supply to "B+" voltage: this is the voltage difference V between the anode and the cathode, and you will read its value from the top (red) scale. Set this voltage to zero and turn on the power supply. Turning on the power supply applies an alternating current to the cathode heater in the glass bulb, which will glow orange after a few seconds. *(Note: "B+" is a bizarre historical term for the voltage between an anode and a cathode. Don't get it confused with the magnetic field, B.)*

1. *Prediction:* You are now going to turn up the voltage and the electron beam will appear. Will it be curved or straight? Why?

- Turn up the voltage until you can see a glowing blue electron beam (the room may have to be dark for this to be visible.)

2. Is the beam curved or straight? Explain.

- Connect the 6 VDC (Volts Direct Current) power supply and the Fluke meter to the Helmholtz coils as shown in figure 3. <u>Be sure to use the 10 amp connection on the Fluke meter to avoid damaging it!</u> This applies 0 – 1.2 amps DC to the coils, creating the uniform magnetic field as per Eq. 2. You can adjust the current using the black knob on the power supply and read its value on the Fluke meter.

3. *Predict:* (i) Will the beam radius increase or decrease if you increase the magnetic field? (ii) What if you increase the anode-cathode (B+) voltage? Explain your reasoning for each conceptually, without simply referring to Eq. 1.

- Increase the magnetic field. Were your predictions correct? If not, explain the correct reasoning here.

- Set the voltage and magnetic field so that you see a well-defined circular electron beam path. You will in a moment measure the radius of the path. But first, develop a radius-measuring technique to avoid the parallax error. (Hint: notice the illuminated scale, or the mirror strip and washers attached to the apparatus.)

4. Explain your method of avoiding parallax errors and why it works.

5. Now measure the radius of the electrons' path. Record your data below, and repeat for four other B-field and voltage combinations. (You may want to make a table so you can calculate your values for e/m right here too.)

6. The electron beam path isn't exactly circular; it spirals slightly inward. Why is this? (Hint: What variables affect the beam's radius?)

7. Calculate the average value of e/m from your four measurements, and compare to the accepted value of 1.76×10^{11} C/kg.

8. What sources of error were present in this experiment? What amount of uncertainty do you estimate each source of error contributed to your final e/m determination? Justify your estimates with words and/or numbers.

NAME: ____________________ SECTION DAY/TIME: ____________________________
GSI: ___________________________ LAB PARTNERS: ___________________________

Lab 6: Introduction to oscilloscope and time dependent circuits

Introduction

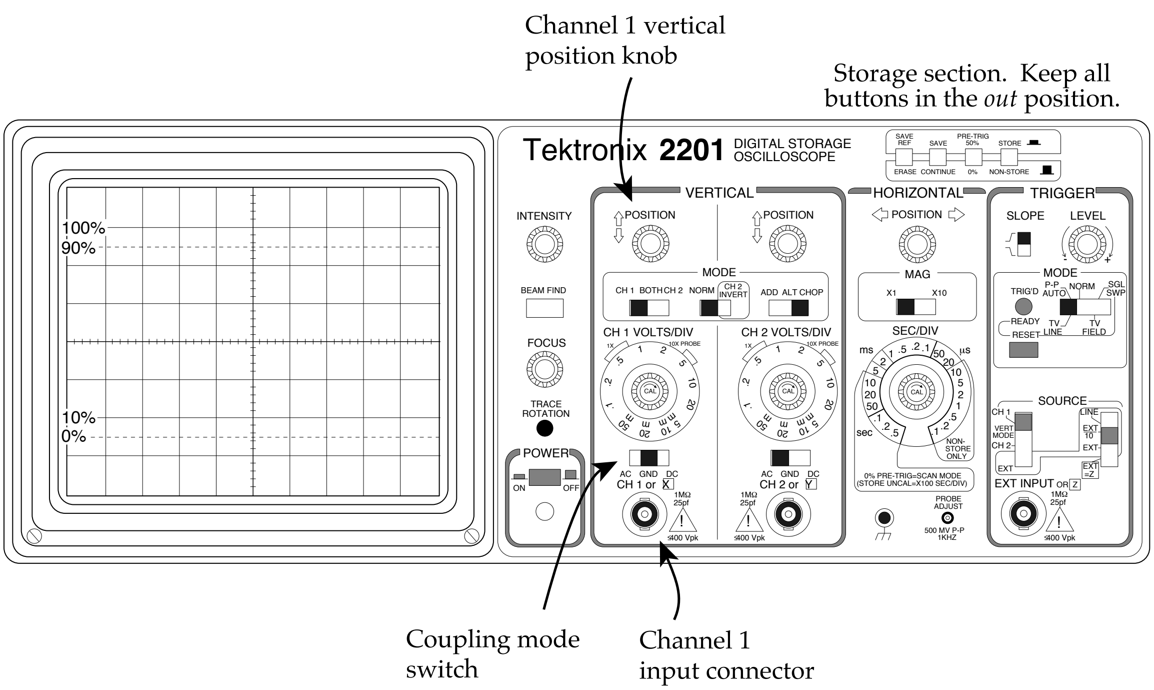
In this lab, you'll learn the basics of how to use an oscilloscope. Then you'll investigate time dependent circuits. When dealing with capacitors and inductors in DC circuits, it's easy to get lost in mathematics, without understanding what's going on conceptually. These questions and lab activities are designed to help you develop an understanding of these circuits, allowing you to address conceptual questions without plugging through unnecessary math. You'll also see what these circuit components look like in real life.

Part I of this experiment, on the basics of the oscilloscope, should take approximately 30 minutes. The rest of your time in lab should be spent working on Part II, on the time dependent RC and LR circuits. (Don't worry if you aren't fully comfortable with the scope by the end of Part I. You'll get more practice in Part II.)

Part I: Oscilloscope Basics

Activity 1: Reset the oscilloscope

- ♦ Turn on the oscilloscope, and disconnect any probes plugged into the "channel 1" (CH 1) input connector.
- ♦ Set all the levers and buttons as indicated here, if they're not already.



- Set the CH 1 coupling mode switch to "ground" (GND).
- Turn down the INTENSITY knob, if necessary, to avoid burning out the screen. The sweeping dot should be clear but not too bright.

Since channel 1 is now "grounded" to zero volts, the oscilloscope should read zero on the vertical axis (using the coordinate axes centered on the screen). If it doesn't. . .

- Adjust the channel 1 vertical POSITION knob so that the oscilloscope reads 0 volts.

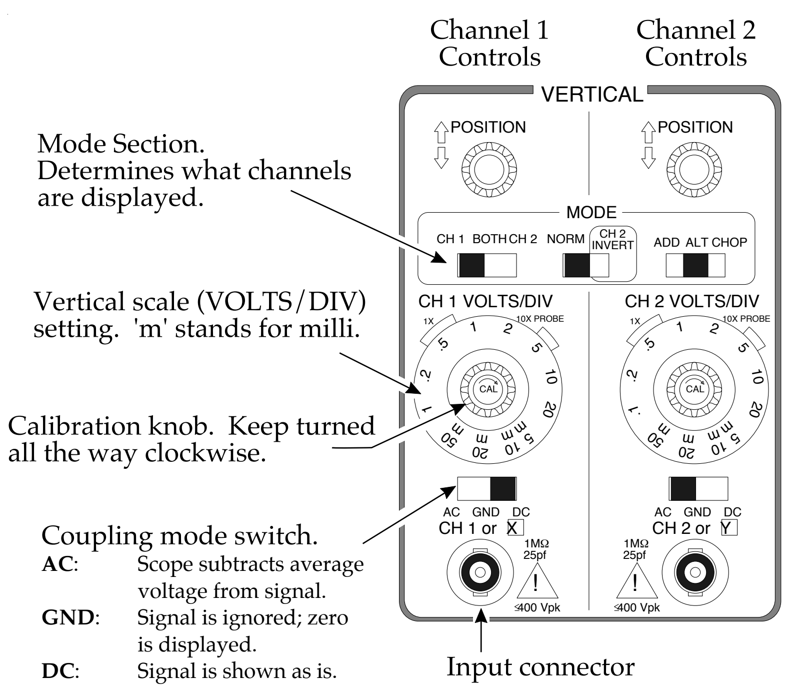
What the oscilloscope does

The oscilloscope graphs voltage vs. time, by sweeping an electron beam across the phosphor screen. Wherever the beam hits the screen, it glows green. For most measurements, the beam sweeps rightward at a constant rate. As you can see, when the beam gets to the right-hand side of the screen, it jumps back to the left-hand side. In this way, the horizontal axis shows time.

When a probe is plugged into the CH 1 input connector, the vertical axis shows the potential difference—i.e., the voltage—between the two wires coming out of that probe. If you're interested, ask your GSI what's going on inside the oscilloscope to deflect the electron beam up or down. Better yet, see if you can figure it out! Hint: Parallel-plate capacitor.

Activity 2: Measuring DC voltages, and using the VOLTS/DIV setting

The point of this brief activity is to practice measuring a voltage with the oscilloscope, and to get a feel for what the VOLTS/DIV control does.



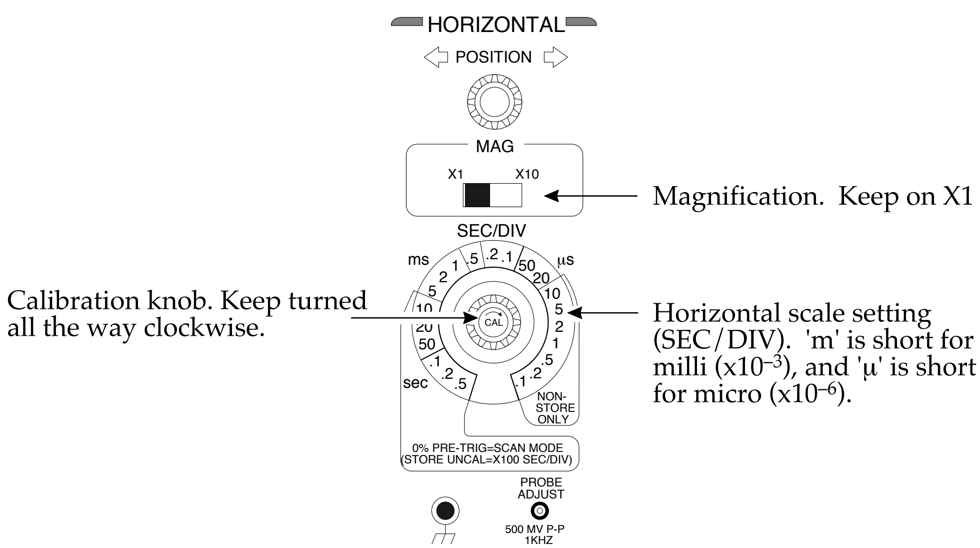
- Set the CH 1 VOLTS/DIV to 2, by aligning the “2” next to the “1X” bracket.
- Set the CH 1 coupling mode switch to DC.
- Now use the oscilloscope to measure the voltage across a 1.5-volt battery.

Make sure you understand what the VOLTS/DIV setting is doing. Students often err in thinking in terms of DIV/VOLT instead of VOLT/DIV.

1. To get a more precise reading of the battery’s voltage, should you turn the VOLTS/DIV knob clockwise or counterclockwise? Why? Try it, to get a feel for how much precision can be gained.

Activity 3: Measuring AC voltages, and the SEC/DIV setting

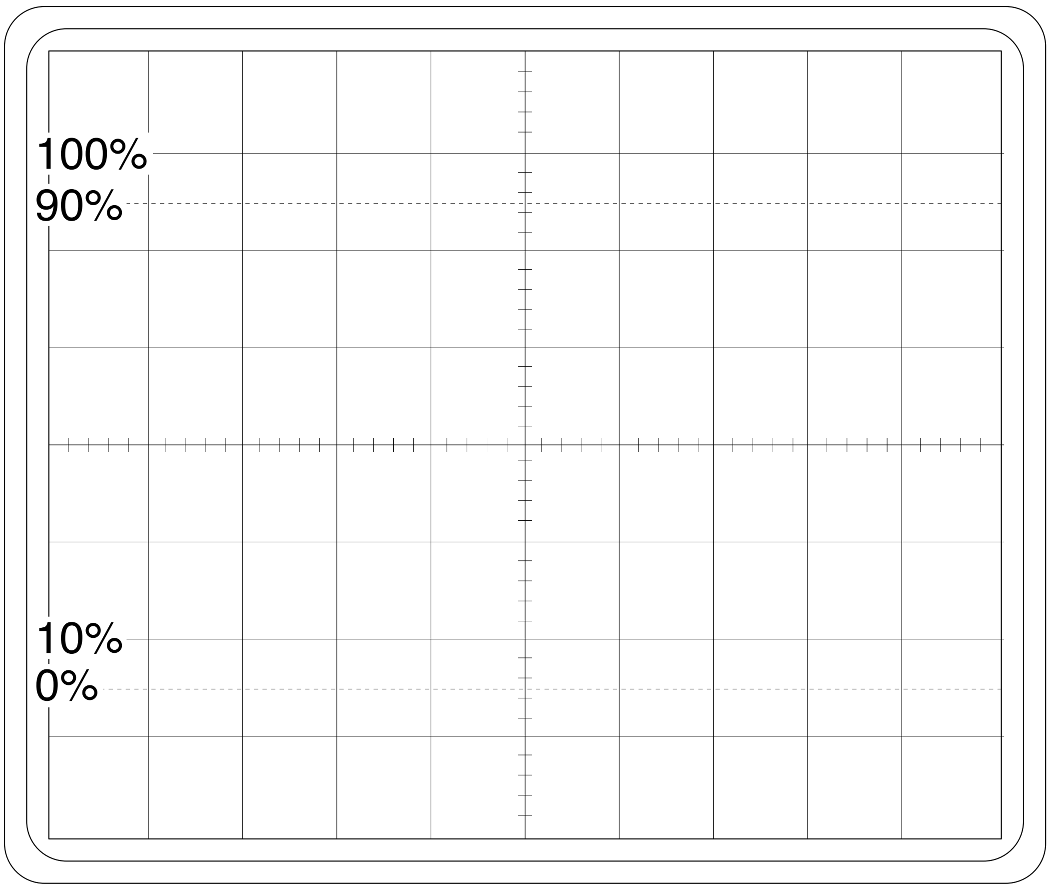
Now you’ll practice using an AC power supply, and you’ll figure out what the SEC/DIV knob does. The “AC” means “Alternating Current”—that is, the voltage put out by the power supply oscillates with a frequency that you set.



- Set SEC/DIV to 0.5 milliseconds.
- Set the CH 1 VOLTS/DIV to 5.

- ♦ Turn on the AC signal generator. Set it to sinusoidal wave, 1000 Hz (i.e., 1.0 kHz). (Note: make sure the sweep width knob is all the way to the left, so it clicks.) But don't connect the AC signal generator to the oscilloscope, until answering this question. . .

2. When you use the oscilloscope to measure the voltage produced by this AC signal generator, what will the screen look like? Sketch your detailed prediction on the next page, paying attention to the amplitude and "wavelength."



- ♦ Now hook up the AC power supply to the oscilloscope.

If your prediction was wrong, see if you can figure out what's going on, or get help from your GSI. Sketch the actual screen display in a different color.

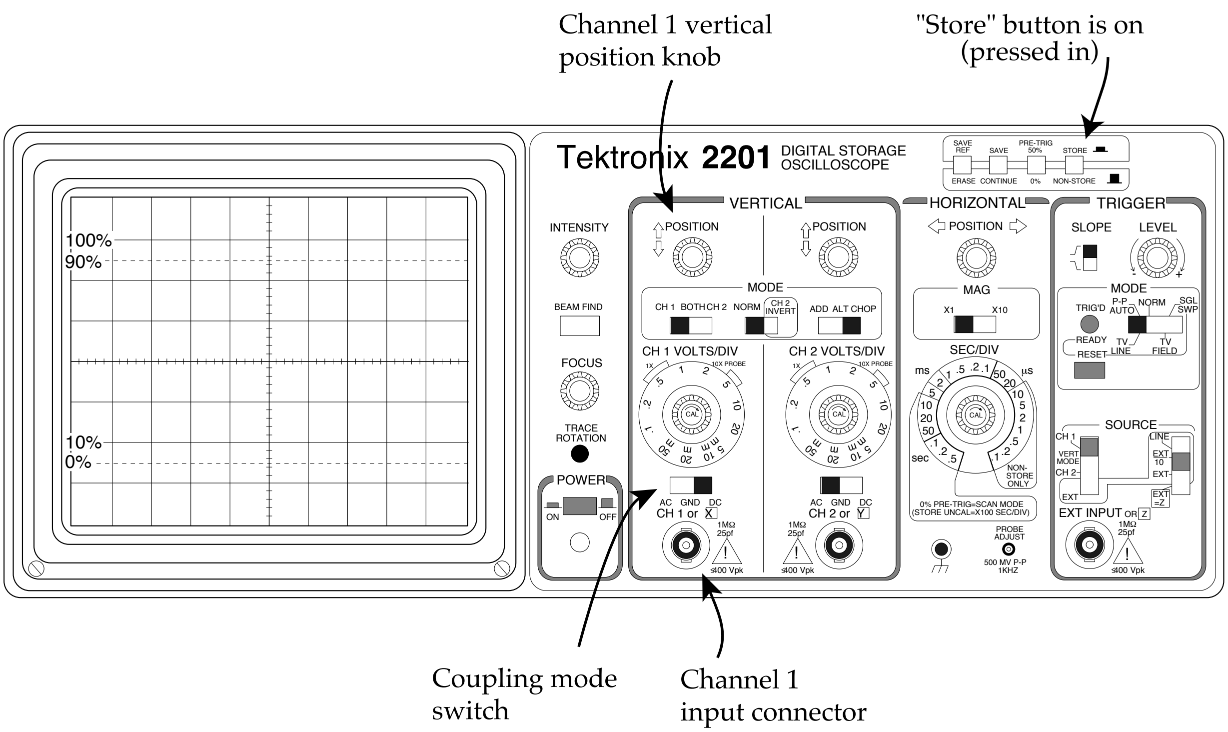
3. To get a more precise measurement of the period of the oscillating voltage, should you turn the SEC/DIV knob clockwise or counterclockwise? Try it.

Part II: Time dependent RC and LR circuits

NOTE: The remainder of the lab is probably too long for the time you have left; your GSI will direct you to which parts of the lab you must complete. Make sure you understand at least questions 1 through 4 before you leave.

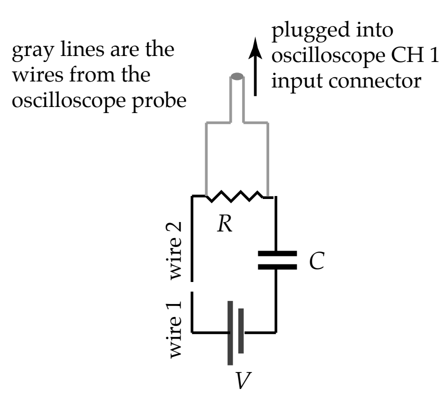
Technical stuff

Adjust the oscilloscope as shown here.



- ♦ Set SEC/DIV to .5 seconds.
- ♦ Set CH 1 VOLTS/DIV to .5 volts.

In all of the experiments, you'll build a simple circuit, and then use the oscilloscope probe to measure the voltage (potential difference) across a circuit element as a function of time. For instance, this set-up shows how you'd measure the voltage across the resistor in an RC circuit. Notice that the circuit starts out "open"; current cannot yet flow around it. You'll "close" the circuit by touching wire 1 to wire 2.

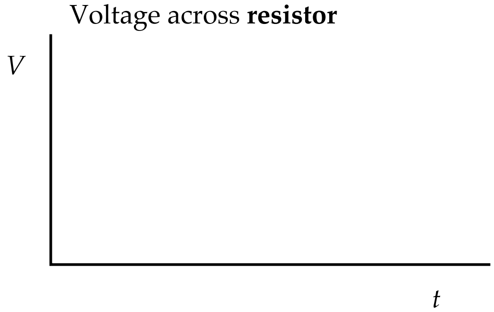


1. Consider a simple RC circuit, with the battery, resistor, and capacitor hooked up in series. Suppose you want to use the oscilloscope to measure the <u>current</u> through this circuit as a function of time. *How can you do it?* (Remember, the oscilloscope can only be used to graph the *voltage* across one or more circuit elements.) We want the graph to have the right general shape; but it need not be scaled properly. In other words, it can be "too tall" or "too short," as long as it has the right shape.

2. For this RC circuit, how can you get the oscilloscope to measure the *charge* on the capacitor as a function of time?

3. Suppose the capacitor is initially uncharged, and the circuit is closed at time $t = 0$. As your prediction, draw a rough sketch of the voltage across the *resistor* as a function of time, and <u>explain</u> your reasoning.

RC CIRCIT WITH BATTERY
Voltage across **resistor**



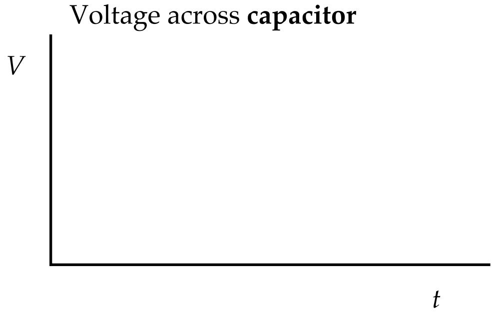
Now do the experiment, using a 1.5-volt battery, a microfarad (10^{-6} F) capacitor, and a megaohm (1 MΩ = 10^6 Ω) resistor. A 1 MΩ resistor is marked with color bands that are **brown, black, and <u>green</u>.** (There are other resistors that have brown, black and brown bands that we will use later in the lab. Don't use this now, since it is only a 100 Ω resistor.)

If the actual result differs from your prediction, sketch it on the graph as a dashed line, and explain what's going on below. **Before closing the circuit, make sure the capacitor is discharged, as demonstrated by your GSI. Each time you redo the experiment, discharge the capacitor again, so that it starts out with zero charge.**

TECHNICAL NOTE: because the oscilloscope has a 1 MΩ resistor at its input, which is in parallel with the 1 MΩ resistor in your circuit, the equivalent resistance of your circuit with the scope attached is (1/2)MΩ. Hence the time constant for your circuit will be half of what you were expecting. We are not concerned with this for the experiment.

4. Same as question 3, but now consider the voltage across the *capacitor* as a function of time. Graph and explain your prediction.

RC CIRCIT WITH BATTERY
Voltage across **capacitor**



Now run the experiment. Re-graph and re-explain, if the results differ from your prediction.

How is the voltage across the capacitor related to the voltage across the resistor as a function of time? Explain.

5. Suppose you place two 1-μF capacitors in series. Is the total capacitance now 2 μF or 0.5 μF? Don't just plug in a formulas; explain your answer conceptually, using diagrams and words. Hint: remember that $Q = C\,\Delta V$.

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6. Now consider an LR circuit, in which a battery, a resistor, and an inductor are hooked up in series. As you saw in question 1 above, graphing the voltage vs. time across the *resistor* tells you the current through the circuit as a function of time. That's because the voltage across the resistor is proportional to the current ($V = iR$). If the circuit is closed at time $t = 0$, what does the voltage vs. time graph across the resistor look like? Sketch and explain your prediction.

LR CIRCIT WITH BATTERY
Voltage across **resistor**

To do the experiment, replace the capacitors with a 4 H inductor, and replace the mega-ohm resistor with a 100 Ω resistor. The 100 Ω resistor is marked with bands that are **brown, black, and brown**.

Remember to put the oscilloscope probe across the resistor, not across the inductor. For best results, you may want to change the SEC/DIV setting to .1 seconds or even 50 milliseconds (ms). Also, lower the VOLTS/DIV setting to 50 millivolts. Does the graph come out as you expected?

7. Your inductor has an inductance of 4 H and a resistance of about 330 Ω. As you saw in question 6, the circuit's current eventually "settles" to some final value. If you replaced this inductor with a 330 Ω resistor, how would the graph of current vs. time differ from the one in question 6? Specifically,

(a) would the current shoot up to its final value more abruptly or less abruptly than it did in question 8? Explain.

(b) Would the current settle at the same final value as it did in question 8? Or would it settle at a higher or lower final value? Explain.

You need not test your predictions.